



TITLE

Renewables Based Power Generation for Kenya  
Pipeline Company

Name

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**Masters Project Report**

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**Prof:**



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## **NOMENCLATURE**

|                       |                                                  |
|-----------------------|--------------------------------------------------|
| <b>MW</b>             | <b>Megawatt</b>                                  |
| <b>HAWT</b>           | <b>Horizontal Axis Wind Turbine</b>              |
| <b>VAWT</b>           | <b>Vertical Axis Wind Turbine</b>                |
| <b>H-VAWT</b>         | <b>H-Shaped Vertical Axis Wind Turbine</b>       |
| <b>V-VAWT</b>         | <b>V-Shaped Vertical Axis Wind Turbine</b>       |
| <b>USD</b>            | <b>United States Dollars</b>                     |
| <b>SG</b>             | <b>Synchronous Generator</b>                     |
| <b>AG</b>             | <b>Asynchronous Generator</b>                    |
| <b>DFAG</b>           | <b>Double Fed Asynchronous Generator</b>         |
| <b>DD</b>             | <b>Direct Driven Electrically Excited SG</b>     |
| <b>VAG</b>            | <b>Asynchronous Generator with Variable slip</b> |
| <b>PM</b>             | <b>Permanent Magnets</b>                         |
| <b>VS</b>             | <b>Variable speed</b>                            |
| <b>FS</b>             | <b>Fixed Speed</b>                               |
| <b>PV</b>             | <b>Photo Voltaic</b>                             |
| <b>MW<sub>p</sub></b> | <b>Peak Megawatts of Solar PV Power</b>          |
| <b>EMF</b>            | <b>Electromotive force.</b>                      |
| <b>MPP</b>            | <b>Maximum power point.</b>                      |
| <b>V<sub>oc</sub></b> | <b>Open Circuit Voltage</b>                      |
| <b>I<sub>sc</sub></b> | <b>Short Circuit Current.</b>                    |
| <b>PWM</b>            | <b>Pulse Width Modulated</b>                     |
| <b>TES</b>            | <b>Thermal energy Storage</b>                    |
| <b>CAES</b>           | <b>Compressed Air Energy Storage</b>             |
| <b>SMES</b>           | <b>Superconducting magnetic energy storage</b>   |
| <b>NiCad</b>          | <b>Nickel Cadmium</b>                            |
| <b>NiMH</b>           | <b>Nickel metal hydride</b>                      |
| <b>η</b>              | <b>Efficiency</b>                                |
| <b>PEMFC</b>          | <b>Proton exchange membrane fuel cell</b>        |
| <b>DMFC</b>           | <b>Direct methanol fuel cell</b>                 |
| <b>AFC</b>            | <b>Alkaline fuel cell</b>                        |
| <b>PAFC</b>           | <b>Phosphoric acid fuel cell</b>                 |
| <b>MCFC</b>           | <b>Molten carbonate fuel cell</b>                |
| <b>SOFC</b>           | <b>Solid oxide fuel cell</b>                     |
| <b>ZAFC</b>           | <b>Zinc air fuel cell</b>                        |
| <b>PCFC</b>           | <b>Photonic ceramic fuel cell</b>                |
| <b>FP</b>             | <b>Fluid power</b>                               |
| <b>SP</b>             | <b>Shaft power</b>                               |
| <b>KPC</b>            | <b>Kenya Pipeline Company</b>                    |
| <b>PS 21</b>          | <b>Pumping Station 21</b>                        |

|                         |                                                                |
|-------------------------|----------------------------------------------------------------|
| <b>UF</b>               | <b>Utilization factor.</b>                                     |
| <b>USD</b>              | <b>United States Dollars.</b>                                  |
| <b>MW<sub>p</sub></b>   | <b>Peak Megawatts of Solar PV Power</b>                        |
| <b>EMF</b>              | <b>Electromotive force.</b>                                    |
| <b>MPP</b>              | <b>Maximum power point.</b>                                    |
| <b>PWM</b>              | <b>Pulse Width Modulated</b>                                   |
| <b>N<sub>PV</sub></b>   | <b>No of PV Modules</b>                                        |
| <b>A<sub>M</sub></b>    | <b>PV Module area</b>                                          |
| <b>G<sub>tt</sub></b>   | <b>Total solar irradiance on tilted plane</b>                  |
| <b>η<sub>pt</sub></b>   | <b>Efficiency of power tracking equipment.</b>                 |
| <b>NOCT</b>             | <b>Nominal operating cell temperature</b>                      |
| <b>T<sub>r</sub></b>    | <b>Module reference temperature.</b>                           |
| <b>G<sub>bt</sub></b>   | <b>Beam component of solar irradiance- tilted Plane</b>        |
| <b>G<sub>dt</sub></b>   | <b>Diffuse component of solar irradiance on tilted plane</b>   |
| <b>G<sub>rt</sub></b>   | <b>Reflected component of solar irradiance on tilted plane</b> |
| <b>t<sub>h</sub></b>    | <b>solar incidence angle on tilted plane</b>                   |
| <b>t<sub>hz</sub></b>   | <b>solar zenith angle</b>                                      |
| <b>φ</b>                | <b>Latitude of the location.</b>                               |
| <b>δ</b>                | <b>Solar declination angle.</b>                                |
| <b>b</b>                | <b>Tilt angle</b>                                              |
| <b>ω</b>                | <b>Solar hour angle</b>                                        |
| <b>g</b>                | <b>Azimuthal angle.</b>                                        |
| <b>r<sub>g</sub></b>    | <b>Ground reflectivity</b>                                     |
| <b>G</b>                | <b>Global solar irradiance on a horizontal surface</b>         |
| <b>F<sub>1</sub></b>    | <b>Sky brightness coefficient.</b>                             |
| <b>Δ</b>                | <b>Sky brightness parameter.</b>                               |
| <b>V</b>                | <b>Wind velocity</b>                                           |
| <b>Z</b>                | <b>Height</b>                                                  |
| <b>x</b>                | <b>Power law exponent.</b>                                     |
| <b>P<sub>w</sub></b>    | <b>Wind turbine power output per sq meter.</b>                 |
| <b>P<sub>r</sub></b>    | <b>Wind turbine rated power per sq meter.</b>                  |
| <b>V<sub>ci</sub></b>   | <b>Turbine Cut in wind speed.</b>                              |
| <b>V<sub>co</sub></b>   | <b>Turbine Cut out wind speed.</b>                             |
| <b>V<sub>r</sub></b>    | <b>Wind Turbine rated wind speed.</b>                          |
| <b>A<sub>w</sub></b>    | <b>Wind Turbine swept area.</b>                                |
| <b>η<sub>WT</sub></b>   | <b>Wind turbine efficiency.</b>                                |
| <b>SOC</b>              | <b>State of charge.</b>                                        |
| <b>C'<sub>bat</sub></b> | <b>Available battery capacity, Ah.</b>                         |

|                    |                                                                 |
|--------------------|-----------------------------------------------------------------|
| $C''_{bat}$        | Nominal battery capacity, Ah.                                   |
| $I_{bat}$          | Battery current                                                 |
| $\sigma$           | Battery self discharge.                                         |
| $\delta_c$         | Battery temperature coefficient                                 |
| $T_{bat}$          | Battery temperature.                                            |
| $V_{bat}$          | Battery Voltage.                                                |
| $Y_{bat\ c}$       | Battery cycle life.                                             |
| $Y_{bat\ f}$       | Battery float life.                                             |
| $c$                | Charging constant.                                              |
| LCE                | Levelised cost of energy                                        |
| $LCE_{CDM}$        | Levelised cost of energy with CDM financing.                    |
| $CO_{PV}$          | Sum of the capital and maintenance costs of the PV system.      |
| $CO_W$             | Sum of the capital and maintenance costs of the wind system.    |
| $CO_{bat}$         | Sum of the capital and maintenance costs of the battery bank.   |
| $E_{an}$           | Annual energy supplied in kWh.                                  |
| CER                | Certified emission reduction.                                   |
| CDM                | Clean development mechanism.                                    |
| UNFCCC             | United Nations Framework Convention on Climate change.          |
| $BE_{electricity}$ | CO <sub>2</sub> baseline emission factor.                       |
| $EG_{baseline}$    | The net quantity of electricity to be displaced by the project. |
| $CER_{sale}$       | Total amount of money obtained from annual sale of CERs.        |
| PIN                | Project Idea Note.                                              |
| PDD                | Project Design Document.                                        |
| $Y_{PR}$           | Lifespan in years of project.                                   |
| LPSP               | Loss of power supply probability.                               |

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**TONY WASHIKA**

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## **1.0. INTRODUCTION**

### **1.1. Background**

The Kenya Pipeline Company is a government owned corporation in Kenya that is involved in Pumping of refined Petroleum products. To achieve this objective, the company has pumping stations spread out all across Kenya. Currently, 8 pumping stations along the Mombasa-Nairobi Pipeline have a peak demand of 3.2 MW each. Whereas, four pumping stations along the Nairobi –West Kenya Pipeline has a peak demand of 1 MW <sup>each</sup>.

During the year 2009, the cost of Electricity Purchased from the national grid increased from about USD 0.14 per kWh (Inc taxes), to about USD 0.2 per kWh (Inc taxes). This shot the company's projected Electricity bill to USD 23 Million for the financial year ending June 2010. The increase has been contributed by two reasons. Firstly, the major component of this increase has been caused by the fact that Hydro Power plants, which account for 70 % of Kenya's electricity were affected by a prolonged drought, and the fact that even when the dams are full, the growth in demand for Electricity outstrips the ability of the state utility to build new power plants. As a consequence, the state utility company in Kenya has resorted to expensive fossil fuel fired power plants in the short term, leading to a steep increase in tariffs. These steep price increases, coupled with the unreliability of the national grid has led to many companies in Kenya initiating their own power plants, an idea which is under consideration in this project. It is intended to build a power plant for one pumping station at Pumping Station 21 (Nairobi) on a pilot basis, and implement other power plants in other stations in phases. This project considers the initial power station and uses renewable sources of energy.

### **1.2. Objectives**

- Determine the load profile of the pumping station and size the power plant appropriately.
- Identify suitable options for power generation based on renewable sources like solar PV, Solar thermal, and Solar-Wind hybrid systems.
- Conduct cost-benefit analysis considering carbon credits along the CDM mechanism of the Kyoto Protocol and any other agreement to replace it.
- Determine which option is feasible, if any.

## **1.3. Literature Review**

### **1.3.1. Wind Turbines**

#### **1.3.1.1. Historical perspective**

Wind power has been utilized for at least three thousand years [1]. It has been used for very many different applications from propelling sailing boats to milling grain, pumping water and sawing wood. From the beginning of the twentieth century, use has also been made of wind power in electricity generation [2]. Wind power is currently one of the fastest growing renewable energy technologies in the world. By the end of 2002, the total installed capacity of wind power was 31,000 MW [3] worldwide. The installed capacity of wind power worldwide doubles every three years and reached 159,213 MW in 2009 [4].

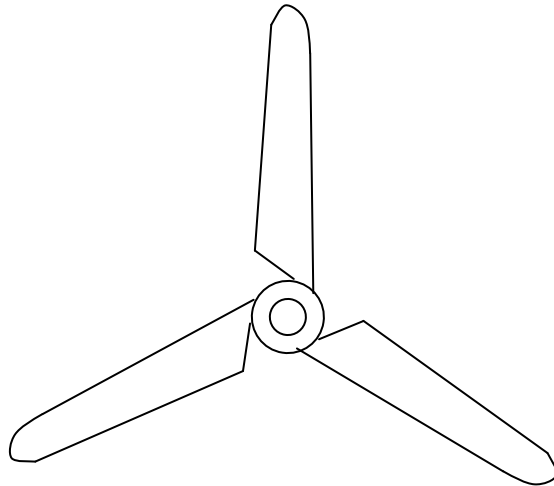
#### **1.3.1.2. Types of wind Turbines**

A wind turbine is a device that converts the kinetic energy of the wind to electrical energy. There are two broad classes of wind turbines: Vertical axis wind turbines (VAWTs) and Horizontal axis wind turbines (HAWTs). They can also be classified based on the principle of operation as Lift machines or drag machines. In this project, the classification of HAWT and VAWT is followed.

##### **1.3.1.2.1. Horizontal Axis Wind Turbines**

In a HAWT, the blades rotate in a vertical plane around a horizontal axis. A yawing mechanism is used to maintain the rotation axis in line with the wind direction. Unlike VAWTs, HAWTs have the disadvantage that they must change direction along with the wind. Both Rotor speed and power output are controlled by use of pitch and stall control. In pitch control, a blade pitch control mechanism is used, which involves pitching the rotor blades along their longitudinal axis. When this happens, rated power remains constant after rated power is reached, irrespective of increasing wind speed.

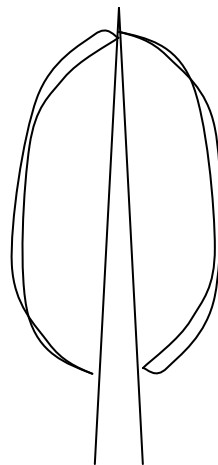
Stall control ensures a constant rotational speed irrespective of wind speed. Active stall control is a combination of pitch and stall control. In low wind speeds, the blades are pitched like in a pitch controlled wind turbine. Upon attaining rated capacity, control switches to stall control. HAWTs usually have two, three, or a larger number of blades. Those turbines having a large number of blades are referred to as high solidity devices and are commonly used for water pumping in rural farming areas. Low solidity HAWTs are commonly used for Electric power generation. Fig 1.1 below shows a 3-bladed HAWT.



**Fig 1.1: Three bladed wind turbine**

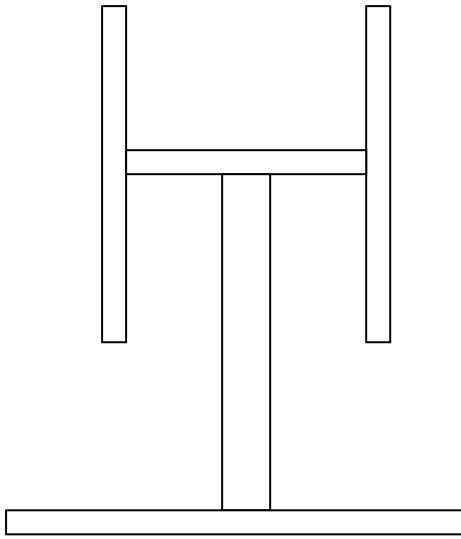
#### **1.3.1.2.2. Vertical Axis Wind Turbines**

In a VAWT, the rotor spins around a vertical axis. They can tap wind power from any direction [5] . There are various types of VAWTs in the market including the Darrieus type, the H-VAWT, and the V-VAWT. The Darrieus type VAWT was invented by the engineer Georges Darrieus in 1925 [6] . It has curved blades, and the ends are attached to the top and bottom of a vertical shaft, as shown below in fig 1.2.



**Fig 1.2: Darrieus type VAWT**

The Darrieus type VAWT has one huge draw back. To manufacture or transport the unusually shaped blades presents technical and logistical difficulties. This problem was addressed by developing straight bladed VAWTs such as H-VAWTs and V-VAWTs. The H-VAWT(see fig 1.3)consists of a tower capped by a hub. Two horizontal cross arms are attached to this hub. The function of the cross arms is to support the blades.



**Fig 1.3: H-VAWT.**

The V-VAWT consists of straight aerofoil blades attached to a vertical shaft in a V-shape. VAWTs have less efficiency than HAWTs and are not as economically competitive. There is currently a lot of research being undertaken on VAWTs.

#### 1.3.1.3. Physics of wind Turbines

The formula for Power extracted from the wind is derived from the expression for kinetic energy of air, which is half of the product of mass( $m$ ) and square of the velocity( $v$ ) of the air,

$$\text{i.e. Power } P = \frac{1}{2} * (\text{Mass flow}) * V^2 \quad (1)$$

Where  $P$  = Power in the moving air (watts)  
 $V$  = Velocity of the air.

$$\text{Mass flow rate} = \rho * A * V \quad (2)$$

Where,  
 $A$  = Swept area  
 $\rho$  = density of air

Combining equations (1) and (2), we get the equation:

$$P = \frac{1}{2} * \rho * A * V^3 \quad (3)$$



In practice, the power extracted from the wind is limited by the power coefficient  $C_p$  due to the fact that the velocity of the wind downstream of the wind turbine can not be zero.

Therefore, the power extracted from the wind is given by:

$$P = \frac{1}{2} \rho A V^3 C_p \quad (4)$$

The maximum theoretical value of  $C_p$  is 0.593 .

#### 1.3.1.4. Aerodynamics of wind Turbines

A wind turbine aerofoil experiences two forces: Drag force and lift force. The drag force is in the direction of the air flow. The lift force is at right angle to the direction of air flow, and is the force that causes modern wind turbines to produce power.

#### 1.3.1.5. Components of wind Turbines

The major components of wind turbines are:

1. Blades.
2. Hub
3. Transmission
4. Gear box
5. Generator
6. Yaw and pitch control systems.

The blades are the most critical component of the wind energy conversion system. The shape should be proper, and they should be strong, elastic, and lightweight. They are usually made from wood, glass reinforced polyester, or epoxy resin.

The blades are connected to the hub. The hub is then attached to a drive mechanism (transmission) that transmits the energy of rotation to a generator housed in an enclosure called a nacelle. The hub also contains blade pitching motors and mechanisms.

The gearbox performs the conversion of the low RPM power of the turbine rotor shaft to high RPM power for the generator shaft. The gear ratio in a modern wind machine is normally high, and the two types of gears used in the drive train are helical (parallel), and planetary (epicyclic).

The generator converts the mechanical motion of the generator shaft to electrical power. The two main types of AC generators are synchronous generators and asynchronous generators. Synchronous generators run in synchronism with the grid. A disadvantage is that they must be specifically designed in order to be used on large grid connected wind turbines. A particular type of synchronous generator is the permanent magnet generator(

PMG) , in which the magnetic field is produced by a permanent magnet. Asynchronous or Induction generators are more commonly used in grid connected wind turbines than the synchronous generator. This is mainly due to their simplicity and robustness. However, because asynchronous generators do not have permanent magnets, they need a separate power source for excitation power. Another disadvantage is that they also require high current during the startup of the generator in order to magnetize the core [7]. Table 1 below illustrates the various generator concepts:

**Table 1. 1: Generator concepts for wind turbines**

| Asynchronous Generators                                                                                                                                                                                                       | Synchronous generators                                                                                           | Others                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Squirrel cage induction generator</li> <li>• Wound rotor induction generator.</li> <li>• Doubly fed induction generator.</li> <li>• Optislip Induction generator(Vestas )</li> </ul> | <ul style="list-style-type: none"> <li>• Wound rotor generator.</li> <li>• Permanent magnet generator</li> </ul> | <ul style="list-style-type: none"> <li>• High Voltage generator.</li> <li>• Salient pole synchronous generator.</li> </ul> |

Adapted from [12].

By pitching the rotor blades about their longitudinal axis, the relative wind conditions and aerodynamic forces are influenced such that the power output does not increase beyond the rated power. In medium and large grid connected turbines, the pitching system is usually hydraulic, and is controlled by a PLC(Programmable Logic Controller). Some wind turbine manufacturers use servomotors to adjust the blade pitch, corresponding to a change in the wind speed in order to maintain fixed power output.

In Stall control, the turbine moves at constant speed irrespective of wind speed. Because of the aerofoil profile, the air stream conditions are affected such that turbulence is created during high wind speed (on the side of the blade not facing the wind) causing stall effect.

The yawing control mechanism has sensors for wind speed and direction, a motor to point the turbine in a specific direction, and a controller which will issue commands to the motor based on wind sensor inputs and control logic. The Yawing control system usually has a power source that is independent of grid supply so as to be able to shut down the turbine and go to stow mode in case of grid power failure.

### 1.3.1.6. Size and weight data for representative turbines

Table 1.2 below shows the size and weight data of representative turbines from several manufacturers.

**Table 1. 2: Size and weight data for representative turbines:**

| Type                  | Control System<br>S-Stall<br>P-Pitch<br>AS-Active stall | Rotor Diameter (m) | No. of Blades | Rated Capacity (kw) | Nacelle and rotor weight (kg) | Generator |
|-----------------------|---------------------------------------------------------|--------------------|---------------|---------------------|-------------------------------|-----------|
| Bonus 300 kW          | S                                                       | 31                 | 3             | 300                 | 14500                         | AG        |
| Bonus 1 MW            | AS                                                      | 54                 | 3             | 1000                | 63000                         | AG        |
| Bonus 2 MW            | AS                                                      | 76                 | 3             | 2000                | 125000                        | AG        |
| Dewind D8             | P                                                       | 80                 | 3             | 2000                | -                             | DFAG      |
| Enercon E-66/1.5      | P                                                       | 66                 | 3             | 1500                | 99590                         | DD        |
| Enercon E-66/1.8      | P                                                       | 70                 | 3             | 1800                | 101100                        | DD        |
| Nordex N-63           | S                                                       | 63                 | 3             | 1300                | 69400                         | SG        |
| Nordex N-80           | P                                                       | 80                 | 3             | 2500                | 119300                        | DFAG      |
| RePower 1000/57       | P                                                       | 57                 | 3             | 1000                | -                             | AG        |
| Vestas V 63-Optislip  | P                                                       | 63.6               | 3             | 1500                | 74000                         | VAG       |
| Vestas V 66-Optislip  | P                                                       | 66                 | 3             | 1650                | 78000                         | VAG       |
| Vestas V 80-Optislip  | P                                                       | 80                 | 3             | 2000                | 95000                         | VAG       |
| Vestas V-52 Optispeed | P                                                       | 52                 | 3             | 850                 | 32000                         | DFAG      |
| Vestas V-66 Optispeed | P                                                       | 66                 | 3             | 1750                | 80000                         | DFAG      |
| Vestas V-80 Optispeed | P                                                       | 80                 | 3             | 2000                | 95000                         | DFAG      |

Adapted from [1]

### 1.3.1.7. Current trends and research on wind turbines

A lot of work is currently underway to manufacture very large turbines, in the range of several megawatts. They are generally based on variable speed operation with pitch control [8]. Also, the wind power industry is working on improving the efficiency of the electrical side of the wind power conversion system. Research work is also underway to increase the generation voltage

on large turbines form 690V to between 3000 and 25000V[9]. Table 1.3 below gives data of large wind turbines whose development was underway in 2002.

**Table 1. 3: Data of large wind turbines under development in 2002.**

| Type                               | Country               | No of blades | Rated Capacity(kW) | Variable speed(VS), or Fixed Speed(FS) |
|------------------------------------|-----------------------|--------------|--------------------|----------------------------------------|
| NEG Micon 3000                     | Denmark & Netherlands | 3            | 3000               | VS                                     |
| DOWEC(Dutch Wind Energy Converter) | Netherlands & Denmark | 6            | 6000               | VS                                     |
| Wincon 2000                        | Denmark               | 3            | 2000               | FS                                     |
| Dewind D9                          | Germany               | 3            | 3500               | VS                                     |
| Jeumont                            | France                | 3            | 1500               | VS+PM                                  |
| Lagerway/ABB                       | Netherlands & Sweden  | 3            | 2000               | VS+PM                                  |
| Mtorres TWT 1500                   | Spain                 | 3            | 1500               | VS                                     |
| Vestas V 90                        | Denmark               | 3            | 3000               | VS                                     |
| Windformer/ABB                     | Sweden                | 3            | 3000               | VS+PM                                  |
| Enercon E112                       | Germany               | 3            | 4500               | VS                                     |
| RePower/N.O.K.5                    | Germany               | 3            | 5000               | VS                                     |
| Pfleiderer/ Multibird              | Germany               | 3            | 5000               | VS                                     |

Adapted from: [11]

The largest wind turbine in the world currently in operation is the 6 MW Enercon E126 , with a rotor diameter of 126m [10]. There are a lot of developments and related research going into off shore wind energy development, with the Norwegian Company Sway AS currently developing what is to be the world's largest wind turbine, a 10 MW ultra light weight gearless wind turbine with a unique downwind floating tower for deep water deployment in 2011 [11].

### 1.3.1.8 Wind Turbine Economy

Studies have identified that the cost of wind turbine manufacture declined by 20% every time the number of manufactured wind turbines doubled, and that the production of large scale grid connected wind turbines doubled every three years.[9]. To conduct an economic appraisal of wind energy, certain factors need to be taken into consideration [12].

- The total energy produced by the turbine annually.
- The capital cost and capital charge rate.
- Period over which the investment in the project is to be recovered.
- Operation and maintenance costs.

The cost of electricity produced by a wind turbine decreases with increase in capacity factor and vice versa. When calculations are done for new wind

turbines installed on land, in the 1.5 to 2 MW range, and wind turbine lifetime of 20 years, the economic appraisal is as follows[13]:

- The costs of investment are in the range of 1400 – 1800 USD/kW.
- Operations and maintenance (O&M) costs are 0.018 USD/kWh.
- Costs of wind energy range from 0.09 USD -0.13 USD/kWh at low wind speed sites, and 0.065 - 0.08 USD/kWh at windy sites.

### **1.3.2. Solar Photovoltaic Systems**

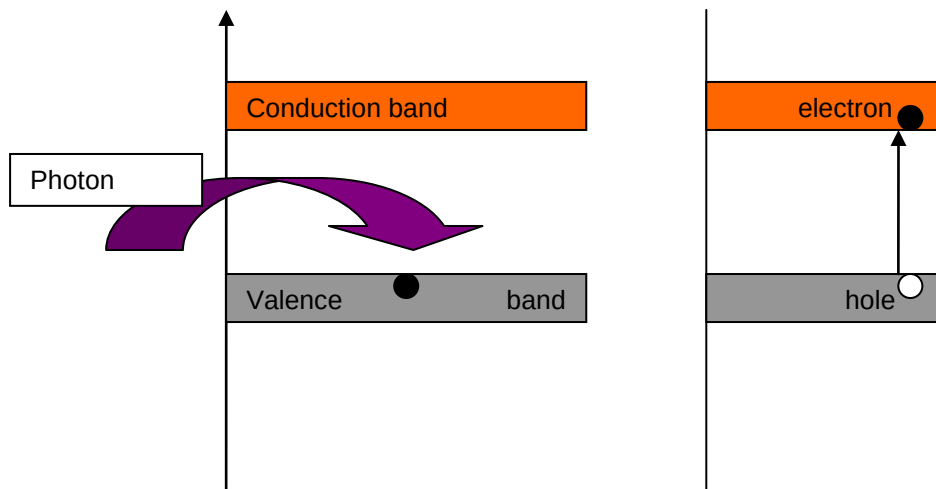
#### **1.3.2.1. Introduction to solar PV**

By solar photovoltaics is meant the generation of electricity directly from solar energy by use of solid state devices. The term photovoltaic is derived from two Greek words, Photos for light, and volt which is the unit of emf. The discovery of the photovoltaic effect is attributed to the French physicist Edmond Becquerel who published a paper in 1839 describing his experiments with this phenomenon[12]. The technology has been developed over the years to the present day with continuous advances in technology, materials, and efficiency. The highest efficiency of the single junction silicon solar cells has now reached 24% [14]. PV Cells are currently used in a very wide array of applications, from providing power for calculators, watches, water pumps, space stations, telecommunications, lighting, and other electrical appliances in remote locations. Because the array of applications is so vast, the demand for solar PV panels has increased tremendously. In 2003, 750 MW<sub>P</sub> of solar PV Panels were sold for terrestrial applications [15].

#### **1.3.2.2. The PV Cell**

The materials from which PV Cells are made is Silicon, Germanium arsenide (GaAs), Copper Indium Diselenide, and several other materials. In all PV Cells, a p-n junction is needed in order to initiate the PV effect. In a silicon PV cell, the p-n junction consists of a layer of p-type silicon and another layer of n-type silicon. The n-type silicon has excess electrons and the p-type silicon has excess holes. When these two dissimilar semiconductors are joined, an electric field is set up in the region of the junction.

When light of a suitable wavelength falls on the p-n junction, the light photons lose some of their energy to the electrons in the material, which become free to escape from the valence band, creating a free electron-hole pair. The electrons that have had their energy levels raised by photons tend to roll backwards to the n-region, creating an electric current flow. The process of creating electron-hole pairs is illustrated in fig 1.4

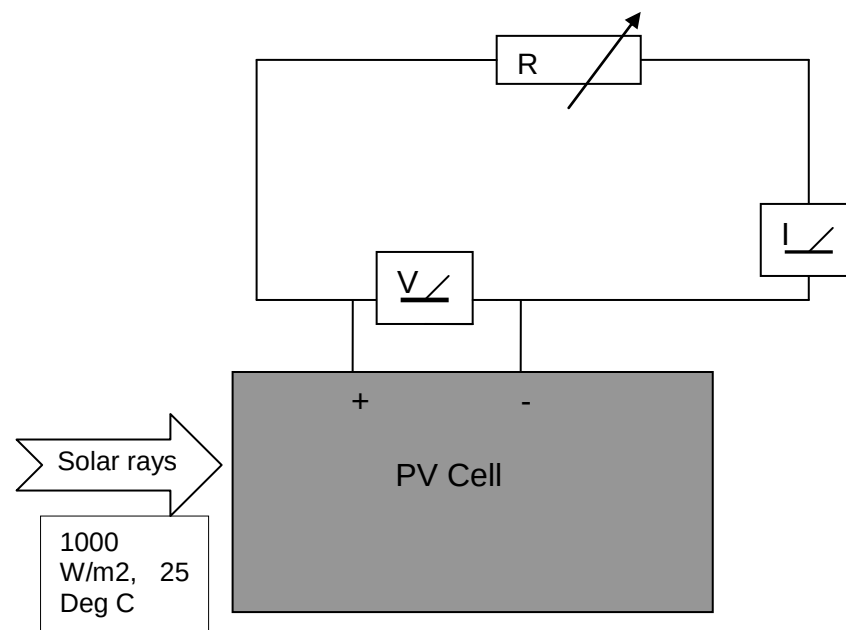


**Fig1.4: An electron in the valence band absorbing photon energy and moving to the conduction band.**

A single crystalline PV cell produces about 0.5V DC, and a current of 3A.

#### 1.3.2.3. I-V Characteristics of Silicon PV Cells and modules

It is possible to experimentally determine the I-V characteristics of PV cells. Here, a voltmeter is connected in parallel with the cell, and an ammeter in series. There is also a variable resistance  $R$  to vary the current.

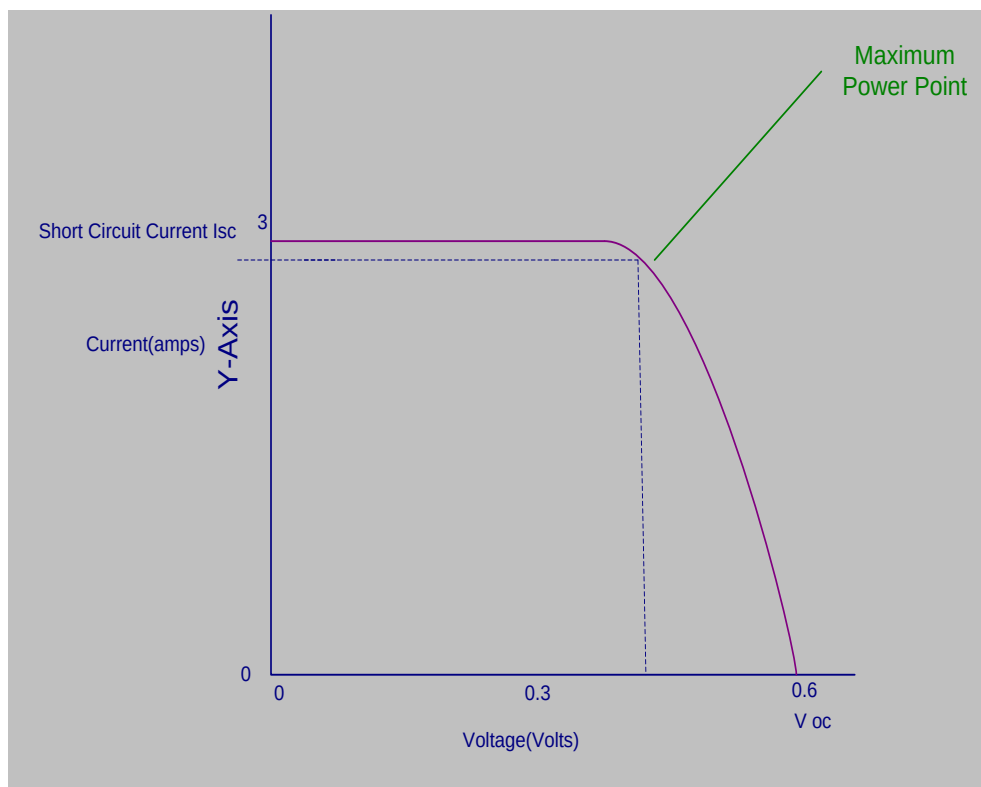


**Fig1.5: Determination of I-V Characteristics of PV Cell.**

When the value of  $R$  is at infinity, the current goes to zero and the PV Cell is at Open circuit voltage,  $V_{OC}$ , which is the highest voltage. When the value of  $R$  is zero, the current is at its highest value, the short circuit current  $I_{SC}$ .

There is a point at which the cell produces the maximum power, called Maximum power point, MPP. The above experiment is carried out at radiation intensity of  $1000\text{W/m}^2$ , and  $25^\circ\text{C}$ , but it could be carried out with less solar intensity. At lower levels, the shape of the I-V graph is the same as for  $1000\text{W/m}^2$ , but the short circuit current decreases proportionally to the solar intensity.

Fig 1.6 below gives the characteristics of a typical silicon PV cell. In real world conditions, solar intensity varies continuously. Some systems therefore incorporate an electronic device called a maximum power point tracking device so that the PV cell will always operate around the maximum power point.



**Fig1.6: Current-Voltage (I-V) Characteristics of a Silicon PV Cell.**

#### 1.3.2.4. PV Modules and arrays

Outputs of PV cells are limited to maxima of about 0.6V and 7A. This necessitates the connection of PV cells in series and parallel so as to increase the output current and voltage. It is typical for a PV module to consist of 36 PV cells connected in series to give a terminal voltage of about 17V. We also other combinations like 48 cells in series, etc. When modules are further combined either in series or parallel so as to increase System Voltage or

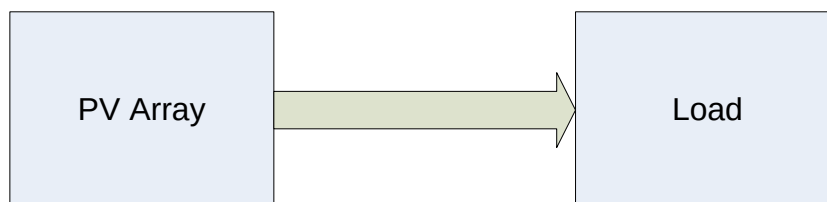
current, the combination is called an array. Fig 1.7 below shows a typical PV array used to power cathodic protection stations at Kenya Pipeline Company. The array has 24 PV Modules connected to give an output Voltage of 24 V DC. Kenya Pipeline Company uses these units to power remote cathodic protection stations along the pipeline. Several of them have been decommissioned due to the availability of mains power, after nearly 30 years of service.



**Fig1.7: BP Solar PV Array on a solar powered cathodic protection station at Kenya Pipeline Company.**

#### 1.3.2.5. PV System Configurations

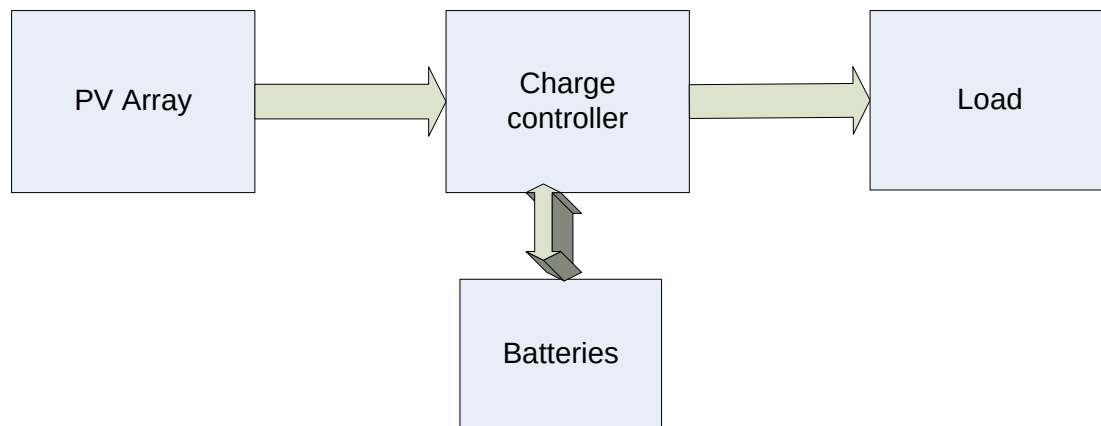
There are at least four possible configurations of solar PV Systems. The direct coupled system shown below in fig 1.8 is the simplest and cheapest.



**Fig1.8: Direct Coupled DC System**

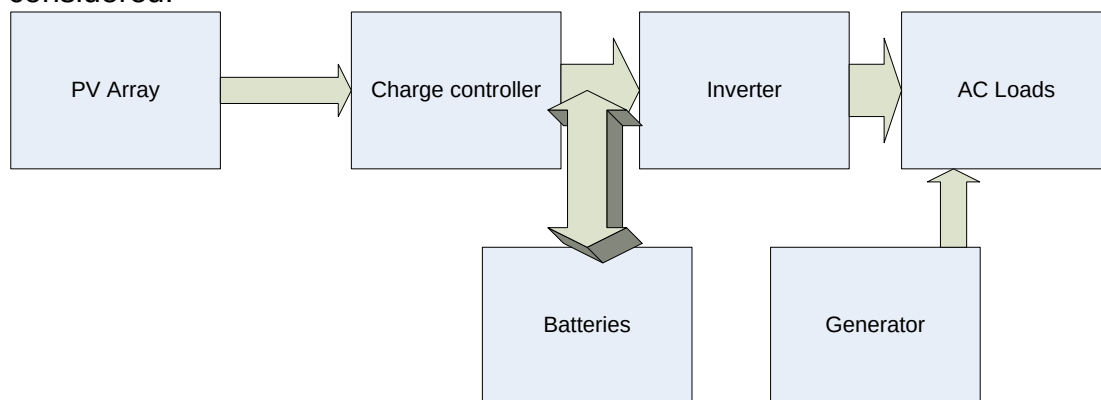
Storage batteries can be incorporated to provide continuous power during times of low solar intensity. A charge controller is incorporated to prevent battery over charge and over discharge as illustrated in fig 1.9.





**Fig1.9: DC System with Storage batteries.**

It is also possible to design a hybrid system incorporating several sources of energy. For example, during periods of low solar intensity, the batteries provide storage power, but if the period is prolonged, battery voltage begins to drop significantly and a generator is started as shown in fig 1.10 below. The inverter converts the DC from the array to AC so that AC loads can be fed. Hybrid systems involving solar and wind, or solar/wind/Fossil fuel can also be considered.



**Fig1.10: Hybrid System with battery storage and Diesel generator.**

The first three systems shown above are standalone systems. Fig 1.11 below is a grid connected system incorporating a grid interactive inverter. The system supplies AC power to the grid, hence the need for the inverter. The inverter monitors grid conditions, and can shut down power supply to the grid when it falls outside certain tolerances.



**Fig1.11: Grid connected system.**

### 1.3.2.6. PV System Components

The main PV System components are listed below

- i. PV Arrays
- ii. Charge controller.
- iii. Inverter
- iv. Batteries

#### 1.3.2.6.1 PV Arrays

PV Arrays have been discussed earlier in this literature survey in section 1.3.2.4. Further, the arrays output varies with temperature and solar intensity and these are considered when defining the power rating of the modules. Testing of PV modules uses Standard test conditions (STC), which is a light intensity of  $1000 \text{ W/m}^2$ , an air mass coefficient of 1.5(dimensionless), and an operating temperature of 25 deg C. These standards are fully defined in the standard IEC 60904.

#### 1.3.2.6.2 Charge Controller

The function of the charge controller is to control charging of the batteries by ensuring that the batteries do not overcharge or over discharge as this will lead to battery damage. The charge regulator (also called Voltage regulator) commonly has got four set points specified [16], these being:

1. Regulation set point- The maximum Voltage. On reaching this voltage, the charge controller either disconnects the charge voltage or regulates charge current. It requires temperature compensation.
2. Regulation hysteresis- The difference between regulation set point and the voltage at which maximum charge current is restored.
3. Low voltage disconnect-This defines the voltage at which the battery is disconnected during discharge.
4. Low voltage disconnect hysteresis-The difference between low voltage disconnect and the value at which reconnection can be allowed.

There are two main control modes for charge controllers, On/off control and constant voltage control. In on/off control, the controller acts as a switch, allowing all available PV charging current to the battery. In constant voltage control the charging current is tapered so that the battery can store all delivered current. In some cases, the controller modifies the voltage regulation set point in order to reduce gassing. Fig 1.12 below shows a photo of a typical BP Solar charge controller used for cathodic protection at Kenya Pipeline Company.



**Fig1.12: A 24 V DC BP Solar Charge controller at Kenya Pipeline Company**

Shunt regulators fix the battery voltage at some preset level by dissipating the array's excess power. In series regulators, the control element is placed in series between the array and battery. Pulse width modulated controllers apply repetitive current pulses with a varying pulse width. The sub array switching Topology is an advancement on the on-off control where instead of switching off the whole array, sub arrays are switched off during control. Self regulating systems work without a controller, and rely on the inherent self regulating features of the PV modules. However, self regulating systems involve too many compromises in design and there is a risk of overcharging the batteries on cooler days. Some charge controllers are maximum power point trackers. Maximum power point trackers are designed such that the array voltage is that which gives maximum power for a given insolation and temperature. The voltage is then converted by a DC-DC converter to a value suitable for the load and batteries. However, in some cases, the energy gains of these MPPT devices may not justify the cost, and therefore a cost-benefit analysis should be done before utilizing them.

#### **1.3.2.6.3 Inverter**

PV systems generate DC. However, some loads require AC. In such a case, an inverter is required to convert the DC to AC. The inverters typically convert the 12, 24, or 48V DC of PV systems to 110 or 240 V AC required by single phase AC loads or convert to much higher voltages for much larger loads or grid connection. Inverters in stand-alone systems need to supply constant voltage and frequency despite varying load conditions. They also need to be able to supply reactive power for Electric Motors and absorb reactive power supplied by capacitors in the load. Inverters commonly available can be classified as follows [17]:

- i. Light duty inverters, typically 100-10,000 W. Mainly for powering small domestic devices and characterized by low efficiency.
- ii. Medium duty inverters, typically 500-20,000 W. Suitable for a wide range of loads, but lack surge capacity required for starting AC induction motors.
- iii. Heavy duty inverters, typically 10,000 to 60,000 W continuous, but can supply surge loads of 30,000 to 200,000 W for startup of AC induction motors.

There are inverters rated up to 1 MW marketed by Companies like Satcon[18]. Traditionally, inverters had efficiencies of about 85%, but the new inverters in the market have got declared efficiencies of up to 97% [19]. There are two main types of inverter output wave forms; Sine waves and square waves. A lot of loads require true sine waves and may overheat if they are supplied with square waves.

#### 1.3.2.7. Recent advancements in Photovoltaics

Various advances have continued to be undertaken in PV technology. This has been on all fronts, i.e. PV cells, PV components, PV materials, PV applications, etc. For example, for a long time, most solar cells were made from mono crystalline Silicon, that is, Silicon with a single continuous lattice structure with almost no impurities. The maximum theoretical efficiency achievable in a single junction PV cell has been calculated to be about 30% if light trapping techniques are utilized to trap as many photons as possible [20]. However, multi junction cells where each junction is designed to trap photons of a different frequency can have a theoretical efficiency of up to 66%. It has been found though, that the highest efficiency achieved in commercially available single junction PV modules is 17% [21].

##### 1.3.2.7.1 Polycrystalline Silicon

Mono crystalline PV is costly because the manufacturing processes are slow, require skilled operators, energy intensive and are fabricated from extremely pure silicon [22]. One approach to solving this problem of cost is through the use of polycrystalline silicon. Polycrystalline silicon consists essentially of small grains of mono crystalline silicon. Polycrystalline silicon can then be made into solar cell wafers. Polycrystalline PV Cells are cheaper than their mono crystalline counterparts, but they are less efficient, achieving up to 14% efficiency. Research is currently going on into thin silicon on ceramics, and thin crystalline silicon on glass [23].

##### 1.3.2.7.2 Other materials

Another crystalline material that can be used for PV is Gallium Arsenide (GaAs). It has a high light absorption coefficient and only a small layer of material is required. It is more efficient than mono crystalline silicon and can operate at high temperatures, finding use in concentrating PV. They are also more expensive than silicon cells.

##### 1.3.2.7.3 Thin film technology

Thin film technology is 2<sup>nd</sup> generation PV. One of the materials used in thin film technology is amorphous silicon. Amorphous silicon (a-si) is a material in which the silicon atoms are less ordered than crystalline silicon. Amorphous silicon cells are cheaper than crystalline silicon. They are also better light absorbers and therefore thin films can be used. They are however very inefficient and have efficiencies between 4-8%.

#### 1.3.2.7.4 Other technologies

Other technologies being developed are photo electro chemical solar cell with module efficiency of 5%, and organic/Polymer solar cell. The longevity and long term stability of these two technologies are not well proven

#### 1.3.2.7. Economics of PV Systems

The cost of manufacturing a PV module consists of labour, material, capital and energy costs. The price of PV modules follows economics of scale and the more the kWp of modules produced per year, the lower the installed cost per kWp. A number of calculations have been carried out to determine the manufacturing cost as a function of the annual PV output. It can be seen from the table that the cost for silicon wafers can reach \$1/W<sub>P</sub> with large scale production. Table 1.4 below shows the calculations.

**Table 1. 4: Module manufacturing costs for various materials.**

| Cell material       | Module manufacturing Cost US\$/W <sub>P</sub> |                   |                    |                     |                     |
|---------------------|-----------------------------------------------|-------------------|--------------------|---------------------|---------------------|
|                     | 1 MW <sub>P</sub>                             | 10MW <sub>P</sub> | 60 MW <sub>P</sub> | 100 MW <sub>P</sub> | 500 MW <sub>P</sub> |
| Mono crystalline Si | 4.7                                           | 2.2               |                    | 1.4                 | 1.0                 |
| Poly crystalline Si | 4.7                                           | 1.9               |                    | 1.2                 |                     |
| Thin film           | 3.3                                           | 1.8               | 1.0                | 0.6                 |                     |

Adapted from [24]

In table 1.4, the column header in MWp refers to the capacity of a single plant in MWp per year.

#### 1.3.3. Energy Storage Systems

There are many definitions of energy storage, one of which is outlined here: "Energy storage in a power system can be defined as any installation or method, usually subject to independent control, with the help of which it is possible to store energy, generated in the system, keep it stored, and use it in the power system when necessary"[25]. Energy storage is required due to the intermittent nature of renewable energy. With the exception of tidal energy, the output of renewable sources of energy is continuously varying, and the output must be regulated by providing storage. Some reasons for employing energy storage in a power system are outlined below:

- To improve the operation efficiency.
- To reduce fuel use by energy conservation.
- No alternative energy source is available.
- To provide security of energy supply.

Some of the parameters to consider when evaluating energy storage are energy density and recovery rate. Energy density is the amount of energy that can be stored per unit mass. The recovery rate is the efficiency at which the energy can be recovered. The main energy storage technologies are outlined below:

- Flywheel

- Pumped hydro
- Compressed air energy storage
- Hydrogen as a synthetic fuel.
- Thermal energy storage
- Batteries.
- Ultra Capacitors.
- Superconducting magnetic energy storage

Table 1.5 compares the various energy storage technologies and shows considerations to make on choice of storage.

**Table 1. 5: Comparison of various energy storage technologies**

| Type              | $\eta$<br>% | Energy density,<br>kWh/kg          | Construction lead time,<br>years | Life time,<br>years |
|-------------------|-------------|------------------------------------|----------------------------------|---------------------|
| Fly wheel         | 85          | Steel-0.05<br>Carbon fiber-<br>0.2 | 3                                | 20                  |
| Pumped hydro      | 80          | 0.3/m <sup>3</sup>                 | 8                                | 50                  |
| CAES              | -           | 2 /m <sup>3</sup>                  | 3                                | 25                  |
| Hydrogen          | 50          | 38                                 | 3                                | 25                  |
| TES               | 75          |                                    | 12                               | 30                  |
| Pb-Acid Batteries | 80          | 0.04                               | 2                                | 10                  |
| Ultra Capacitor   | 80          |                                    | 2                                | 10                  |
| SMES              | 90          |                                    | 12                               | 30                  |

Adapted from [26]

### 1.3.3.1 Flywheel Energy Storage

A flywheel stores energy in the form of rotational kinetic energy. It uses an electric motor to accelerate the flywheel shaft to high speed and the same motor can be used as a generator to recover the energy. Their application in energy storage has been limited to where short bursts of high energy are required. The rotational energy of a rotating body is given by:

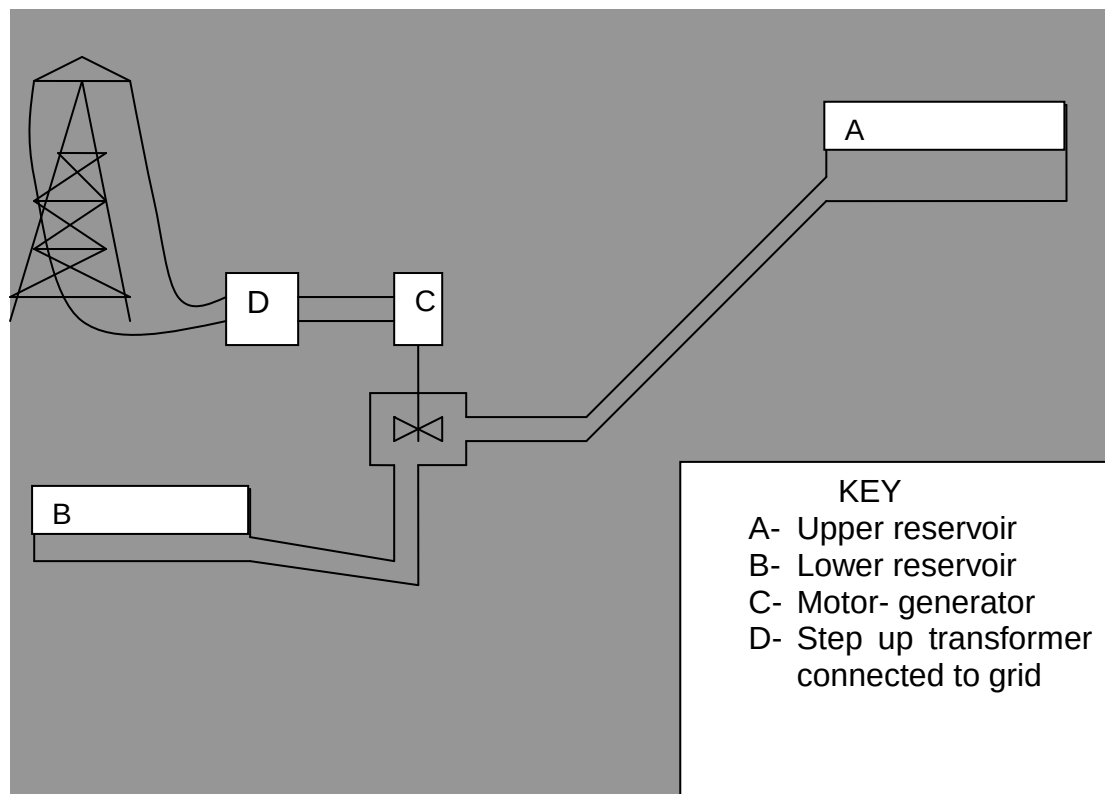
$$E = 0.5 I \omega^2 \quad (5)$$

Where  $I$  is the moment of inertia and  $\omega$  is the angular velocity. In a flywheel, dense material stores more energy, but it can fall apart as angular speeds increase due to centrifugal forces. As seen from table 1.5, fly wheels have an efficiency of about 85%. Some materials used for manufacturing fly wheels are glass fiber epoxy, mild steel, etc. Flywheels are widely used in automobile applications.

### 1.3.3.2 Pumped Hydro Energy Storage

Pumped hydro is a technique of energy storage where during off peak periods for a hydro electric power plant, the excess power is used to pump the water that has already gone through the turbines back into upper reservoir. It is a storage system based on potential energy. During peak load periods, the stored water is then discharged through the pumps, then acting as turbines, to meet the load demand.

It is very suited to storing renewable energy, especially wind, and is suited to mountainous areas. The choice of suitable storage sites for pumped hydro is determined by Topographical and geological characteristics. It is very common for a reservoir to be formed by damming a river, with the design allowing for the release in a controlled fashion of floodwater and the passage of fish. Currently, soil and rock fill dams are preferred over concrete due to appearance and construction cost [27]. A typical pumped hydro storage system is illustrated in Fig. 1.13 below:



**Fig1.13: Typical pumped hydro storage system.**

Pumped hydro storage has been successfully employed worldwide, with the Dinorwig Pumped hydro storage power station, rated 1800MW, and has the fastest response time worldwide among Pumped hydro schemes, being able to supply 1320 MW in 10 s of demand. [28]. Pumped hydro is the best developed and most reliable energy storage technology. The biggest problem with it is that it is not always possible to find two reservoirs with a head difference of 100m which are near the load. It also requires massive civil works, with associated environmental concerns like interference with aquatic life. For this reason, there have been proposals to site one of the reservoirs underground.

### 1.3.3.3. Compressed Air Energy Storage

Along with pumped hydro energy storage, CAES is also a potential energy storage system. It involves using excess energy to drive an air compressor. The compressed air is then cooled and stored in a pressure vessel. When the energy is required, the compressed air is fed into a gas turbine after being heated in a combustor. The gas turbine drives a generator to generate electricity. The volume of the air reservoir is determined by the amount of energy to be stored. The compressor rating is dependent on the time period taken to recharge the compressed air vessel. Along with TES and pumped hydro, CAES is suited to large scale energy storage applications. It has been found that large scale CAES where the air is stored in natural or artificial underground Caverns is very feasible.

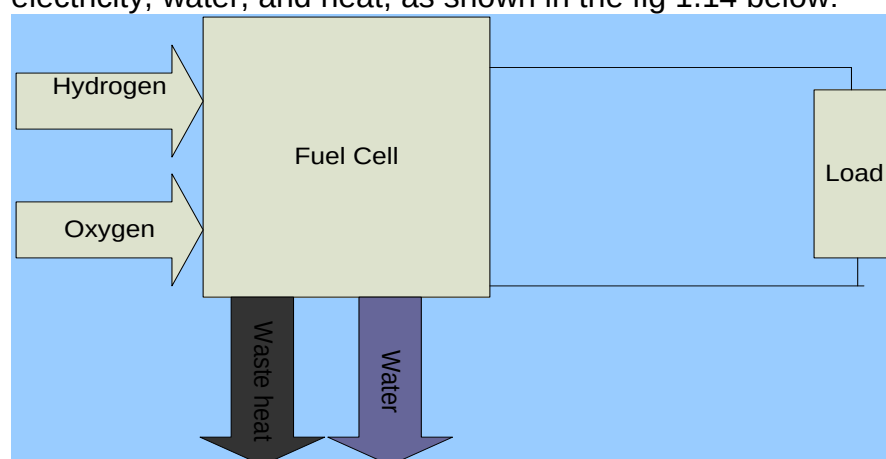
### 1.3.3.4. Hydrogen Storage

- Among all the methods of storage, hydrogen has the highest energy density at 38 kWh/kg. TES also has a similar energy density. The hydrogen has to be produced during periods of excess energy production by chemical means or an electrolyser. The hydrogen has to be stored in a central store with proper considerations taken as regards safety.

The cost of liquefaction is high, and the best option is as compressed gaseous hydrogen in underground caverns. The hydrogen could then be fired in a combustion chamber of a power plant, or fed into a fuel cell to produce electricity.

#### 1.3.3.4. 1. Fuel cells

A fuel cell is a device that converts chemical energy from reactants into electric power. It uses fuel, e.g. Hydrogen and an oxidant to produce electricity, water, and heat, as shown in the fig 1.14 below:



**Fig1.14: Block diagram of a fuel cell.**

Fuel cells are usually classified according to the type of electrolyte. These include, among others, PEMFC (Proton exchange membrane fuel cell) DMFC



(Direct methanol fuel cell), AFC (Alkaline fuel cell), PAFC (Phosphoric acid fuel cell), MCFC (Molten carbonate fuel cell), SOFC (Solid oxide fuel cell), ZAFC (Zinc air fuel cell), and PCFC (Photonic ceramic fuel cell). The properties of the various types of fuel cells are shown below in fig 1.6 below.

**Table 1. 6: Properties of fuel cells.**

| Fuel cell Type | Electrolyte                     | Charge carrier                | Operating temp, Deg C. | Fuel                                 | Electric efficiency, % | Power Range/ Application           |
|----------------|---------------------------------|-------------------------------|------------------------|--------------------------------------|------------------------|------------------------------------|
| AFC            | KOH                             | OH <sup>-</sup>               | 60-120                 | H <sub>2</sub>                       | 35-55                  | < 5 KW, niche markets.             |
| PEMFC          | Solid polymer                   | H <sup>+</sup>                | 50-100                 | H <sub>2</sub>                       | 35-45                  | Auto, CHP, 5-250 Kw                |
| PAFC           | Phosphoric acid                 | H <sup>+</sup>                | 220                    | H <sub>2</sub>                       | 40                     | CHP, 200Kw                         |
| MCFC           | Lithium and Potassium carbonate | CO <sub>3</sub> <sup>2-</sup> | 650                    | H <sub>2</sub> , CO, CH <sub>4</sub> | >50                    | 200kW to MW range CHP, standalone. |
| SOFC           | Solid oxide electrolyte         | O <sup>2-</sup>               | 1000                   | H <sub>2</sub> , CO, CH <sub>4</sub> | >50                    | 2 kW to MW range, CHP, standalone  |

Adapted from [29]

Fuel cell technology is just coming of age. Currently, the biggest disadvantages with fuel cells is the cost and slow transient response[30]. However, there is a lot of research into fuel cells, and it is just a matter of time before prices drop. Currently, a lot of fuel cells are used for transportation applications, and hybrid systems have been proposed that utilize wind and PV as energy sources and an electrolyser fuel cell system for energy storage.

#### 1.3.3.5. Thermal Energy Storage

TES is the direct storage of energy as heat in insulated solids or liquids. TES has been utilized since ancient times. Although it is often used in applications where cheap, low quality, low temperature energy is required, like space heating, it can also be used in conventional power plants. One key difference between TES and other forms of energy storage is that for the other forms of storage, the energy is generally recovered from the storage as electrical energy, but for TES, it is recovered as thermal energy. Broadly, there are two TES mechanisms:

- Sensible heat storage, based on the heat capacity of the storage medium; and
- Latent heat storage based on phase change energy in the storage medium.

Water is good for sensible heat storage because it is cheap and has a high heat capacity(4180J/Kg/K). It's disadvantage is it is only suitable as a storage medium between 5 and 95 deg C.

A decision on whether to use TES is mainly based on practical considerations, as TES is the only storage option where there isn't a separate storage plant, and this means that at times, it may not be possible to use the stored thermal energy. This means that TES in a plant reduces flexibility in operations.

#### 1.3.3.6. Battery Storage

There are many types of batteries presently used in standalone PV applications. These are indicated below:

- Lead acid
- Nickel-Cadmium(NiCd)
- Nickel –metal- hydride.
- Rechargeable alkaline manganese
- Lithium-ion
- Lithium-Polymer
- Redox.

The most commonly used batteries are lead Acid. Batteries can be used for both short term storage to effectively redistribute the load over a 24-hor period, or long term storage to cater for seasonal variations. Due to battery maintenance issues, it is required that batteries meet the following requirements in order to be used for long term storage [31]

- Long life
- Very low self discharge
- Long duty cycle
- High charge storage efficiency
- Low cost
- Low maintenance.

The battery capacity is defined as the maximum energy that can be extracted without battery voltage falling below a prescribed minimum value. The battery capacity is measured in Ampere hours (Ah) at constant discharge rate. (It could also be measured in kWh). Depth of discharge is the percentage of the rated battery capacity that can be withdrawn.

The performance factors of a battery storage system are outlined below:

- Lifetime-Measured as max No. Of charge discharge cycles.

- Cycle efficiency.
- Unit cost of energy stored.
- Depth of discharge per cycle.

The properties of various types of batteries are outlined in table 1.7.

**Table 1. 7: Properties of various types of batteries.**

| Battery type | Energy density(Wh/kg) | No. of cycles | Energy efficiency(%) |
|--------------|-----------------------|---------------|----------------------|
| Pb-Acid      | 40                    | 500           | 82.5                 |
| NiCad        | 60                    | 1350          | 72.5                 |
| NiMH         | 70                    | 1350          | 70                   |
| Li-ion       | 125                   | 1000          | 90.0                 |
| NaNiCl       | 125                   | 1000          | 92.5                 |

Adapted from [32]

Table 1.6 shows that despite being available for a long time, and with a lot of expertise available on them, lead acid batteries have the lowest energy density at 40 Wh/Kg. This is not a major drawback for stationery application like power plants. Nickel cadmium batteries have higher energy density and very high cycle life at 1350, but the drawback is high cost of purchase, being two to three times the cost of Pb-Acid batteries. There are also environmental concerns caused by the presence of cadmium. Lithium batteries have a potential for energy densities up to 2000 Wh/kg, but further optimization is still required as to life and system safety [33]. The other battery technologies are promising, but are still at various stages of development to make them more commercially viable. Despite the limitations of Lead acid batteries, like low cycle life and environmental concerns caused by lead, they still are widely used in PV applications due to low cost. Fig 1.15 below shows a photo of a lead acid battery bank typically used in a Kenya Pipeline Company remote solar PV installation used for cathodic protection.



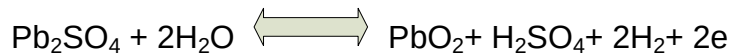
**Fig1.15: 24 V DC Battery bank used for Solar PV Storage at Kenya Pipeline Company.**

The battery bank consists of 12 No. series connected battery cells, each with a cell voltage of 2V, giving a total voltage of 24V. Lead Acid batteries have been used in 1 MW-2 MW Installations worldwide. The capital cost for NiCad batteries is about USD 250 – USD 300 per kWh, while that of lead Acid batteries is about USD 60 per kWh . The open circuit cell voltages for the two battery technologies are 1.35 V and 2.05 V respectively.

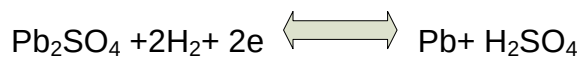
#### 1.3.3.6.1. Lead Acid Batteries

Lead Acid batteries are the most commonly used batteries in standalone Solar PV systems. They consist of an anode, cathode, and electrolyte. The battery consists of alternate pairs of plates one made of lead and the other of lead dioxide. The electrolyte is dilute sulphuric acid. During discharge, both electrodes are converted into lead sulphate. Charging restores the positive electrode to lead dioxide and the negative one to lead.

Reaction at positive plate



Reaction at negative plate



As previously stated, the Pb-Acid battery is the most popular battery for stationery power applications, and has been used in applications exceeding 1 MW.

#### 1.3.3.7. Ultra Capacitors

Energy in capacitors is stored in the electric field. A capacitor is a charge storage device. An ultra capacitor, also known as a double layer capacitor polarizes an electrolytic solution to store energy electrostatically. It is an electrochemical device, but no chemical reactions are involved in its storage mechanism. The process of charge and discharge is highly reversible, and can occur hundreds of thousands of times[35]. An ultra Capacitor stores much more energy than ordinary capacitors due to the much higher surface area created by the porous carbon electrodes and the small charge separation. However, they can not store as much energy as batteries, but can respond much faster to demand. Because of these characteristics, ultra capacitors find many applications in electric vehicles, particularly in storing energy recovered from regenerative braking.

### 1.3.3.8. Superconducting Magnetic Energy Storage

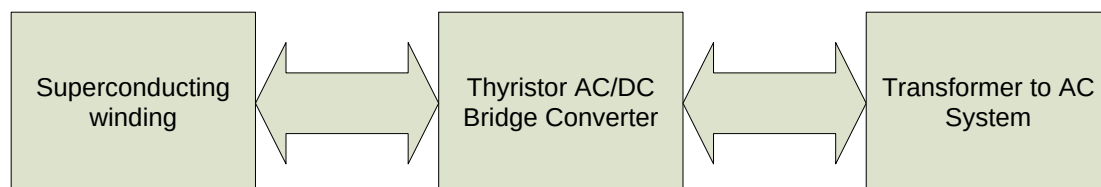
SMES is a technology which stores energy in the magnetic field created by the flow of DC in a cryogenically cooled superconducting material. The energy stored in the magnetic field can then be discharged when required. SMES can store large amounts of energy, and typical parameters for a large SMES storage system are given in table 1.8 below:

**Table 1. 8: Typical parameters for a large SMES storage system.**

|                          |                         |
|--------------------------|-------------------------|
| Total stored energy      | 10,000-13,000 MWh       |
| Available energy         | 9,000-10,000 MWh        |
| Discharge time           | 5-12 h                  |
| Max power                | 1000- 2500 MW           |
| Max Current              | 50-300 kA               |
| Max field                | 4-6 T                   |
| Mean diameter of winding | 300 m                   |
| Total height of windings | 80-100 m                |
| Mean depth below surface | 300-400 m               |
| Efficiency               | 85-90 %                 |
| Converter losses         | 2% of mean power rating |
| Refrigerator drive power | 20—30 MW                |

Adapted from [36]

A block diagram for a typical SMES system is shown below in fig 1.16.

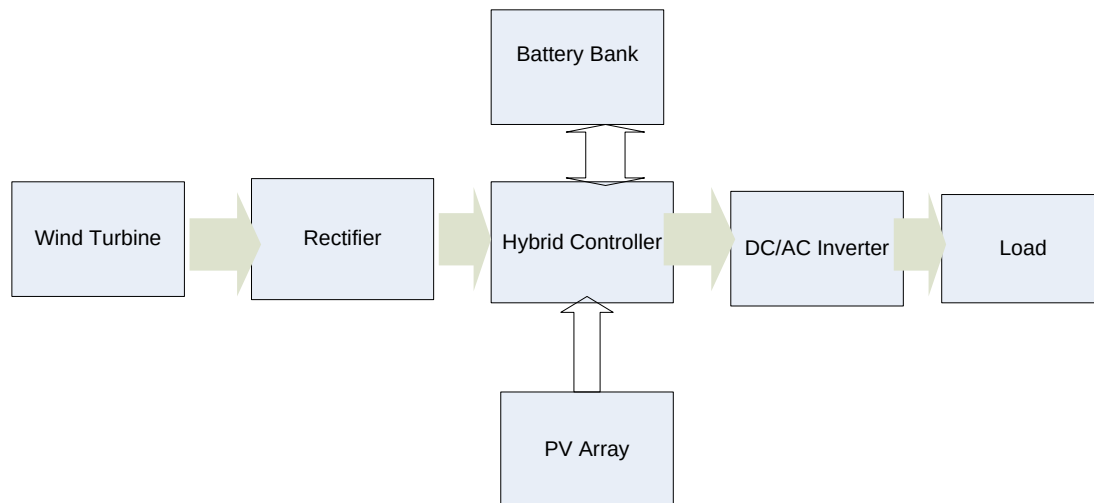


**Fig1.16: Block diagram for a typical SMES system.**

Technical challenges with the SMES system include compensating for stray fields and sudden appearance of normal conducting zones. This introduces complexity in the design of SMES systems. At present, most SMES projects are still concepts, and a lot of research is still on-going into reducing costs for full commercialization. One area of research is in the use of high temperature super conducting materials.

### 1.3.4. Hybrid Solar-Wind-Battery Renewable Energy Systems

Despite the fact that wind and solar energy are freely available and hold a lot of promise, they experience a major drawback in that they are both intermittent in nature. This is caused by the fact that they depend on weather and climatic conditions which are continuously changing. It has been found that hybrid systems which combine both wind and solar power generation sources are very suitable for remote standalone power supplies because the two sources of energy complement one another and reduce the cost of energy [37] and increase power supply reliability. A typical Solar-Wind-Battery hybrid energy system for an AC load is illustrated in Fig 1.17.



**Fig1.17: Block Diagram of hybrid Solar PV –Wind-Battery System**

In order to efficiently use Solar PV-Wind hybrid energy systems, the respective solar and wind Generators as well as the storage batteries have to be optimally sized. The process involves modeling the various components of the system, and then various sizing methods can be used to determine the optimal size of the system. For this, an evaluation is carried out on the basis of power supply reliability and system life cycle cost. Various techniques are then employed for sizing and optimization, some of which employ software [38]. A lot of research is currently underway into PV-Wind Hybrid Energy system optimization techniques [38].

### 1.3.5. Pumps

Pumps may be classified based on the applications they serve, the materials of construction, the fluid pumped, or orientation. However, the two basic groups of pumps are kinetic and positive displacement pumps.

#### 1.3.5.1.Kinetic Pumps

Kinetic pumps impart velocity and pressure to the fluid as it passes the pump impeller. Kinetic pumps are further divided into two groups of centrifugal pumps and vertical (or turbine) pumps. Centrifugal pumps are equipped with a volute, or casing, whose function is to collect the liquid discharged by the

impeller and convert some of its kinetic energy to pressure. Vertical pumps are equipped with an axial diffuser instead of a volute.

#### 1.3.4.1.1. Centrifugal Pumps

The capacity of a pump, or flow rate  $Q$  is the volume of fluid pumped per unit time. It is measured in cubic meters per second, Liters per second etc. Head,  $H$ , is the elevation of a free surface of water above or below a reference. Head is given in meters.



**Fig1.18: Flowserve Multistage Centrifugal Pump at Kenya Pipeline Company. The drive consists of a 1.6 MW ABB Motor and a Voith variable speed coupling.**

If the pressure difference built across the pump is  $\Delta p$ , the fluid power (Energy per second carried in the fluid as pressure and quantity),

$$FP = Q \times \Delta p \quad \text{Watts} \quad (6)$$

Where  $Q$  is the flow rate in  $\text{m}^3/\text{s}$  and  $\Delta p$  is the pressure change across the pump in  $\text{N/m}^2$ . Shaft power (the power transmitted to the pump by the shaft) is given by the formula

$$SP = \omega \times T \quad (7)$$

Where  $\omega$  is the angular velocity in  $\text{rad/s}$  and  $T$  is the torque transmitted in  $\text{Nm}$ . Substituting  $\omega$  with  $N$ , the speed in  $\text{Rev/min}$ , we get

$$SP = 2\pi NT/60 \quad \text{watts.} \quad (8)$$

$$\text{The overall efficiency } \eta = FP/SP. \quad (9)$$

Pumps having more than one impeller are called multistage pumps. This can be done either to increase the discharge pressure or the flow rate.

#### 1.3.5.2. Positive displacement Pumps

In a positive displacement pump, the moving element displaces the fluid from the pump casing (or cylinder), raising the pressure in the process. The three main groups of positive displacement pumps are reciprocating pumps, rotary pumps, and pneumatic pumps. In reciprocating pumps, the piston moves up and down. During suction stroke, the pump fills with fresh liquid, which is discharged through a non return valve. These pumps are known to develop very high pressures. In rotary pumps, the pump rotor displaces the fluid by rotating motion or rotating with orbiting motion. Compressed air is used to move the liquid in pneumatic pumps.



## 2.0. MODELLING

The work involves modeling of the entire hybrid system components, determining an algorithm indicating how the sub-models will work together and the logic of the simulation, to identify the main inputs and outputs of the models, to select the numerical values of the physical parameters and constants, and identify the software for model simulation.

### 2.1. Mathematical models of Components

Mathematical modeling of the components of the hybrid PV system has been done. The models are listed here below:

- The solar PV Model
- The wind turbine model
- The battery model
- The load demand model
- The Power supply reliability model
- The economic model

Each of these has been modeled separately as a sub model, and then all the sub models integrated together. The rest of this chapter is on the modeling, integration, and choice of simulation software.

#### 2.1.1. Solar PV Model

The equations governing the power generated by the solar PV panels are described in [34] and are given by,

$$P_{pv} = \eta_g * N_{pv} * A_m * G_{tt} \quad (2.1)$$

Where  $\eta_g$  is the instantaneous PV generator efficiency,  $A_m$  is the area of a single PV module,  $G_{tt}$  is the global solar irradiance incident on the tilted plane ( $W/m^2$ ), and  $N_{pv}$  is the number of PV modules.

The formula for  $\eta_g$  is given in the equation 2.2 below:  $\eta_g = \eta_r * \eta_{pt} * \{ 1 - \beta_t * (T_a - T_r) - \beta_t * G_{tt} * [(NOCT - 20)/800] * (1 - \eta_r * \eta_{pt}) \}$  (2.2)

$\eta_{pt}$ ( efficiency of power tracking equipment),  $\beta_t$ ( array efficiency temperature coefficient),  $NOCT$ ( nominal operating cell temperature), and  $A_m$  are

parameters dependent on the module type and are given by the module manufacturer.  $\eta_r^*$  is the PV module reference efficiency and  $T_r$  is the module reference temperature.

The global solar irradiance on a tilted plane,  $G_{tt}$  is described in [41] as:

$$G_{tt}=G_{bt}+G_{dt}+G_{rt} \quad (2.3)$$

Where  $G_{bt}$  is the beam component of solar irradiance on a tilted plane,  $G_{dt}$  is the diffuse component of solar irradiance on a tilted plane, and  $G_{rt}$  is the reflected component of the solar irradiance on a tilted plane.

### 2.1.2. The wind Turbine model

The power output of a wind turbine depends on the wind speed at the hub height and speed characteristics of the turbine. The wind speed at hub height can be calculated using the power law equation:

$$V_z= V_i *[Z/Z_i]^x \quad [39] \quad (2.15)$$

Where  $V_i$  and  $V_z$  are the wind speed at hub and reference heights  $Z$  and  $Z_i$ , and  $x$  is the power law exponent. The power law exponent  $x$ , varies with elevation, type of terrain, etc. The power output,  $P_w$  ( $\text{kW/m}^2$ ) from a wind turbine can be calculated as follows [40]:

$$\begin{aligned} P_w &= 0, & V < V_{ci} \\ P_w &= a \cdot V^3 - b \cdot P_r, & V_{ci} < V < V_r, \\ P_w &= P_r, & V_r < V < V_{co} \\ P_w &= 0, & V > V_{co} \end{aligned} \quad (2.16)$$

$$\text{Where } a = P_r / (V_r^3 - V_{ci}^3), \quad b = V_{ci}^3 / (V_r^3 - V_{ci}^3), \quad (2.17)$$

$P_r$  is the rated power per square meter ( $\text{Kw/m}^2$ ),  $V_{ci}$ ,  $V_{co}$ , and  $V_r$  are the cut-in, cut-out, and rated wind speeds of the wind turbine. The actual power output of the wind turbine is given by [40]:

$$P_{WT} = P_w \cdot A_w \cdot \eta_{WT} \quad (2.18)$$

Where  $A_w$  is the swept area of the turbine,  $\eta_{WT}$  is the efficiency of the wind turbine generator and corresponding converters.

### 2.1.3. The Battery model

It is proposed to use Lead Acid batteries for energy storage in this project. When the total output of the PV Arrays and the wind turbine is greater than the load, the batteries charge until the maximum battery state of charge is achieved, at which stage the battery charge controller disconnects the batteries from the hybrid charging system. One of the parameters used to model battery performance is state of charge. This has been adopted in this project.

The Battery's state of Charge (SOC) is represented during the charging process by [41]:

$$SOC(t+1) = SOC(t) * (1 - (\sigma * \Delta t) / 24) + \frac{I_{bat}(t) * \Delta t * \eta_{bat}}{C'_{bat}} \quad (2.19)$$

Where  $\sigma(t)$  is the daily self discharge rate.  $C'_{bat}$  is the available battery capacity in Ampere Hours,  $\Delta t$  is the time elapsed in hours from time  $t$  to time  $(t+1)$ , and  $\eta_{bat}$  is the battery charge/discharge efficiency.

The battery current  $I_{bat}$  is given by the formula below for a hybrid wind solar PV system (with no DC load):

$$I_{bat}(t) = \frac{P_{PV}(t) + P_{WT}(t) - P_{load}(t)}{V_{bat} \eta_{inverter}} \quad (2.21)$$

For purposes of model simplification,  $V_{bat}$  is taken as being equal to the nominal battery Voltage.

There are limitations on battery life. These are battery Cycle life  $Y_{bat c}$  and the battery float life  $Y_{bat f}$ .

### 2.1.4. The load Demand model

The total load consumed per month at Kenya Pipeline Company's Pumping Station 21 at Nairobi is indicated in Table 2.1. The consumption details cover a period of 1 year from January 2009 to December 2009.

**Table 2. 1: Monthly Load Consumption at KPC PS 21**

|   | Month         | Consumption(kWh) | Max kW |
|---|---------------|------------------|--------|
| 1 | January 2009  | 778700           | 1320   |
| 2 | February 2009 | 776500           | 1274   |
| 3 | March 2009    | 688500           | 1274   |
| 4 | April 2009    | 771800           | 1290   |
| 5 | May 2009      | 757700           | 1340   |
| 6 | June 2009     | 847700           | 1330   |
| 7 | July 2009     | 708900           | 1320   |
| 8 | August 2009   | 796500           | 1190   |

|    |                |        |      |
|----|----------------|--------|------|
| 9  | September 2009 | 784000 | 1300 |
| 10 | October 2009   | 755392 | 1311 |
| 11 | November 2009  | 767616 | 1297 |
| 12 | December 2009  | 754728 | 1285 |

Adapted from [42]

The pump operating hours during the year 2009 are as indicated in table 2.2 .

**Table 2. 2: Pump Operating hours at Pump Station 21**

| Month    | Down time Hours | Run hours |
|----------|-----------------|-----------|
| Jan 09   | 12.62           | 731.38    |
| Feb 09   | 26.10           | 645.90    |
| Mar 09   | 41.27           | 702.73    |
| April 09 | 31.77           | 688.23    |
| May 09   | 68.13           | 675.87    |
| June 09  | 19.67           | 700.33    |
| July 09  | 29.77           | 714.23    |
| Aug 09   | 35.08           | 708.92    |
| Sept 09  | 30.47           | 689.53    |
| Oct 09   | 34.10           | 709.9     |
| Nov 09   | 22.43           | 721.57    |
| Dec 09   | 27.13           | 716.87    |
| Total    | 378.54          | 8405.46   |

Adapted from [43]

At Pump Station 21, only one pump was run at a go and the other on standby during the year 2009 [44]. From table 2.2 above, the total pumping time in 2009 was 8405.46 hours. The pump Utilisation factor was given by

$$UF = \frac{\text{Total Operating hours}}{8760} = 96\% \quad (2.22)$$

The summarised load list for Pump Station 21 is as shown in table 2.3.

**Table 2. 3: Suumarised load list for PS 21:**

|                  |          | Continuous | Intermittent | Standby |
|------------------|----------|------------|--------------|---------|
| Item Description | Rated kW | kW         | kW           | kW      |
| 3.3 kV Motors    | 2014.2   | 954.55     | 0            | 954.55  |
| Bus 1 (415Vac)   | 188.75   | 120.01     | 5.27         | 62.75   |
| Bus 2 (415Vac)   | 24.57    | 26.35      | 28.59        | 0       |
| Sub total        | 2227.52  | 1100.91    | 33.86        | 1017.3  |

Adapted from [45]

From table 2.3 above, it is clear that PS 21 has a continuous load of 1100.91 kW and an intermittent load of 33.86kW. The maximum load at PS 21 is therefore given by :

$$P_{load\ max} = \text{Continuous Load} + \text{Intermittent load.} \quad (2.23)$$

Inserting the values from table 2.3 above, this gives a value of  $P_{load\ max} = 1134.77\text{Kw}$ . This value differs from the max kW Values indicated in table 2.2. This is because the figures in table 2.1 include power supplied to PS 10( a separate pump station), which at the time was having a common power supply with PS 21. The focus of this project is power supply to PS 21, and therefore a maximum load of 1134.77kW Will be considered.

The following information is used to model the load:

- The maximum continuous load is 1134.77 kW.
- 96% of the time, the load fluctuates between 1100.91 kW and 1134.77 kW.
- 4% of the time, the load fluctuates between 0 kW and 33.86Kw.
- From table 2.3, the rated load is 2227.52Kw.( The inverters should be designed to meet this load demand as there are occasions when the standby pump is started while the first one is still operating for about 5 min).

The load is modelled as follows:

- For simplicity, the load is taken as 1135kW Continuous, i.e  $P_{load(t)} = 1135\text{ kW}$ . This is not far from the actual situation because that is the load 96% of the time. Also, if unplanned and unscheduled stoppages were eliminated, the entire pumping station would be off for only 2hours per month for stock taking, and 8 hours per year for High Voltage maintenance, which works to a utilisation factor of 99.6%. The utilisation factor drops to 96% due to equipment breakdowns and other unforeseen factors [43].

### 2.1.5. The Power Supply Reliability Model

Due to the intermittent nature of wind and solar energy, power reliability analysis plays a key role in the overall Simulation. Various concepts can be adopted, e.g Loss of power supply Probability (LPSP) and Loss of Load Probability in the modelling. LPSP is defined as the probability that an insufficient power supply results when the hybrid power supply system is unable to satisfy the load demand [46].

However, in this project, various constraints have been proposed to model power supply reliability. These are outlined below;

- i. Maximum annual capacity shortage. This is the maximum allowable value of the capacity shortage fraction, which is the total capacity shortage divided by the total annual electric load. The capacity shortage is the shortfall between the required operating Capacity and the actual operating Capacity. The required operating capacity is the sum of the electric load and the required operating reserve.
- ii. Required operating reserve, the minimum amount of operating reserve that the system will be capable of providing.

- iii. Unmet Load, i.e the electrical load that the power system is unable to serve. A minimum value of unmet load for the year is set as a constraint in the power supply reliability model.

If the system does not meet the any of constraints provided above, it is declared infeasible.

#### 2.1.6. The Economic model

During Optimization, the hybrid systems must satisfy both reliability and economic requirements. A key requirement is to keep the cost of energy as low as possible. In this project, the concept of Levelised cost of energy, LCE is adapted for cost analysis.

The Levelised cost of energy is defined as the total cost of the hybrid system divided by the energy supplied from the hybrid system[49].

This is illustrated in equation 2.29 below:

$$LCE = \frac{(CO_{PV}/Y_{PV}) + (CO_W/Y_W) + (CO_{Bat}/Y_{Bat})}{E_{an}} \quad (2.29)$$

Where LCE is the levelised cost of energy in \$/kWh,  $CO_{PV}$  is the sum of the capital cost and indirect costs in the lifespan of the PV System;  $Y_{PV}$  is the lifespan in years of the PV system;  $CO_W$  is the sum capital cost and replacement /maintenance cost in the lifespan of the whole wind power generation system;  $Y_W$  is the lifespan in years of the wind system;  $CO_{Bat}$  is the sum cost of the capital cost and maintenance cost in the lifespan of the Battery bank;  $Y_{Bat}$  is the lifespan in years of the battery bank; and  $E_{an}$  is the annual energy supplied from the hybrid energy system.

It is also proposed to pursue sell of Certified emission reduction (CER) credits under the Kyoto Protocol's Clean development mechanism to parties based in Annex 1 (developed Countries) countries. A preliminary analysis shows that the project would most likely qualify for CDM financing under renewable energy generation. The clean development mechanism, CDM is one of the mechanisms that was set up under the Kyoto Protocol. Under the CDM Mechanism, a country with an emission reduction or emission limitation commitment (Annex 1 country) implements an emission reduction Project in developing countries. The credits obtained from this emission reduction Project are then used by the developed (Annex 1) Country to meet its obligation under the Kyoto Protocol. The Project developers in the developing country would then be paid for the CERs, thus obtaining CDM financing. The current Project would qualify because it would replace some electricity from Kenya's national grid, of which 29% is generated from fossil fuel oils [55]. It also qualifies under the Key requirement of financial additionality. The additionality criteria as stated in the Kyoto Protocol is that reductions in emissions should be additional to any activity that would occur in the absence of the certified project activity [56]. The purpose of the additionality criteria is to ensure that CDM does not fund projects that would otherwise go ahead. In the current Project, the barrier to implementation shall be financial, as the cost

of hybrid solar wind energy is still not yet comparable to traditional sources of energy[57].

However, the process for approval for CDM financing is long and tedious, as outlined below [58]:

- i. Initial feasibility assesment.
- ii. Development of a project design document.
- iii. Host country approval.
- iv. Project validation.
- v. Registration.
- vi. Emission reduction verification and credit issuance.

The entire project cycle shown above could take more than a year [59], and the project must meet the following conditions to be eligible[60]:

- Complies with the eligibility criteria, (e.g sustainable development criteria) of the host country and receives project approval by the host country.
- Provides real, measurable and long term benefits related to the mitigation of climate change using an approved baseline and monitoring methodology.
- Delivers emissions reductions that are additional to any that would occur in the absence of the project activity.
- Does not result in significant environmental impacts and undertakes public consultation.
- Does not result in the diversion of official development assistance.

The project meets the above conditions, and is a likely candidate for CER credits.

The economic model assumes two scenarios, and models the two. The first scenario, which presumes no approval for CER's has already been modelled. The economic model for the scenario involving payments for CERs is shown below:

The baseline emissions due to displacement of electricity from the national grid during the year are given by[61]:

$$BE_{\text{electricity}} = EF_{\text{electricity}} * EG_{\text{baseline}} \quad (2.30)$$

Where  $BE_{\text{electricity}}$  are the Baseline emissions due to displacement of electricity from the national grid during the year in tons of  $CO_2$ .

$EG_{\text{baseline}}$  (MWh) is the net quantity of electricity generated during the year due to project activity.

$EF_{\text{electricity}}$  is the  $CO_2$  baseline emission factor for the electricity displaced from the national grid during the year. From equation 2.29, the annual energy generated is given by  $E_{an}$ , (kWh) therefore,

The value of  $EF_{\text{electricity}}$ , the  $CO_2$  baseline emission factor for the electricity displaced from the national grid used in this model is the combined margin emission factor,  $EF_{\text{grid, CM, y}}$ . It's value is given by equation 2.32 below[62].

$$EF_{\text{grid, CM},y} = EF_{\text{grid, OM},y} * W_{\text{OM}} + EF_{\text{grid, BM},y} * W_{\text{BM}} \quad (2.32)$$

Where:

$EF_{\text{grid, OM},y}$  is operating margin CO<sub>2</sub> emissions factor in year y ( tons CO<sub>2</sub> per MWh),

$EF_{\text{grid, BM},y}$  is the build margin CO<sub>2</sub> emissions factor in year y ( tons CO<sub>2</sub> per MWh),

$W_{\text{OM}}$  is the weighting of operating margin emissions factor,

$W_{\text{BM}}$  is the weighting margin of the build margin emissions factor.

For wind and solar power generation activities,  $W_{\text{OM}} = 0.75$  and  $W_{\text{BM}} = 0.25$  for the first crediting period[62].

The process for calculating  $EF_{\text{grid, OM},y}$  and  $EF_{\text{grid, BM},y}$  shown in equation 2.3.2 is tedious and forms part of the PDD to be submitted to the Executive board of the UNFCCC. For this project, the value of emission factor used is that which is available with the US Department of Energy's Energy information administration, which gives the emission factor(combined) for Kenya as 0.393 as shown below[63].

$$EF_{\text{grid,CM},y} = 0.393 \text{ tons CO}_2 \text{ per MWh} \quad (2.33).$$

The amount of money paid from CERs annually is calculated from:

$$CER_{\text{sale}} = CER_{\text{price}} * BE_{\text{electricity}} \quad (2.34)$$

Where  $CER_{\text{sale}}$  is the total amount of money obtained from the sale of CERs per year, and  $CER_{\text{price}}$  is the market price of each CER sold to a party in an Annex 1 country, and  $BE_{\text{electricity}}$  is the baseline emission of CO<sub>2</sub> which will be saved under this project. The price of CERs on the market is not fixed and keeps fluctuating. Typically, prices have varied from a bottom price of about 6 USD(5 Euro) to about 24 USD (20 Euro) [64]. The lower margin is mainly dictated by the huge demand of the chinese market, and the upper is the European union allowance(EUA). For this project, the worst case of bottom price of USD 6 per CER Shall be adopted. The money shall be paid out over a period of 10 years, non renewable.

There are certain CDM Specific Project costs, which would not be incurred if it was not a CDM Project. The costs are outlined below, for small scale projects (Up to 15 MW Capacity):

[65]

- i. For Planning phase, i.e PIN, PDD, New methodology, validation, and registration, the costs could vary from USD 18,500 to 117, 000. For project analysis purposes, we take the average value of USD 67,750.
- ii. In the construction phase, the costs are negligible.
- iii. In the operations phase, the cost is 2% of CERs per year, i.e  $2\% * CER_{\text{sale}}$ , plus USD 5000 per year assuming annual verification.

The total of the CDM specific costs, over the lifetime is given by:

$$CO_{\text{CDM}} = CO_{\text{PL CDM}} + (2\% * CER_{\text{sale}} * Y_{\text{CDM}}) + 5000 * Y_{\text{CDM}} \quad (2.35)$$



Where:

$CO_{CDM}$  is the total costs related to CDM over the project lifespan,

$CO_{PL\ CDM}$  is the total CDM specific planning costs,

$Y_{CDM}$  is the period chosen for CDM credit sales in years, in this case 10 years.

Where  $CO_{annual\ CDM}$  is the annual cost of CDM financing, and  $Y_{PR}$  is the project lifespan for the solar wind hybrid PV system in years.

The levelised cost of energy with CDM financing is given by:

$$LCE_{CDM} = LCE(\text{without CDM}) + \left( \frac{\text{Annual CER Sales} - \text{Annual CDM Cost}}{\text{Annual energy produced}} \right)$$

Substituting the symbols,

$$LCE_{CDM} = LCE + \left( \frac{CER_{sale} - CO_{annual\ CDM}}{E_{an}} \right) \quad (2.37)$$

The report only presents the situation involving CDM financing for evaluation purposes and does not pursue it further in the model as the process of obtaining CERs and subsequent financing is a very tedious exercise.

## 2.2. Interconnection of the Sub-Models

The various sub-models used in the optimization process can be interconnected in various ways. For this project, the Hybrid Solar-Wind System Optimization Sizing Model tool (HSWSO) has been implemented[50]. The flow chart of the model is illustrated in Fig 2.1. As illustrated, the evaluation and optimization approach is as follows:

- i. The long term weather data for Pump Station 21 Nairobi is analyzed.
- ii. The three main parts of the system are modelled as illustrated in section 2.1; these are Solar PV System, The wind turbine generation system, and the battery bank.
- iii. The sizing is optimized using the power supply reliability model for reliability analysis, and LCE technique for economic analysis.

The logic of the simulation is as follows:

- i. Yearly weather data and model inputs and parameters are input into the various models. The PV model gives an output of the number of PV modules which is an input to the power supply reliability model.
- ii. The wind turbine model gives an output of  $P_{WT}$ , which becomes an input to the power supply reliability model.
- iii. The battery model gives out an output,  $C_{bat}$ , which also becomes an input to the power supply reliability model.

- iv. The load demand model outputs the load demand to the power supply reliability model.
  - v. Inputs to the power supply reliability model are fed by the designer.
  - vi. The power supply reliability model calculates the maximum annual capacity shortage, operating reserves, and annual unmet load. If any of these constraints are not met, the various inputs of the models are changed, e.g tower height in wind turbine model, capacity of the battery bank, number of PV modules, tilt angle, etc. If the constraints are met, the configuration is stored.
  - vii. All configurations that meet power supply reliability requirements are then subjected to the LCE model.
  - viii. The configuration with the least cost of energy is the optimal choice.
- The flow chart of the optimization model is given in fig. 2.1.

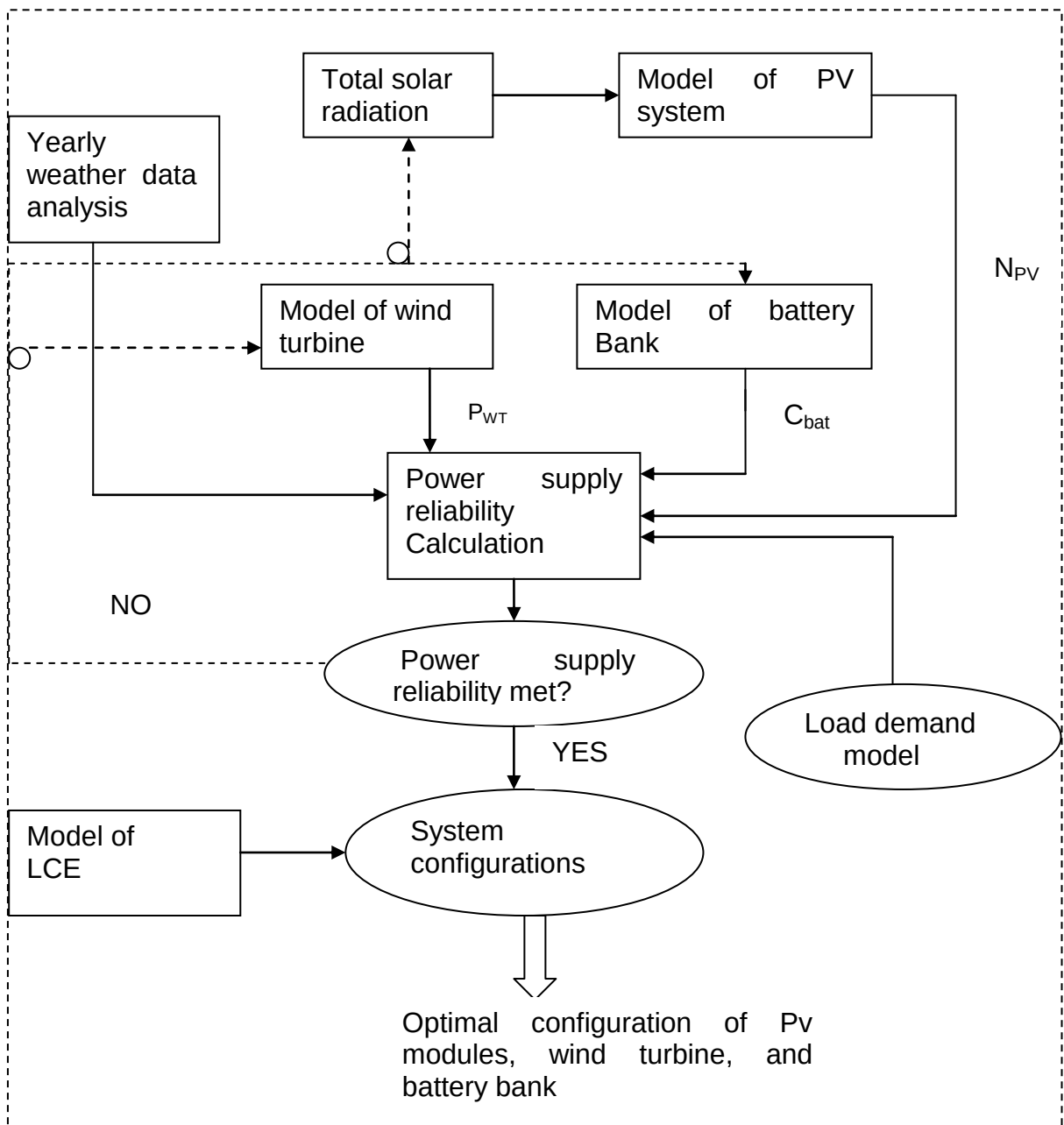


Fig 2. 1: Flow chart of the optimization model. Adapted from [50].

## 2.3. Inputs, outputs and model Parameters

### 2.3.1. Solar PV Model inputs, outputs, and parameters.

The solar PV module chosen is the Yingli YL 185 P series[51].The inputs of the Solar PV Modules are given below:

- $G$ - The global direct solar irradiance on horizontal plane( $W/m^2$ )
- $G_{bh}$ - The horizontal direct radiation.
- $G_{dh}$ - The global diffuse radiation on horizontal plane
- $T_a$ - Ambient temperature(K).
- $b$ - Tilt angle.

The outputs from the PV Model are outlined below:

- $N_{PV}$ - The number of PV modules.

Table 2.4 indicates the parameters of the Solar PV model:

**Table 2. 4: Input parameters used in the solar PV Model**

| Parameter                                   | Value        |
|---------------------------------------------|--------------|
| PV Module reference efficiency, $\eta_r$    | 14.3%        |
| Efficiency of MPPT equipment, $\eta_{PT}$   | 100%         |
| PV Cell reference temperature, $T_r$        | 298.15 K     |
| Nominal operating cell temp, NOCT           | 47 Deg C     |
| The area of a single PV Module, $A_m$       | 1.287 $m^2$  |
| Power coefficient of tempearture, $\beta_t$ | -0.0045/K    |
| Latitude, $\phi$                            | -1.2638 Deg  |
| Extraterrestrial Irradiance, $G_0$          | 1367 $W/m^2$ |

Adapted from [51] and [52].

### 2.3.2. Wind turbine Model inputs, outputs, and parameters.

The inputs to the wind turbine model are:

- $V_i$ , the wind speed at reference height  $Z_i$ .

The output of the wind turbine model are:

- $N_{WT}$ , the number of wind turbines.

Two different types of wind turbines are proposed for this project, the 1650 kW Vestas V82 and the 330 kW Enercon E33. The parameters of the two wind turbine models are indicated in table 2.5:

**Table 2. 5: Input parameters used in the wind turbine models:**

| Parameter                               | Vestas V82          | Enercon E33 |
|-----------------------------------------|---------------------|-------------|
| Hub height Z                            | 80 m                | 50          |
| Power law exponent x                    | 0.14                | 0.14        |
| Reference height $Z_i$                  | 10 m                | 10 m        |
| Swept area $A_w$                        | 5281 m <sup>2</sup> | 876.2       |
| Wind turbine efficiency $\eta_{WT}$     | 93.5%               | 93          |
| Cut in wind speed of turbine, $V_{CI}$  | 3.5m/s              | 3 m/s       |
| Cut out wind Speed of turbine, $V_{CO}$ | 25m/s               | 25 m/s      |
| Turbine rated wind speed, $V_r$         | 13m/s               | 13 m/s      |
| Rated power output of turbine, $P_r$    | 1650 kW             | 335kW       |

Adapted from [53] and [54]

### 2.3.3. Battery model Inputs, Outputs, and Parameters.

The inputs to the battery model are given below:

- Initial SOC.
- Elapsed time  $\Delta t$ .
- Battery temperature,  $T_{bat}$ .
- Wind turbine power output,  $P_{WT}(t)$ .
- Solar PV System power output  $P_{PV}(t)$ .
- Load at time t,  $P_{load}(t)$ .

The outputs from the battery model are as outlined below:

- The number of parallel battery strings,  $N_{bat}$

The battery chosen for the application is the Vision battery model 6FM 200D. It has a nominal capacity of 200 Ah , and a nominal voltage of 12 V DC. The parameters of the battery model are indicated in table 2.6.

**Table 2. 6: Battery model parameters.**

| Parameter                                     | Value                      |
|-----------------------------------------------|----------------------------|
| Battery hourly self discharge rate, $\sigma$  | 0.0000417                  |
| Battery nominal capacity, $C''_{bat}$ , in Ah | 200                        |
| Inverter efficiency $\eta_{inverter}$         | 97%                        |
| Battery cycle life, $Y_{bat c}$               | 650 cycles @ 50% discharge |
| Battery float life, $Y_{bat f}$               | 10 years                   |
| Depth of discharge DOD                        | 80%.                       |
| Nominal battery voltage, $V_{bat}$            | 12 V DC.                   |

#### 2.3.4. Power supply Reliability Model Inputs, Outputs, and Parameters.

The input values to the LPSP model are :

- Load  $P_{load}(t)$ .
- Photovoltaic power produced  $P_{PV}$ .
- Wind turbine power  $P_{WT}$ .
- Battery state of charge,  $SOC(t)$ .

The output from the Power supply reliability model is:

- Annual Capacity Shortage
- Operating reserve
- Unmet Load.

The parameters of the Power supply reliability model are indicated in table 2.7.

**Table 2. 7: Model Parameters for the Power supply reliability Model.**

| Parameter                                     | Value                                                                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Total No. Of hours, T                         | 8760                                                                                                                    |
| Nominal Battery Voltage, $V_{bat}$            | 12 V DC                                                                                                                 |
| Inverter efficiency , $\eta_{inverter}$       | 97%                                                                                                                     |
| Battery nominal capacity, $C''_{bat}$ , in Ah | 200                                                                                                                     |
| Required Operating reserve                    | 5% of hourly load, 5% of annual peak load, 5% of wind power output for the hour, 5% of solar power output for the hour. |
| Maximum annual capacity shortage              | 10%                                                                                                                     |

#### 2.3.5 Economic Model Inputs, Outputs, and Parameters.

The input values to the economic model are:

- Annual energy generated,  $EG_{baseline}$  in MWh.

The outputs are as follows:

- Levelised cost of energy, LCE in USD per kWh.

The parameters of the economic model are outlined in table 2.8:

**Table 2. 8: Model parameters for the economic model.**

| <b>Parameter</b>                                                       | <b>Value</b>                                                                             |
|------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Capital, replacement and maintenance cost of PV System.                | Capital- 1500 USD per kW<br>Replacement- 1500 USD per kW.<br>O&M- 0 USD per kW           |
| Cost of capital replacement, and maintenance cost of wind power system | Capital-800 USD per kW<br>Replacement- 800 USD per kW.<br>O&M- 8 USD per kW.             |
| Cost of capital , replacement and maintenance cost of batteries.       | Capital- 150 USD per 200 Ah.<br>Replacement- 150 USD per 200 Ah<br>O&M- 5 USD per 200 Ah |
| PV module lifespan, $Y_{PV}$                                           | 25 years                                                                                 |
| Wind system lifespan, $Y_W$                                            | 25 years.                                                                                |
| Battery system float life, $Y_{bat}$                                   | 10 years                                                                                 |
| Project lifespan, $Y_{PR}$                                             | 25 years.                                                                                |

Adapted from [38]

### **2.3.6 Load Model Inputs, Outputs, and Parameters.**

The inputs are:

- The load  $P_{load}(t)$

The outputs are:

- The annual energy output,  $E_{an}$ .

The parameters are,

- $P_{load}$ = 1135 kW Continuous
- Time,  $T$ = 8760 Hrs.

## **2.4. Optimization Software**

The simulation software chosen for this project is HOMER, developed by National Renewable Energy Laboratory(NREL), USA.

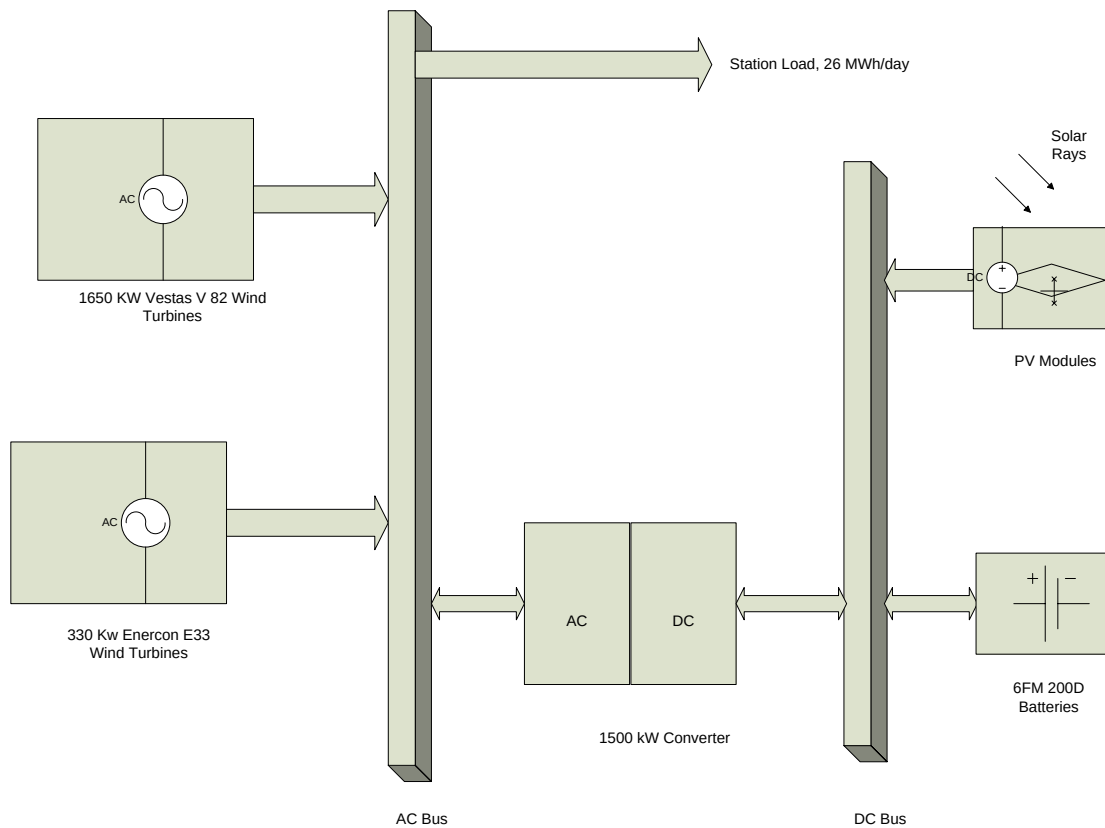
## **3.0. RESULTS**

### **3.1. Short notes on the findings**

The location for KPC Pumping Station 21(PS 21) is Nairobi, 1<sup>0</sup> 16'S, 36<sup>0</sup> 48'E, and Altitude 1624 m above sea level. The meteorological data used for the wind and solar resource are the TMY data for the Kenya Meteorological department station at the Jomo Kenyatta international Airport, which is the nearest Meteorological Station to PS 21, about 2 KM away. As mentioned earlier in the report, the main objective is to determine if any of the renewable power generation options are feasible. For this the system was modeled and simulated using software.

### **3.2. The Model and Setup of the Hybrid System**

The system was modelled using Hybrid Optimization Model for Electric Renewables,(HOMER version 2.68), a software developed by the US National Renewable Energy Laboratory(NREL). HOMER models a power system's physical behaviour and it's life cycle cost. Homer allows the modeler to compare various design options based on their technical and economic merits. It also assists in analysing the effects of uncertainty. HOMER can model both grid connected and off grid systems serving electric and thermal loads and comprising any required combinations of PV modules, wind turbines, small hydro, batteries, Biomass power, generators, etc. HOMER performs three tasks mainly: simulation, optimization, and sensitivity analysis. In the simulation process, HOMER models the performance of a particular hybrid micropower system configuration to determine it's technical feasibility and life-cycle cost. In the optimization process, HOMER simulates many different configurations in search of the one that satisfies the technical constraints at the lowest life-cycle cost. Finally, in the sensitivity analysis process, HOMER performs multiple optimizations under a range of input assumptions to gauge the effects of uncertainty. All the various components for HOMER were selected. The wind and solar resource inputs from TMY data for the area were input into the resource windows. Simulation, Optimization and sensitivity analysis was then carried out and the results are presented in this report. The Schematic of the Solar-wind-Battery hybrid system for the current project is shown in Fig 3.1.



**Fig 3. 1: Schematic of the hybrid system.**

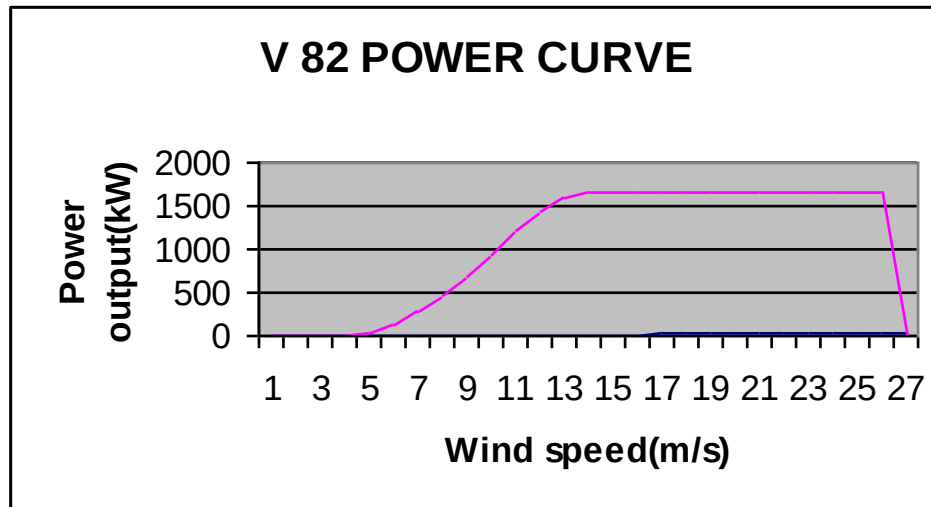
HOMER requires various inputs for the solar, wind resources and Components. Some of the inputs for the components are summarized in Table 3.1 .

**Table 3. 1: Inputs to the Software**

|                                 | PV                  | Wind Turbine 1       | Wind Turbine 2     | Battery (Vision 6FM 200 D)   | Converter                              |
|---------------------------------|---------------------|----------------------|--------------------|------------------------------|----------------------------------------|
| Rating (kW)                     | 0.185 kW per module | 1650 kW Per turbine. | 330 kW Per turbine | 200 Ah per battery.          | 1500 kW                                |
| Capital Cost (\$ per component. | USD 1500 per kW.    | USD 800 per kW.      | USD 800 per kW     | USD 150 per 200 Ah battery   | USD 600,000 for the 1500 kW Converter. |
| Replacement Cost(\$)            | USD 1500 per kW.    | USD 800 per kW.      | USD 800 per kW     | USD 150 per 200 Ah battery   | USD 600,000 for the 1500 kW Converter. |
| O&M Cost(\$/year)               | 0                   | USD 8 per kW/yr.     | USD 8 per kW/yr.   | USD 5 per 200 Ah battery /yr | 10,000                                 |
| Life time                       | 25 years            | 25 years             | 25 years           | Float life 10 years          | 15 years                               |

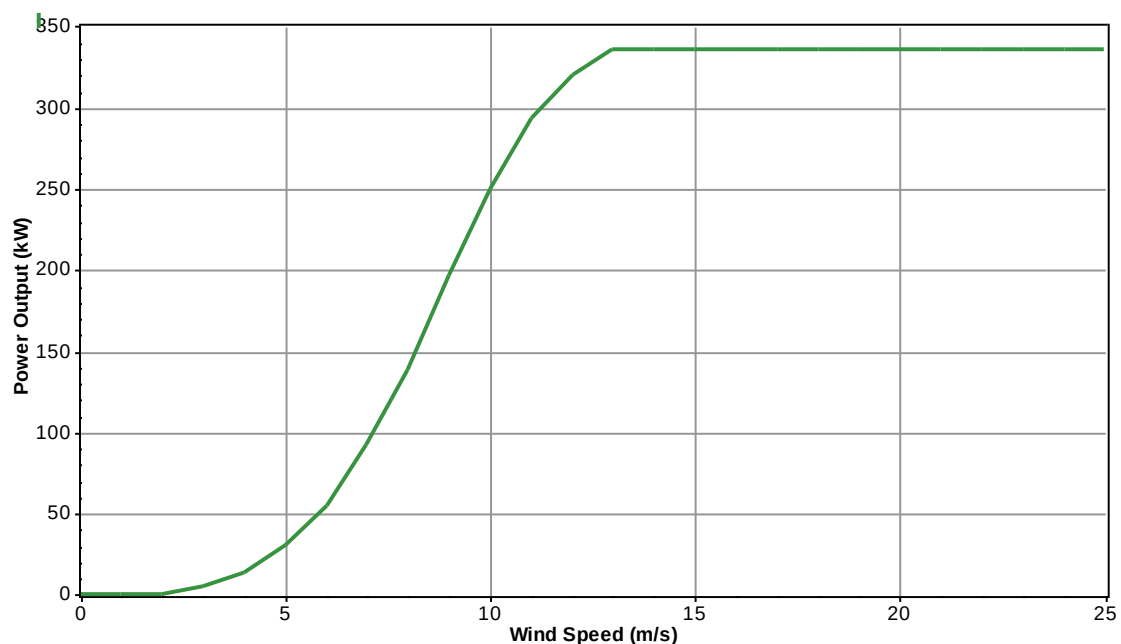


The wind turbines were selected from those suggested by HOMER on the consideration of low cut-in speed as the site is not very windy. The power curve of the 1650 kW Vestas V 82 wind turbine is shown in Fig 3.2.



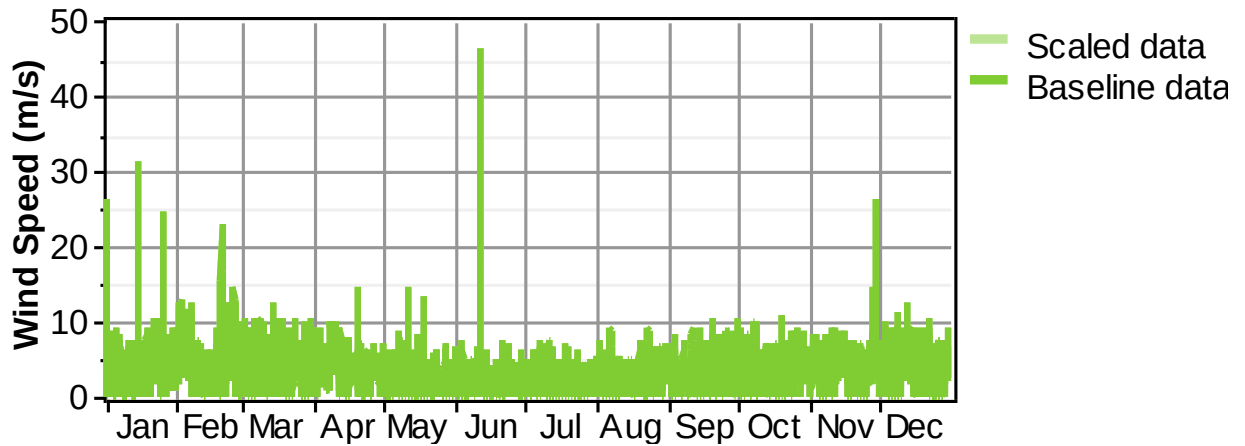
**Fig 3. 2: Power Curve for the GE 1,5sl wind turbine. Adapted from HOMER[66].**

The power curve for the 330 kW Enercon 33 wind turbine is shown in Fig 3.3.



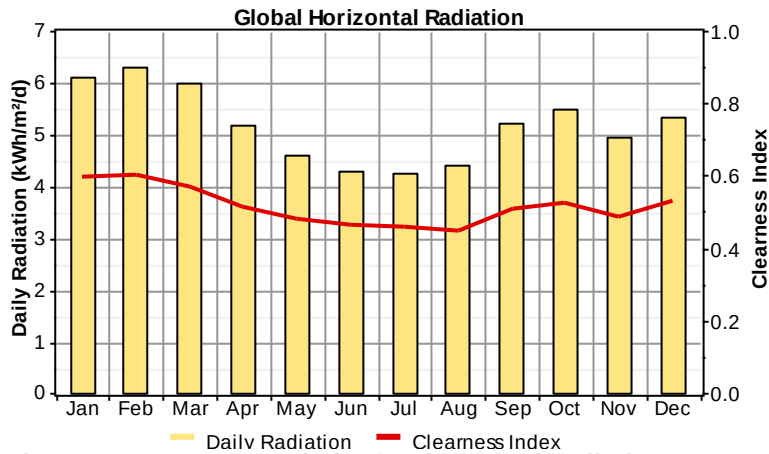
**Fig 3. 3: Power Curve for the Enercon 33 wind turbine. Adapted from HOMER [66].**

The Solar and wind resource data used is the hourly TMY data( 8760 hours) obtained from the the Solar and wind Energy Resource Assessment(SWERA) program[67]. Aplot of the hourly wind speeds is shown in fig 3.4.



**Fig 3. 4. HOMER Hourly plot of wind speeds for PS 21 area in Nairobi for the whole year..**

The plot shows that the wind regime is not very good.  
The HOMER plot for the Solar irradiation is shown in Fig 3.5.



**Fig 3. 5. HOMER Annual plot for the solar irradiation.**

For Economic modelling, in order to determine the net present cost of the various configurations, a real interest rate of 7.5% was adopted along with a Project life of 25 years. The HOMER system control inputs chosen were a simulation step time of 60 minutes, a dispatch strategy of cycle charging with set point for state of charge of 80%, and a system that allows two types of wind turbines (to allow more combinations).

The inputs considered for the PV modules were a derating factor of 100%, a tracking system incorporating vertical axis continuous adjustment, with the effect of temperature being considered in order to determine PV output. The Enercon E33 wind turbine hub height is 50m while that of the Vestas V 82 is 80m. Because the wind speeds were taken at anemometer heights of 10m, the HOMER input for wind speed variation with height has a power law exponent of 0.14 according to the one seventh power law as the site is far away from tall buildings and tall trees[68]. The efficiency of the converter was set at 97%, which is the approximate efficiency of modern rectifiers and inverters as was seen earlier in [18].

### 3.3. Optimization Results

The optimization found 91,584 feasible combinations. The various combinations can be put in three broad categories. The first category has got PV modules and Vetast V82 wind turbines as power sources. The second category has got PV modules alone as power sources, while the third category has got Only Enercon E 33 wind turbines as power sources. The most optimum result from each of the three categories are shown in table 3.2 as categorized optimization results.

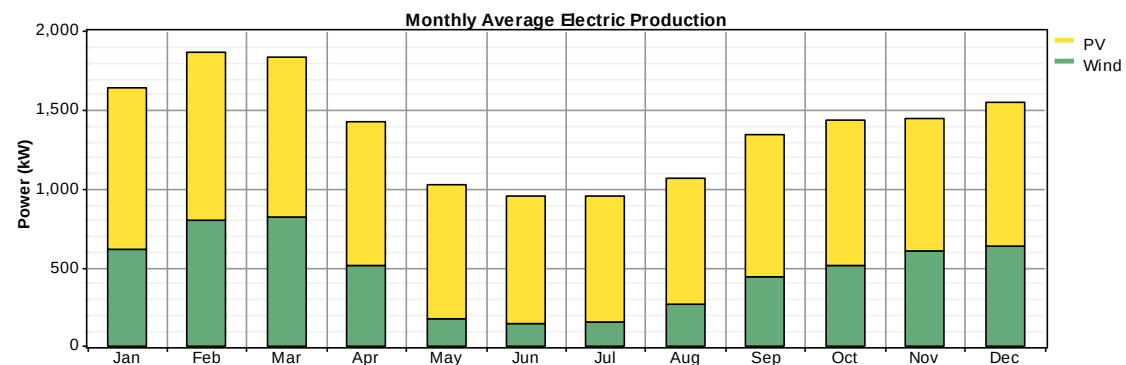
**Table 3. 2. Categorized Optimization results for the hybrid system for PS 21.**

| PV (kW) | V82 Wind Turbine Quantity (No.) | E33 Wind Turbine Quantity (No.) | 6FM 200D Batteries Quantity (No.) | Initial capital | Operating cost (\$/yr) | Total NPC     | COE (\$/kWh) |
|---------|---------------------------------|---------------------------------|-----------------------------------|-----------------|------------------------|---------------|--------------|
| 4070    | 2                               | 0                               | 15000                             | \$ 11,584,000   | 731,572                | \$ 19,738,794 | 0.201        |
| 6660    | 0                               | 0                               | 15000                             | \$ 12,822,000   | 983,440                | \$ 23,784,346 | 0.229        |
| 0       | 0                               | 100                             | 20,000                            | \$ 30,000,000   | 721,532                | \$ 38,042,876 | 0.386        |

The table listing the 200 most feasible simulations is found in the appendices( table 6.1).

The results show that the best optimal combination is 4070 kW of PV modules,two Vestas V 82 1650 kW wind turbines, a total of 15,000 batteries model Vision 6FM 200D. For this, the initial capital would be USD 11,584,000, the net present cost would be USD 19,738,794, and the levelized cost of energy would be 0.201 USD per kWh of energy.

For this optimal combination, the monthly average electrical power generation is shown in Fig 3.6.



**Fig 3. 6. HOMER plot of Monthly average Electric production for PS 21.**

Table 3.3 gives the tabulated figures for the electricity production in one year.

**Table 3. 3. Annual Electricity Production**

| Component | Production | Fraction |
|-----------|------------|----------|
|           | (kWh/yr)   |          |
| PV array  | 7,928,915  | 66%      |

|              |            |      |
|--------------|------------|------|
| Wind turbine | 4,129,618  | 34%  |
| Total        | 12,058,533 | 100% |

The table shows that 66% of the energy produced is from PV, and 34% is from the wind turbines. The electricity production can further be broken down to indicate the excess electricity and unmet load, etc. This is shown in table 3.4.

**Table 3. 4: Excess Electricity, Unmet load, and Capacity shortage.**

| Quantity           | Value     | Units  |
|--------------------|-----------|--------|
| Excess electricity | 2,116,023 | kWh/yr |
| Unmet load         | 834,255   | kWh/yr |
| Capacity shortage  | 970,577   | kWh/yr |
| Renewable fraction | 1         |        |

The PV system output was also analysed and presented in table 3.5

**Table 3. 5: PV System output**

| Quantity           | Value     | Units  |
|--------------------|-----------|--------|
| Rated capacity     | 4070      | kW     |
| Mean output        | 905       | kW     |
| Mean output        | 21,723    | kWh/d  |
| Capacity factor    | 22.2      | %      |
| Total production   | 7,928,915 | kWh/yr |
| Minimum output     | 0.00      | kW     |
| Maximum output     | 4,592     | kW     |
| Hours of operation | 4,339     | hr/yr  |
| Levelized cost     | 0.0689    | \$/kWh |

The wind turbine output as simulated by HOMER is presented in table 3.6.

**Table 3. 6: Wind turbine output.**

| Variable             | Value     | Units  |
|----------------------|-----------|--------|
| Total rated capacity | 3300      | kW     |
| Mean output          | 471       | kW     |
| Capacity factor      | 14.3      | %      |
| Total production     | 4,129,618 | kWh/yr |
| Minimum output       | 0.00      | kW     |
| Maximum output       | 2,815     | kW     |
| Hours of operation   | 6,215     | hr/yr  |

|                |        |        |
|----------------|--------|--------|
| Levelized cost | 0.0637 | \$/kWh |
|----------------|--------|--------|

The cost summary of initial capital costs is shown in fig 3.7.

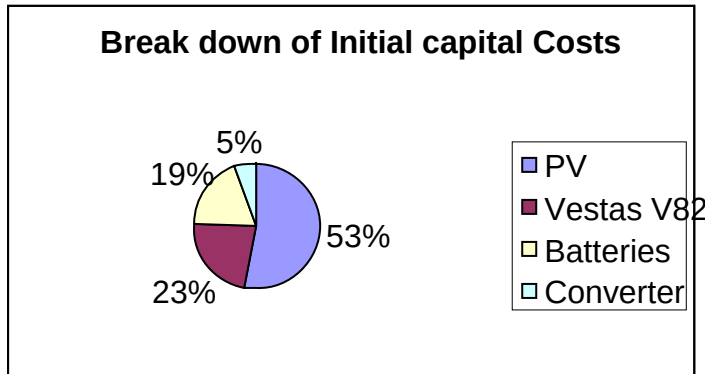


Fig 3. 7: Cost summary of initial capital costs for the hybrid system.

The optimal configuration uses 15,000 No. 200Ah batteries.

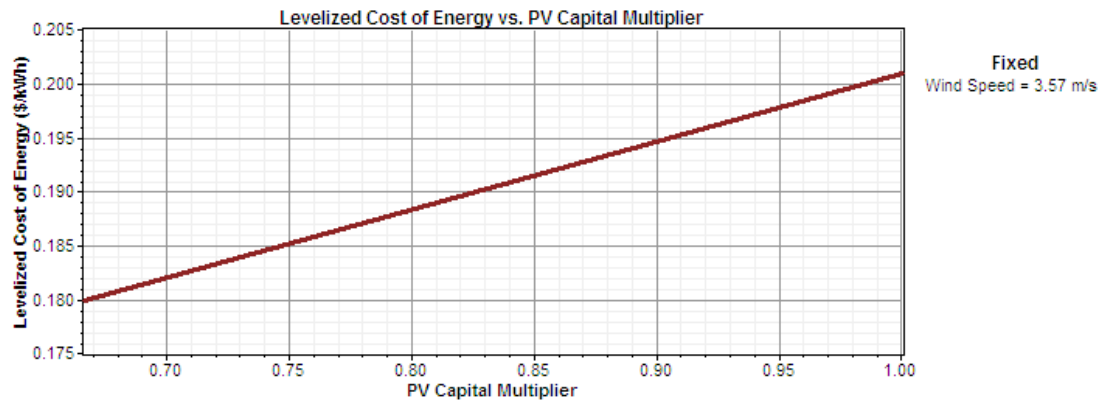
### 3.4. Sensitivity Analysis Results

A sensitivity analysis was then carried out with the HOMER software to determine the effect of uncertainty in some inputs. These were wind speed and PV Capital cost. The two inputs were selected for the following reasons:

- The cost of PV modules is on a continuous decline and is dependent on the MWp of PV power produced in a given year by a single plant [24]. Sensitivity analysis would determine the most economic PV module price to generate power.
- Wind speed. The site of the Project has got a low annual average wind speed of 3.57 m/s.. A sensitivity analysis was done so as to determine whether it might be prudent to consider locating the wind turbines at a more windy site , e.g the Ngong Hills about 30 km away and transmitting the power to PS 21.

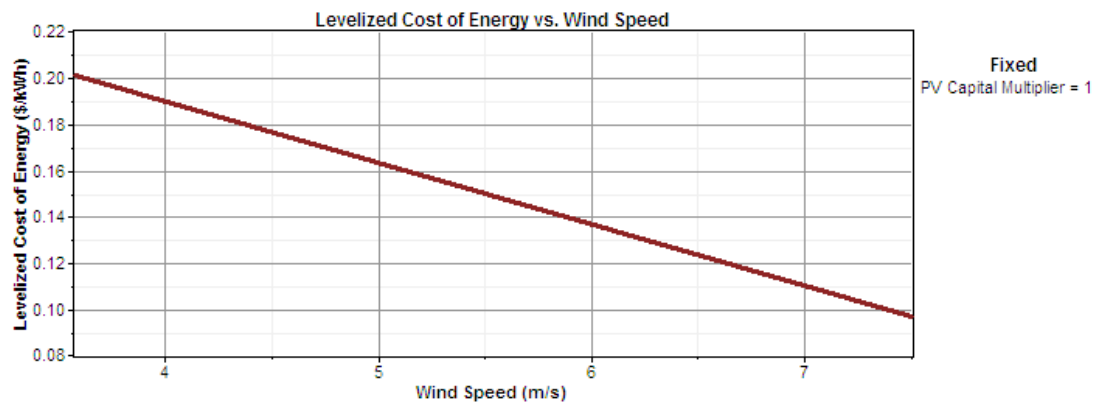
The economic results of the most optimal solution of a sensitivity analysis with PV module cost at two thirds, i.e with a PV capital multiplier of 0.67 and other inputs unchanged show that the levelised cost of energy would reduce to USD 0.18 per kWh from USD 0.201 per kWh.

Fig 3.8 shows a plot for the sensitivity analysis for the cost of PV modules.



**Fig 3. 8: Sensitivity analysis for cost of PV modules**

The optimization results for a sensitivity analysis with a wind speed of 7.5 m/s ( all other inputs unchanged) show that the levelized cost of energy reduces to USD 0.097 per kWh from USD 0.201 per kWh, which is a very significant reduction. This is illustrated in Fig 3.9.



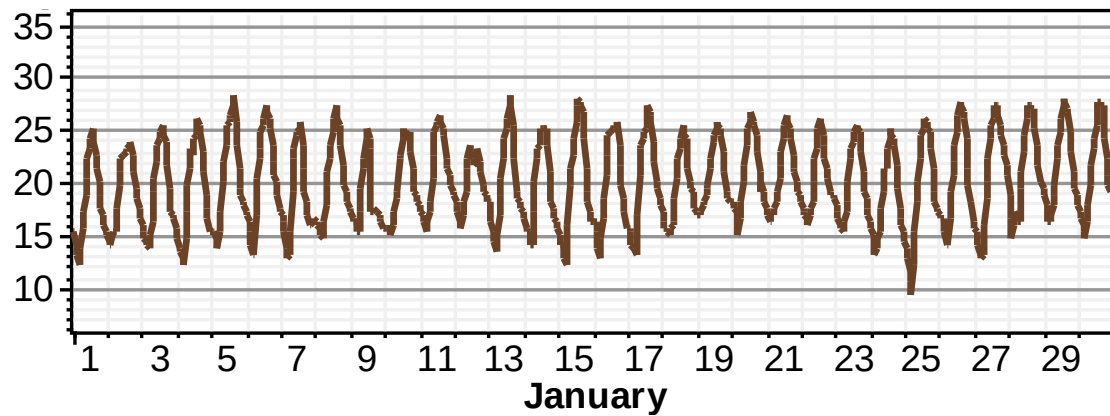
**Fig 3. 9: Sensitivity analysis for wind speed.**

### 3.5. Plots for Summer, winter , and Economic results

Various plots were produced using the HOMER software. The plots were produced for a summer month(January), and a winter month(July). Plots of economic results were also produced.

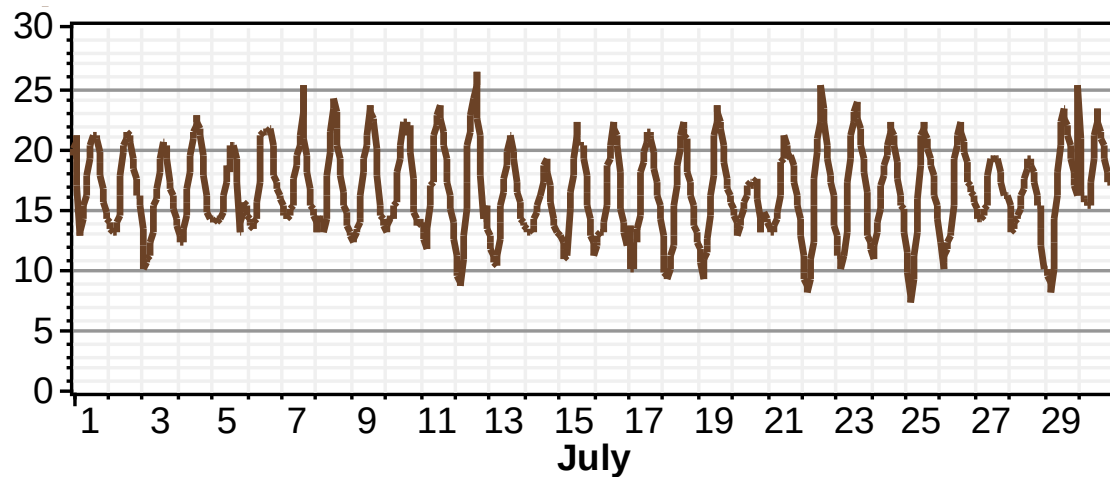
#### 3.5.1. Summer and Winter months.

A typical summer month in Kenya is January. The hourly plot for ambient temperature in Deg C in summer(January) is shown in Fig 3.10.



**Fig 3. 10: HOMER hourly plot of ambient temp- Deg C for Summer month**

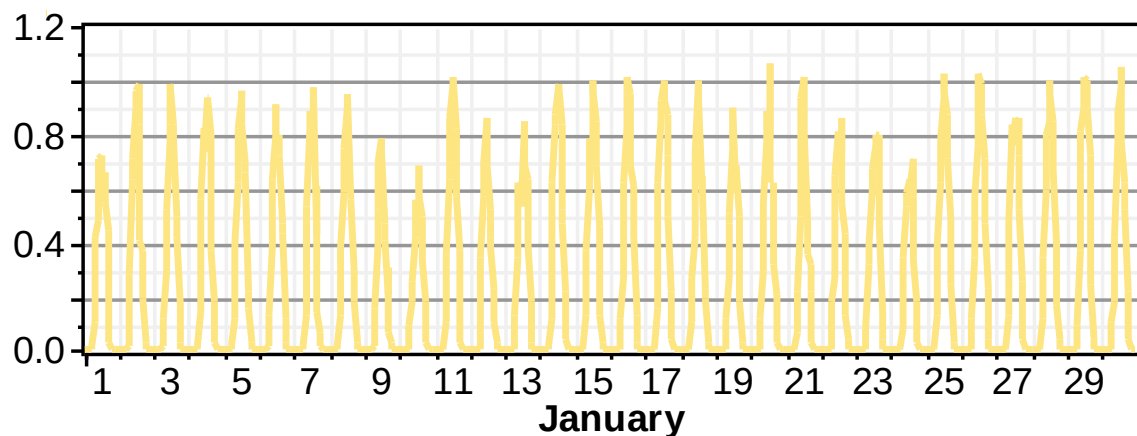
A typical winter month in Nairobi, Kenya is July. The hourly plot for ambient temperature in Deg C in winter(July) is shown in Fig 3.11.



**Fig 3. 11: HOMER hourly plot for winter ambient temperature.**

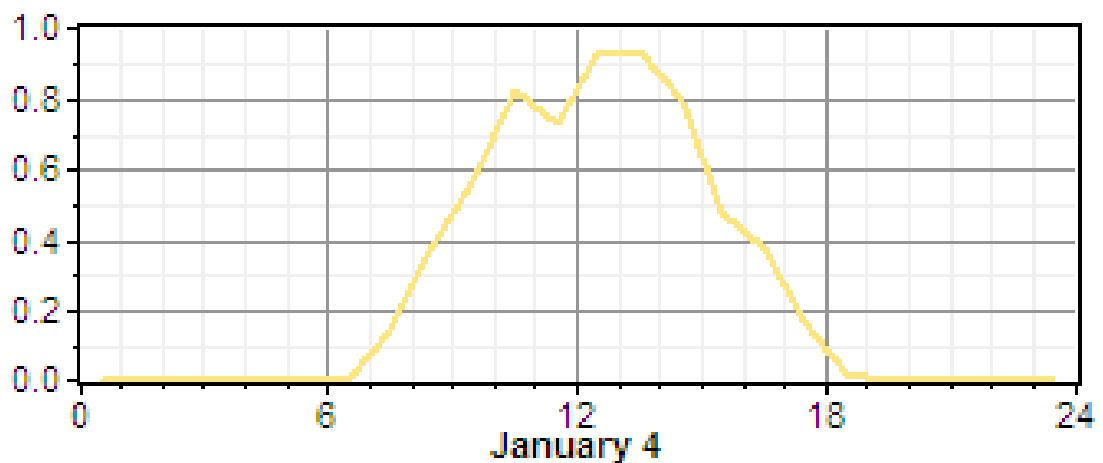
The winter temperatures are only slightly lower than those for summer, with July average being 16.3 Deg C, and January being 19.8 Deg C.

The hourly plot of global solar irradiance in  $\text{kW/m}^2$  for a typical summer month is shown in Fig 3.12.



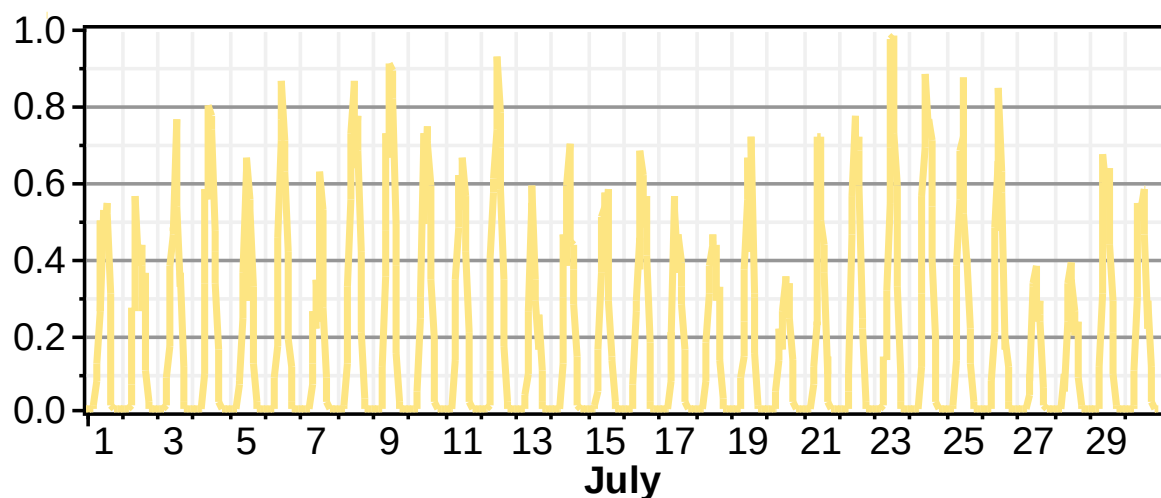
**Fig 3. 12: HOMER hourly plot of Global solar irradiance in kW/m<sup>2</sup> for a summer month**

The figure shows that the global solar irradiance reaches a daily peak of about 1kW/m<sup>2</sup> during summer. This is clearly seen in a plot of hourly global solar irradiance over a typical summer day in fig 3.13 .



**Fig 3. 13: HOMER Plot of hourly Global solar irradiance over a typical summer day.**

The corresponding hourly Global solar irradiance plot for the entire winter month of July is shown in Fig 3.14



**Fig 3. 14: HOMER plot of Global solar irradiance for a winter month.**



The global solar irradiance in the winter month is slightly lower than the summer month, with the average daily irradiation for the summer month of January being 6.094 kWh/m<sup>2</sup>/day, while that of the winter month is 4.258 kWh/m<sup>2</sup>/day.

The wind speed for a summer month, typically January is shown in Fig 3.15.

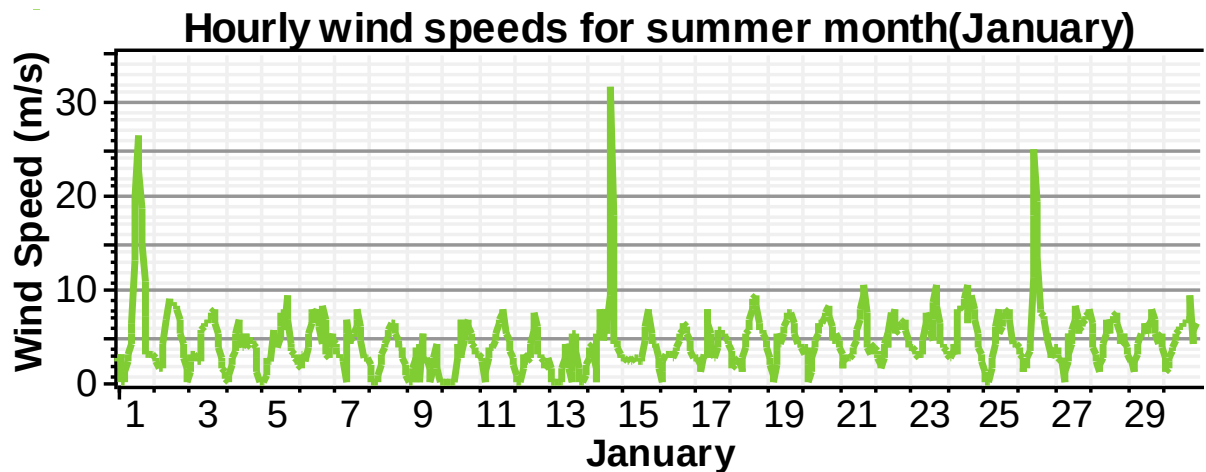


Fig 3. 15: HOMER plot for hourly wind speeds for typical summer month.

The plot shows that wind speeds are average, with peaks of up to 10 m/s, and occasional gusts of up to 30 m/s. The average wind speed for January is 4.242 m/s. The plot for hourly wind speeds for a typical summer day is shown in Fig 3.16 .

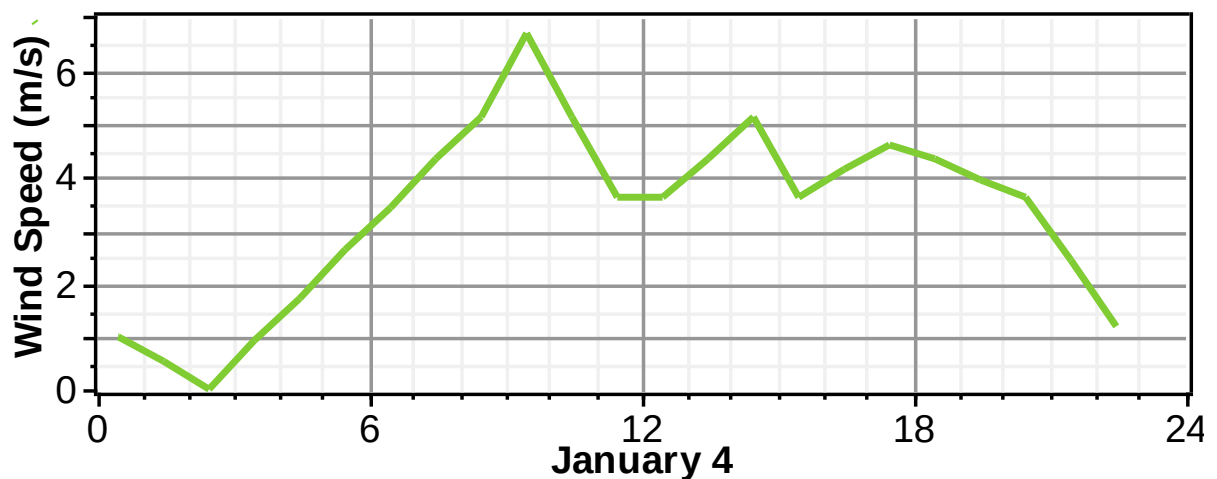


Fig 3. 16: HOMER Plot of hourly wind speed variation over a typical summer day.

The HOMER plot of hourly wind speeds for the winter month of July is shown in Fig 3.17.

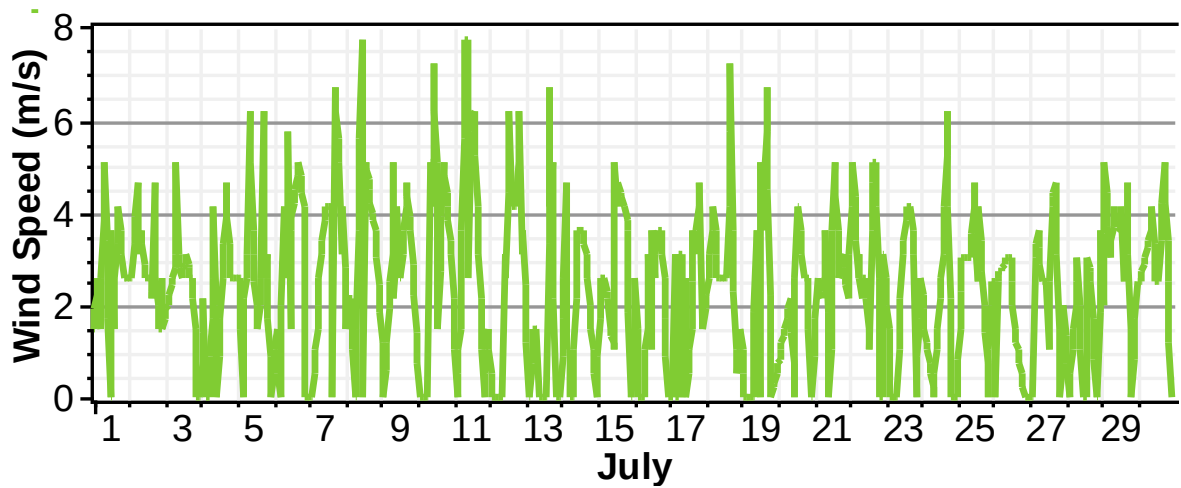


Fig 3. 17:HOMER plot of hourly wind speeds for the winter month of July.

The average wind speed for the winter month is 2.413 m/s, whereas for the summer month of January, it is 4.242 m/s.

The AC Primary load profile for the summer month of January is shown in fig 3.18.

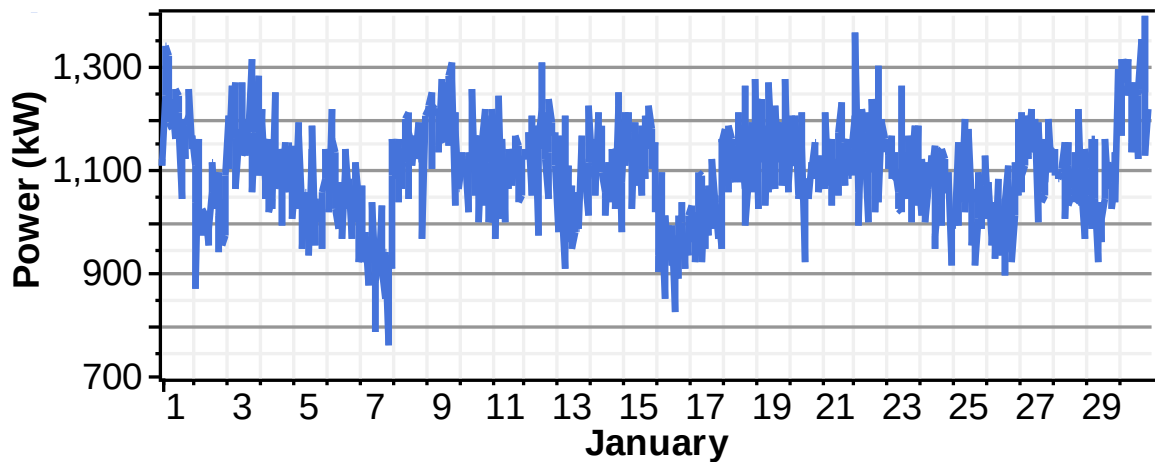


Fig 3. 18: AC primary load Profile for the summer month of January.

The load profile for a typical January day is shown in Fig 3.19.

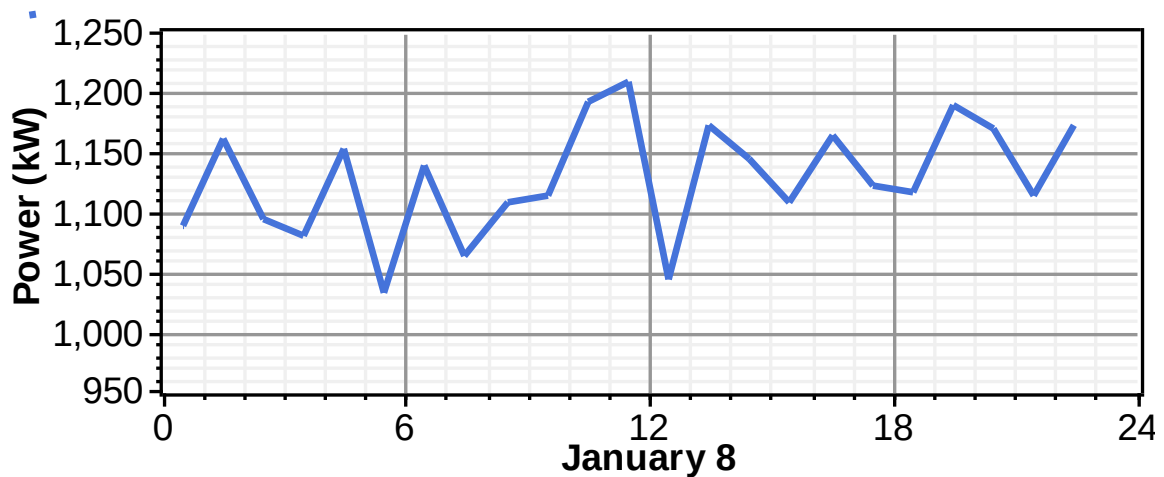
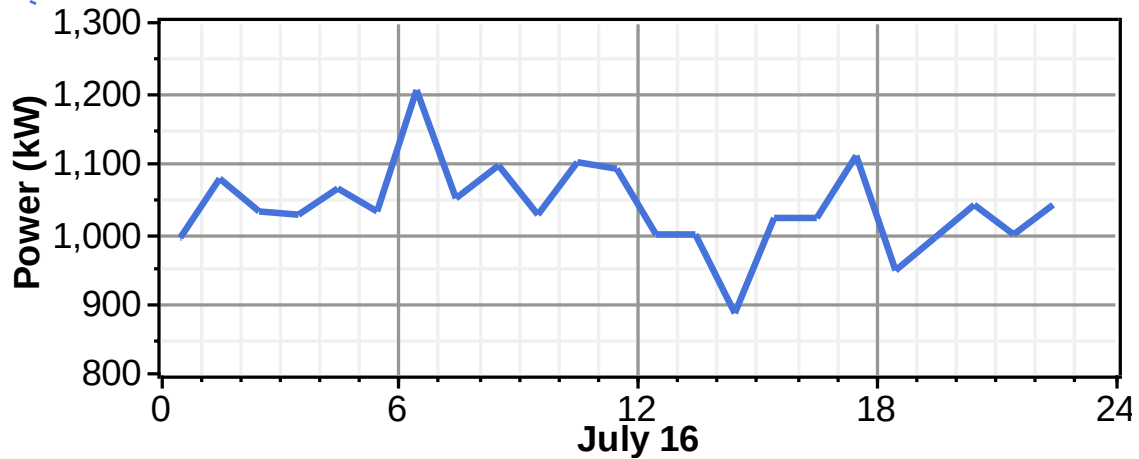


Fig 3. 19: AC power load profile for a typical summer day.

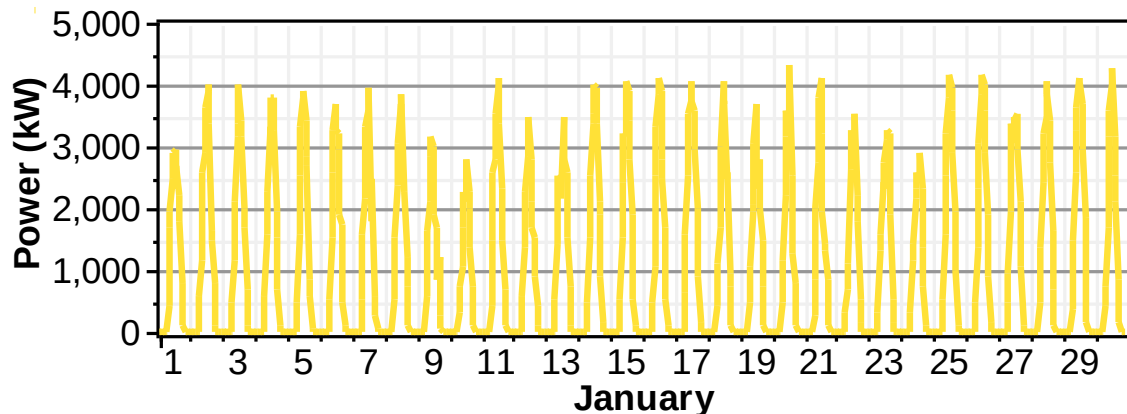
On the date shown, the maximum load was about 1200 kW and the minimum was about 960 kW, i.e. the load variation was minimal.  
In the winter month of July, the AC power load on a typical day is as shown in figure 3.20.



**Fig 3. 20: AC power load on a typical winter day.**

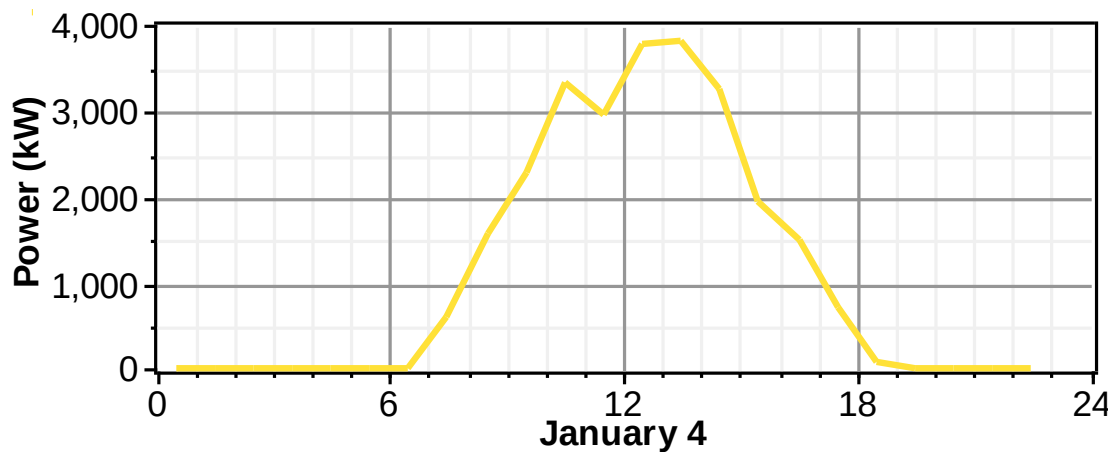
The hourly AC power load profiles for a typical summer and winter day are not very different because the load in question is mostly fuel pumping, and is independent of temperature.

The profile of the PV power generated during summer by the system on an hourly basis is shown in Fig 3.21.



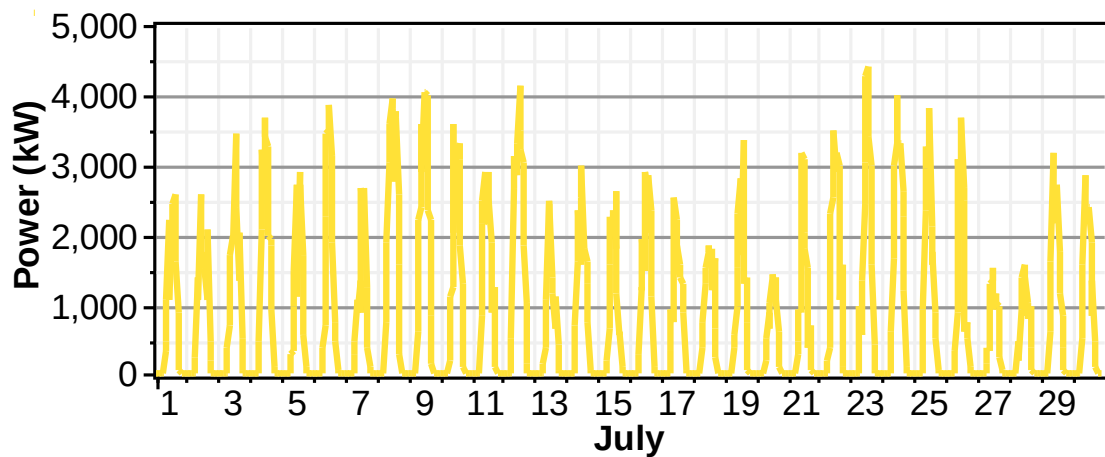
**Fig 3. 21: HOMER hourly Plot of PV power generated by the system in summer month.**

The plot shows that the PV power generated follows a fairly predictable pattern each summer day, as shown in the hourly plot of PV power generated on a typical day, in Fig 3.22.



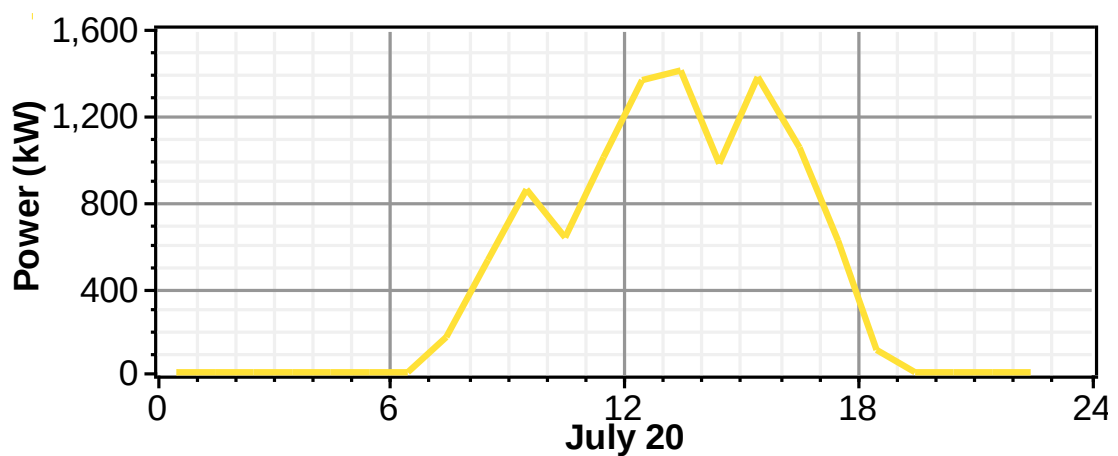
**Fig 3. 22: HOMER hourly plot of PV power generated on a typical Summer day.**

During winter, the hourly PV power generated by the system is as shown in Fig 3.23 for the entire month of July.



**Fig 3. 23: Hourly PV power generated in the winter month of July.**

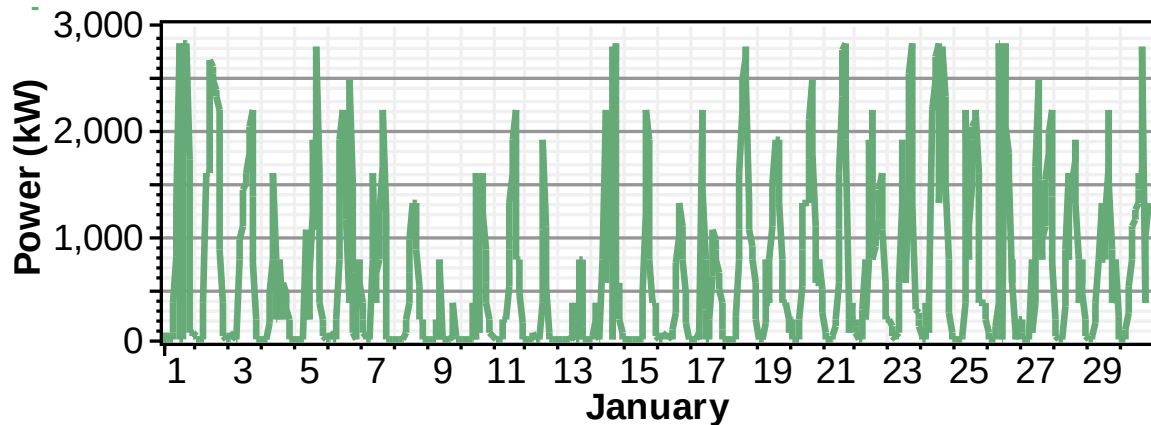
During this time, the typical hourly PV power plot for a winter day is given below in Fig 3.24.



**Fig 3. 24: HOMER plot of hourly PV Power in a typical winter day.**

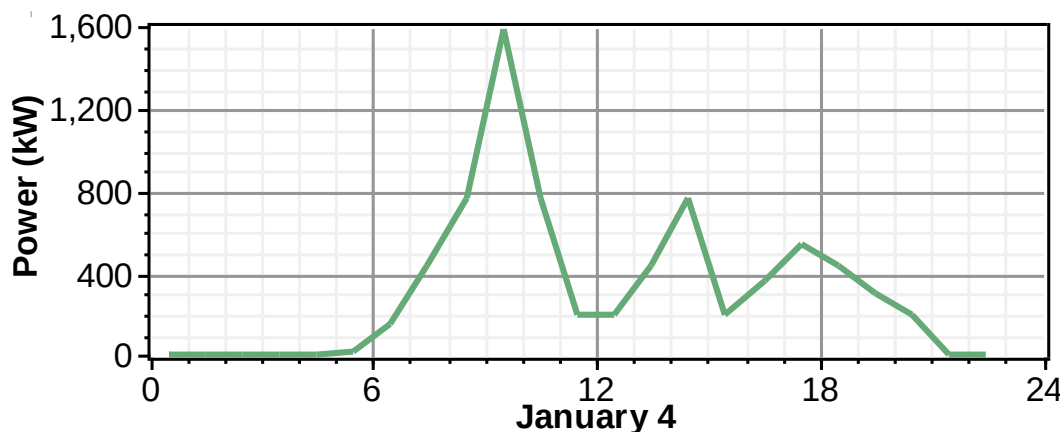
The plot shows that on a typical winter day, a peak of about 1400 kW of solar PV power was generated compared to a peak of 4000 kW of PV power for a typical summer day.

The hourly Homer plot for wind power generated by the Vestas V82 wind turbines over the summer month of January is shown in Fig 3.25.



**Fig 3. 25: Hourly plot of power generated by V 82 wind turbines in summer.**

The daily profiles of the power generated are fairly similar, and the profile of a typical summer day is shown in fig 3.26.



**Fig 3. 26: HOMER plot of power generated by the V 82 wind turbines on a summer day.**

During winter, the power generated by the Vestas V82 wind turbines for a typical day is as shown in Fig 3.27.

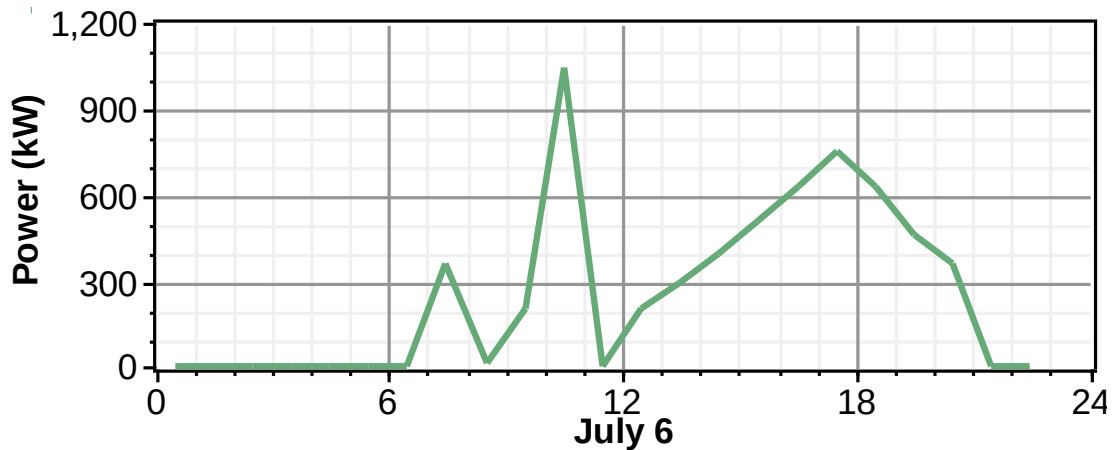


Fig 3. 27: Hourly Plot of Wind power generated by V 82 turbines on a typical winter day.

The plot for wind turbine power generated in the winter month of July is as shown in Fig 3.28.

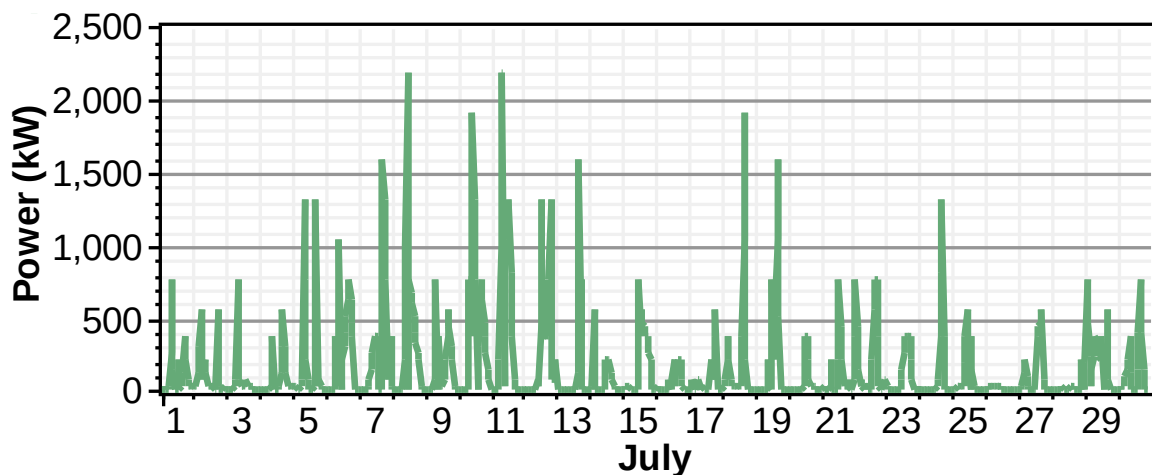


Fig 3. 28: Hourly plot of wind power generated by V 82 turbines in the winter month.

The power generated by the wind turbines in winter is less than that generated in Summer, the daily peak being about 1000 kW during winter, and about 1600kW during Summer.

The system uses a 1500 kW Inverter/Rectifier. The power output of the inverter in a summer month is shown in Fig. 3.29.

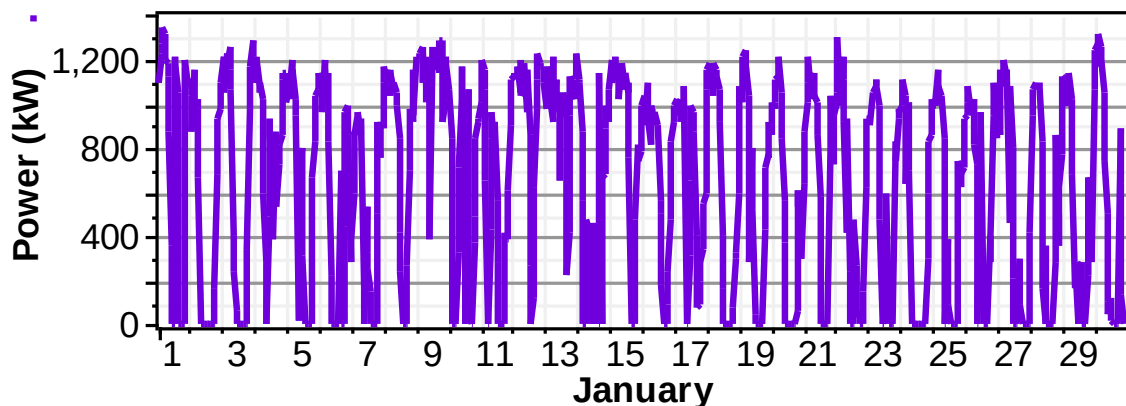
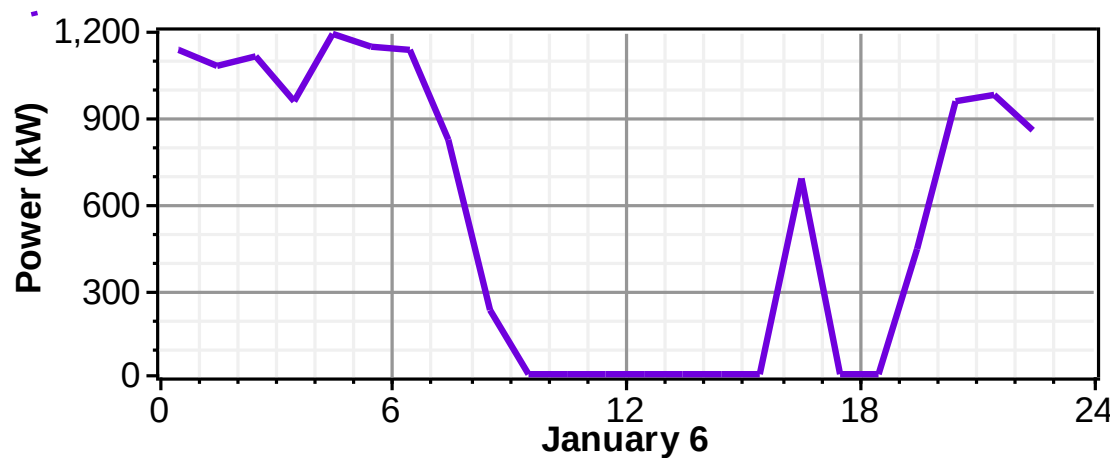


Fig 3. 29: Hourly plot of Inverter power output in Summer month.

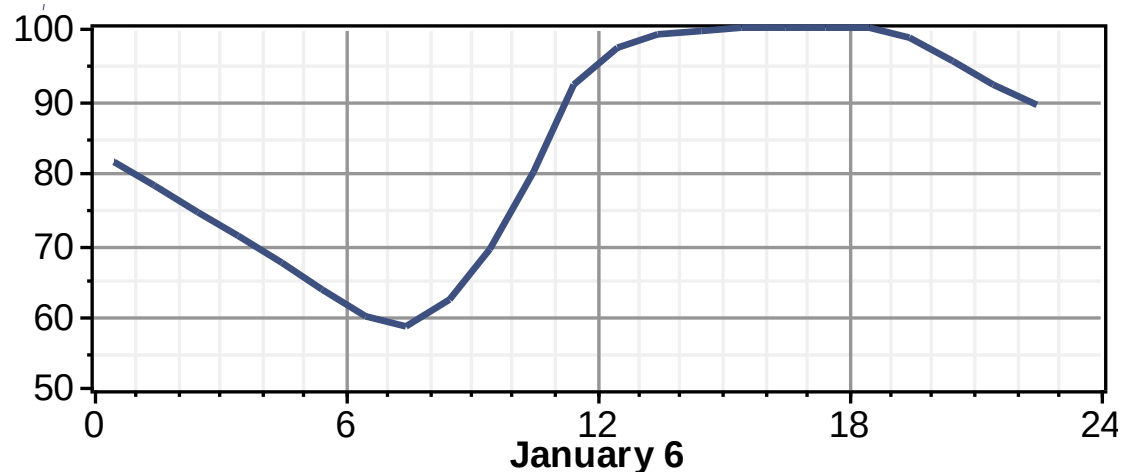
The profile is fairly repetitive each day and follows the daily profile shown in Fig 3.30.



**Fig 3. 30: Hourly plot of inverter power output over a typical summer day.**

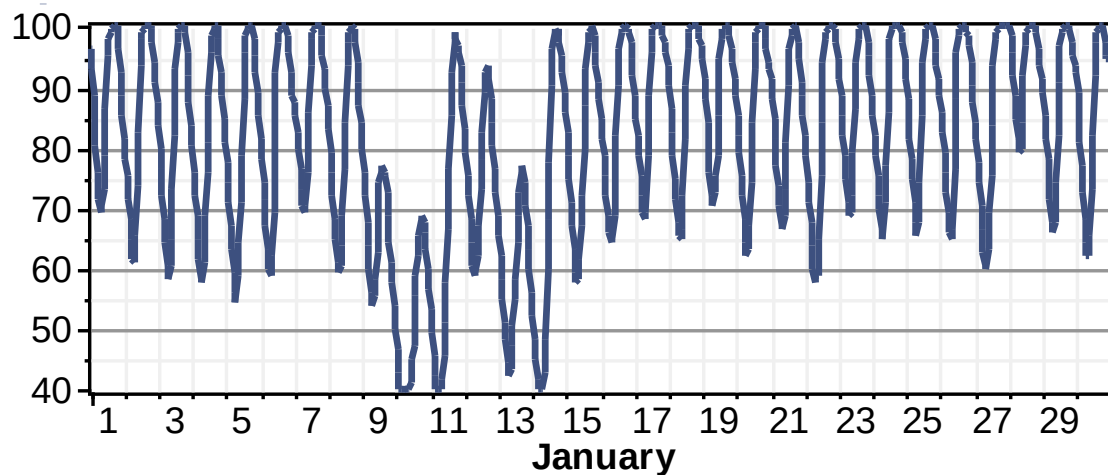
An observation of fig 3.27(plot of Vestas wind turbine hourly power output for a typical day) and fig 3.30(plot of inverter power output) shows that the inverter power output is inversely proportional to the wind turbine power output (all other factors remaining constant). It can be seen that when the Wind turbine power output reduces (usually in the night), the power from the batteries is converted from DC to AC by the inverter, and therefore inverter power increases.

Thus, during the night, the battery SOC reduces, while during the day, the batteries are charged by combined wind and solar Power, and the battery SOC increases, as shown in the plot of battery SOC in fig 3.31.



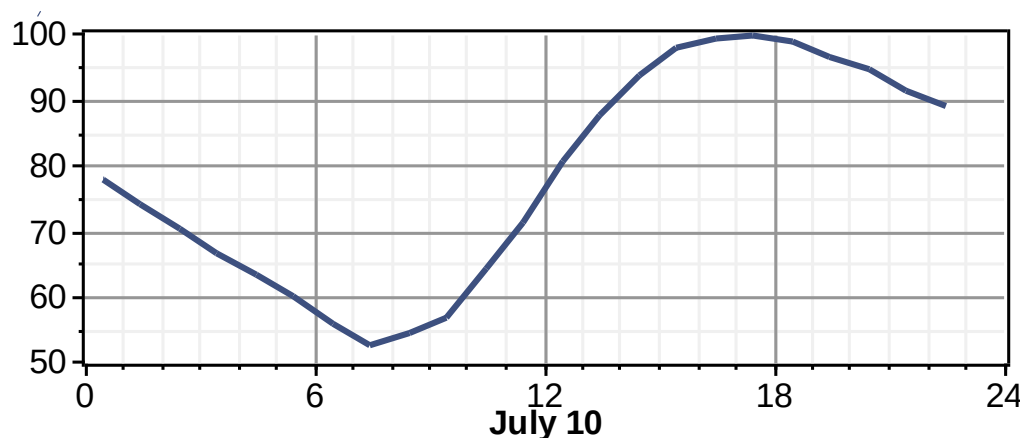
**Fig 3. 31: HOMER hourly plot of Battery SOC(%) for a typical summer day.**

This process is repeated throughout the entire month, as can be seen in fig 3.32.



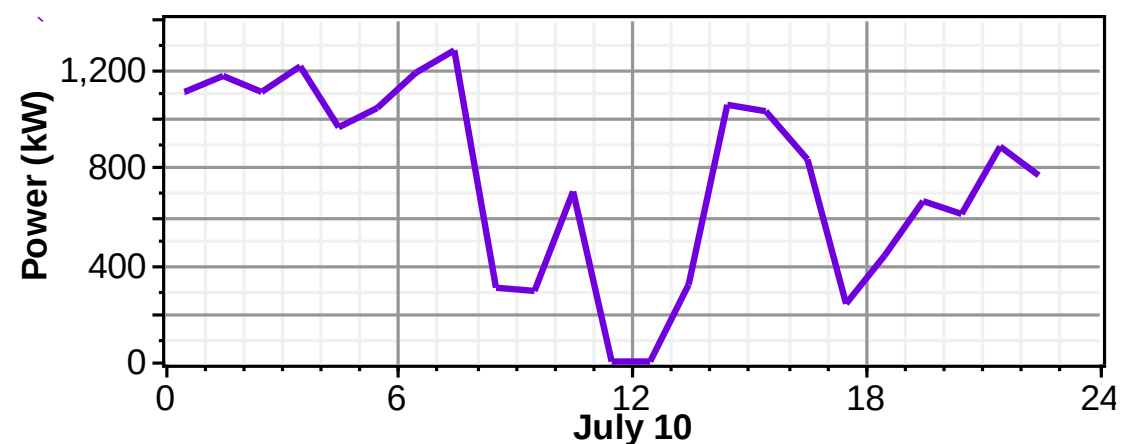
**Fig 3. 32: HOMER hourly plot of Battery SOC during a typical summer month.**

The corresponding plot of battery state of charge for a typical day in the winter month of July is shown in Fig 3.33.



**Fig 3. 33: Hourly plot of battery SOC for a typical winter day.**

Similarly, the plot for the Inverter power output is shown in Fig 3.34.



**Fig 3. 34: Hourly plot of Inverter power output for a typical winter day.**

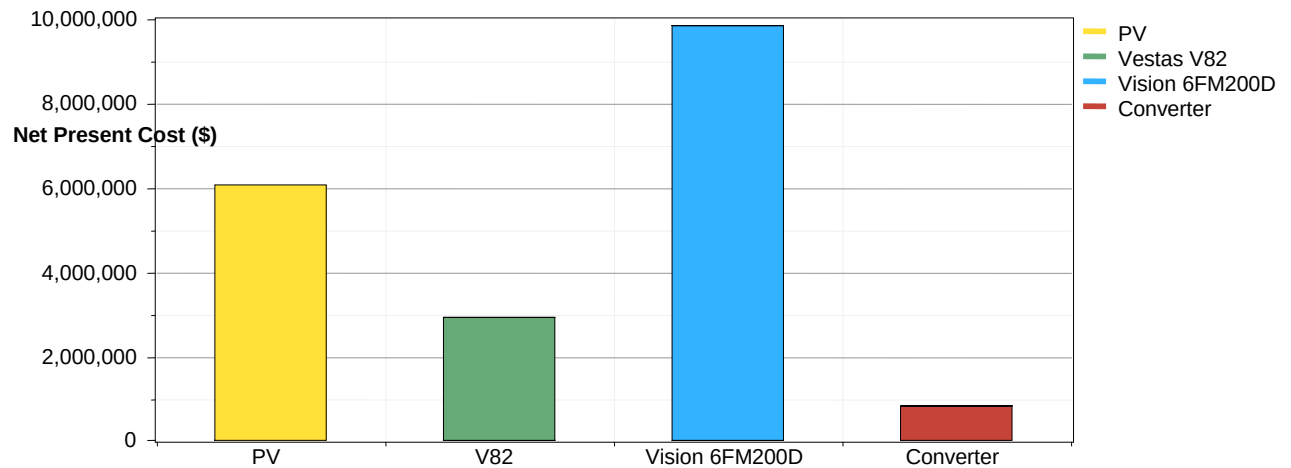
It can be seen from the plots for SOC and inverter power output that on a typical winter day, the inverter output power is high is when the battery SOC is



decreasing, i.e. the battery is being discharged (Usually when there is no wind power being generated).

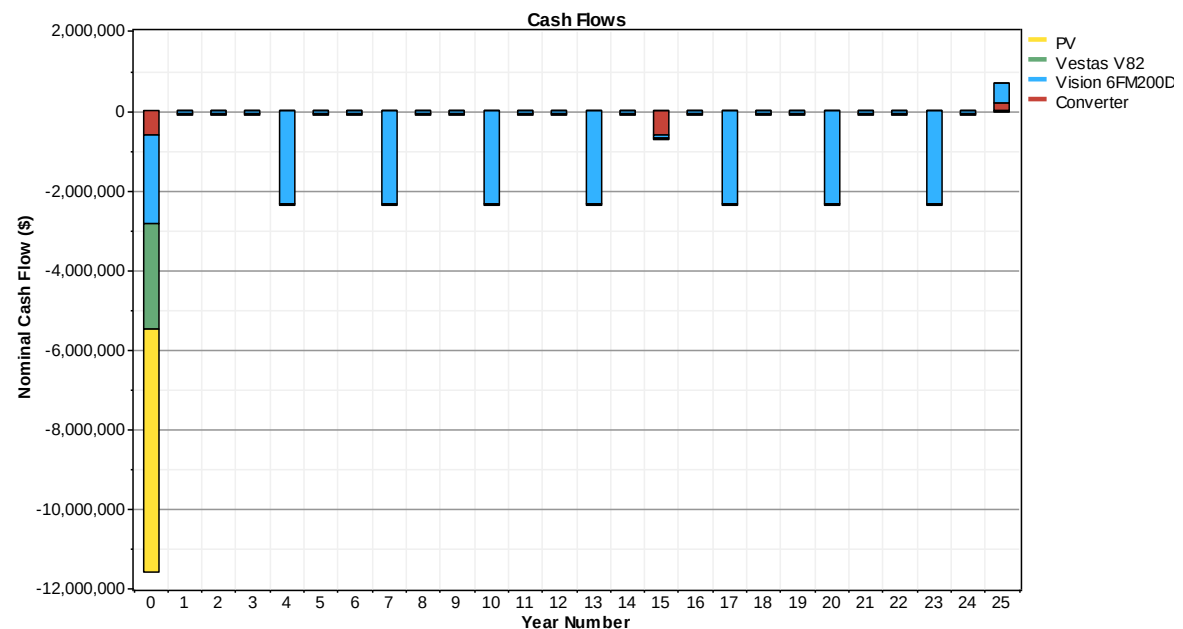
### 3.5.1. Economic results.

The summary of the costs is given in Fig 3.35



**Fig 3. 35: Plot of Cost summary by component.**

From the plot for cost summary, it can be seen that the highest component of the net present cost is attributable to the Vision 6FM200D batteries. The plot for cash flow is shown in Fig 3.36.



**Fig 3. 36: Plot showing cash flow over project life time.**

From the plot, it can be seen that the capital costs of PV modules wind turbines are only incurred at the commencement of the project, while the

inverter /converter is replaced in the 15<sup>th</sup> year, while the batteries are replaced every four years, leading to a high net present cost attributable to batteries.

## 4.0. DISCUSSION

According to the Simulation and Optimization results, the most Optimal solution is a combination of 4070 kW of solar PV modules, two No. 1650 kW Vestas V82 wind turbines and 15,000 No. 200 Ah batteries with a levelized cost of Energy of \$0.201/kWh. This matches the cost of energy purchased from Kenya's national grid which varies from \$0.14 to \$0.2 per kWh.

The results of the sensitivity analysis show that if the wind turbines had been located in an area with an average annual wind speed of 7.5 m/s, the cost of energy would have dropped to \$0.097/kWh. This is a very significant drop, meaning that a good wind speed regime would lead to a very huge reduction in LCE. The other sensitivity analysis done shows that if the PV module cost reduced to 2/3, the levelized cost of energy would have been \$0.18 per kWh.

The value of Levelized cost of energy of US \$ 0.201 per kWh shows that the project is feasible. This is because even though the LCE just matches the cost of energy purchased from the national grid, (meaning an almost infinite payback period), the benefits of renewable energy generation( e.g. sustainability and global warming reduction) mean that the project meets eligibility criteria for the environment. The government of Kenya is very positive about renewable energy projects, meaning that the project is also politically feasible.

The cost of PV modules is also on a decline and the LCE value will decrease even further in the very near future.

The project also explored the possibility of selling CER's to a party in an Annex 1 country under the Kyoto Protocol. It was noted that this is possible and should be explored as it would bring down the Net Present Cost of the Project ( and subsequently the LCE) even further.

It is also worth noting that the largest component in the net present cost of the system is the batteries. 50 % of the net present cost is attributable to batteries.

## 5.0.CONCLUSION

The solar and wind energy potential at Kenya Pipeline Company's PS 21 station has been identified. It has been shown that it is indeed technically and economically feasible to generate sufficient power using a solar PV-Wind-Battery hybrid system. The solar and wind resource data used are those provided by the UNEP Solar Wind Energy Resource Assessment (SWERA) Project for the nearby Jomo Kenyatta International Airport, Meteorological Station 5 KM away. The Meteorological station is operated by the Kenya Meteorological department and the data used was hourly Typical Meteorological Year (TMY) data.

From the results, it can be seen that the hybrid Solar-Wind renewable energy potential of the site good. Various technically feasible solutions were determined by the Simulation and Optimization software used (HOMER), but the most optimal solution had a levelized cost of energy of \$0.201 per kWh.

A preliminary study was carried and it was determined that the project could benefit from the sale of Certified Emission Credits (CERs) under the CDM mechanism of the Kyoto Protocol. However, a more detailed feasibility study should be carried out to determine the actual amounts of money that may be obtained and the extent this may have towards reducing the levelized cost of energy. It has also been observed in the report that the cost of Solar PV modules is on a decline with time. That being the case, the Levelized cost of energy could reduce further to \$ 0.18 per kWh.

An option exists where the wind turbines can be located on hilly windy sites around Nairobi, e.g. the Ngong hills, around 30 KM away, where many wind farms are located. A separate study can be carried out to determine the feasibility of this, especially considering the Energy transmission costs.

In conclusion, the project is technically and economically feasible.

## REFERENCES

- [1] Ackermann.T., Soder.L.(2002), An overview of wind energy status. Elsevier science.
- [2] Ackermann.T., Soder.L.(2002), An overview of wind energy status. Elsevier science.
- [3] Boyle,G.(2004), Renewable energy,Power for a sustainable future. Oxford University press in association with the open university,Milton Keynes, England.
- [4] World wind energy association(2009),World wind energy report 2009.Report available at [www.wwindea.org](http://www.wwindea.org) , as accessed 2-4-2010.
- [5] Boyle,G.(2004), Renewable energy,Power for a sustainable future. Oxford University press in association with the open university,Milton Keynes, England.
- [6] Boyle,G.(2004), Renewable energy,Power for a sustainable future. Oxford University press in association with the open university,Milton Keynes, England.
- [7] Ackermann.T., Soder.L.(2002), An overview of wind energy status. Elsevier science.
- [8] Ackermann.T., Soder.L.(2002), An overview of wind energy status. Elsevier science.
- [9] Ackermann.T., Soder.L.(2002), An overview of wind energy status. Elsevier science.
- [10] Renewable energy sources(2009),7 MW Enercon E-126 wind turbine now in operation. News report available at [www.renewable-energy-sources.com](http://www.renewable-energy-sources.com), as accessed 4/4/2010, updated 2-2-2009.
- [11] Renewable energy sources(2010),Sway to bild 10MW offshore wind turbine. News report available at [www.renewable-energy-sources.com](http://www.renewable-energy-sources.com), as accessed 4/4/2010.
- [12] Boyle,G.(2004), Renewable energy,Power for a sustainable future. Oxford University press in association with the open university,Milton Keynes, England.
- [13] European Wind Energy Association(2009), Wind Energy-The Facts: A guide to the technology, economics, and future of wind power. Earthscan, London, England.
- [14] Boyle,G.(2004), Renewable energy,Power for a sustainable future. Oxford University press in association with the open university,Milton Keynes, England.
- [15] Kreith,F.,Goswami,Y.D.(2007), Handbook of energy efficiency and renewable energy. CRC Press, Boca Raton, USA.
- [16] Wenham,S.R.,Green,M.A.,Watt,M.E.,Corkish,R.(2007),Applied PhotoVoltaics 2nd Ed. Book,Earthscan, retrieved [www.knovel.com](http://www.knovel.com) 24-3-2010.
- [17] Wenham,S.R.,Green,M.A.,Watt,M.E.,Corkish,R.(2007),Applied PhotoVoltaics 2nd Ed. Book,Earthscan, retrieved [www.knovel.com](http://www.knovel.com) 24-3-2010.
- [18] Satcon(2010),Powergate Plus 1MW.Download available at [www.satcon.com/downloads](http://www.satcon.com/downloads) as accessed 11-4-2010, updated 2010.

- [19] Satcon(2010),Powergate Plus 1MW.Download available at [www.satcon.com/downloads](http://www.satcon.com/downloads) as accessed 11-4-2010, updated 2010.
- [20] Boyle,G.(2004), Renewable energy,Power for a sustainable future. Oxford University press in association with the open university,Milton Keynes, England.
- [21] Boyle,G.(2004), Renewable energy,Power for a sustainable future. Oxford University press in association with the open university,Milton Keynes, England.
- [22] Boyle,G.(2004), Renewable energy,Power for a sustainable future. Oxford University press in association with the open university,Milton Keynes, England.
- [23] Kreith,F.,Goswami,Y.D.(2007), Handbook of energy efficiency and renewable energy. CRC Press, Boca Raton, USA
- [24] Kreith,F.,Goswami,Y.D.(2007), Handbook of energy efficiency and renewable energy. CRC Press, Boca Raton, USA
- [25] Ter-Gazarian,A.G.(1994), Energy Storage for power Plants.Institution of Engineering and Technology,Lodon, England.
- [26] Ter-Gazarian,A.G.(1994), Energy Storage for power Plants.Institution of Engineering and Technology,Lodon, England.
- [27] Ter-Gazarian,A.G.(1994), Energy Storage for power Plants.Institution of Engineering and Technology,Lodon, England.
- [28] Ter-Gazarian,A.G.(1994), Energy Storage for power Plants.Institution of Engineering and Technology,Lodon, England.
- [29] Hoogers G.(2003),Fuel cell technology.CRC Press,Boca Raton, USA.
- [30] Duarte J.L., Hendrix M., Simoes M.G. (2007), Three-Port bidirectional converter for hybrid fuel cell systems. IEEE Transactions on Power Electronics, Vol 22(2).
- [31] Wenham,S.R.,Green,M.A.,Watt,M.E.,Corkish,R.(2007),Applied PhotoVoltaics 2nd Ed. Book,Earthscan, retrieved [www.knovel.com](http://www.knovel.com) 24-3-2010.
- [32] Van den Bosche P.,Vergels F., Van Mierlo J.,Matheys J.,Van Autenboer W.(2005), An assessment of sustainable battery technology.Elsevier Press.Available at <http://www.battery-electric.com/subatdocs/WP6-032.pdf>. Accessed 17-4-2010.
- [33] Van den Bosche P.,Vergels F., Van Mierlo J.,Matheys J.,Van Autenboer W.(2005), An assessment of sustainable battery technology.Elsevier Press.Available at <http://www.battery-electric.com/subatdocs/WP6-032.pdf>. Accessed 17-4-2010.
- [34] Diaf S., Diaf D., Belhamel M., Hadadi M., Louche A.( 2007), A methodology for optimal sizing of autonomous hybrid PV/Wind system. Hyper Articles en Ligne, available at <http://hal.archives-ouvertes.fr>, last modified 18th June 2008, as accessed 7th June 2010
- [35] National renewable energy laboratory(2009), Ultra Capacitors, Available at <http://www.nrel.gov/vehiclesandfuels/energystorage/ultracapacitors.htm> ], accessed 17-4-10, updated 30-9-2009

- [36] Ter-Gazarian, A.G. (1994), Energy Storage for power Plants. Institution of Engineering and Technology, London, England.
- [37] Karki R., Billinton R.(2001), Reliability/Cost implications of PV and Wind energy utilization in small isolated power systems. IEEE transactions on energy Conversion 2001;16(4):368-373.
- [38] Zhou W., Lou C., Li Z., Lu L., Yang H.(2010), Current Status of research on Optimum Sizing of stand-alone hybrid Solar-wind power generation systems. Applied Energy 87(2010) 380-389
- [39] Deshmukh M.K., Deshmukh S.S.(2008),Modelling of hybrid renewable energy systems.Renewable and Sustainable Energy Reviews 12(2008) 235-249.
- [40] Chedid R., Akiki H., Rahman S.(1998), A decision support technique for the design of hybrid solar-wind power systems. IEEE Transactions on Energy Conversion 1998;13(1):76-83.
- [41] Yang H., Zhou W., Lu L., Fang Z.(2008), Optimal sizing method for stand-alone hybrid solar wind system with LPSP technology by using genetic algorithm. Solar Energy 2008;82(4):354-67.
- [42] Power Utility bills for Pumping Station 21, January to December 2009.
- [43] Sambu V.(2009), Email Communication with Victor Sambu, Operations Engineer, Kenya Pipeline Company, Nairobi, June 25 2010.
- [44] Kimaiyos D. (2010), Telephone conversation with David Kimaiyos, Depot Manager, Kenya Pipeline Company, Nairobi Terminal, June 25, 2010.
- [45] Okombo S. (2010), Email communication with Stephen Okombo, Assistant Engineer, Kenya Pipeline Company, Nairobi, June 3 2010.
- [46] Zhou W., Lou C., Li Z., Lu L., Yang H.(2010), Current Status of research on Optimum Sizing of stand-alone hybrid Solar-wind power generation systems. Applied Energy 87(2010) 380-389.
- [47] Yang H., Zhou W., Lu L., Fang Z.(2008), Optimal sizing method for stand-alone hybrid solar wind system with LPSP technology by using genetic algorithm. Solar Energy 2008;82(4):354-67.
- [48] Yang H., Lu L., Zhou W.(2007), A novel optimization sizing model for hybrid solar-wind power generation system.Solar Energy 81(2007) 76-84.
- [49] Yang H., Lu L., Zhou W.(2007), A novel optimization sizing model for hybrid solar-wind power generation system.Solar Energy 81(2007) 76-84.
- [50] Yang H., Lu L., Zhou W.(2007), A novel optimization sizing model for hybrid solar-wind power generation system.Solar Energy 81(2007) 76-84.
- [51] Yingli Green Energy holding Company Ltd(2009), Product datasheet for solar module Yingli YL 185 P. Available at [www.yinglisolar.com](http://www.yinglisolar.com), last modified 2009, as accessed 26th June 2010.
- [52] University of Oregon Solar radiation monitoring Laboratory (2002), Solar radiation basics. Available at <http://solardat.uoregon.edu/solarRadiationBasics.html>, last modified march 2002, as accessed 27th June 2010.

- [53] Enercon(2009), Wind turbine download brochures. Available at [http://www.enercon.de/en/\\_home.htm](http://www.enercon.de/en/_home.htm) last modified 2009, as accessed 16th September 2010.
- [54] Takala R. (2006), Advanced MW-Class wind turbine converter technology. Available at <http://www.univ-lehavre.fr/recherche/greah/documents/ecpe/takala.pdf>, last modified Feb 9 2006.
- [55] International Energy agency (2010), Electricity , heat in Kenya, 2007. Available at [http://www.iea.org/stats/electricitydata.asp?COUNTRY\\_CODE=KE](http://www.iea.org/stats/electricitydata.asp?COUNTRY_CODE=KE), as accessed 28th June 2010, modified 2010.
- [56] UNFCCC. Kyoto Protocol, article 12.5C
- [57] Nema P., Nema R.K., Rangnekar S(2009), A current and future state of art development of hybrid energy system using wind and PV-Solar: A review. Renewable and sustainable energy reviews 13(2009) 2096-2103.
- [58] Capacity development for CDM project, UNEP RISOE Centre(2007), Guidebook to financing CDM Projects. Available at <http://www.cd4cdm.org/Guidebooks.htm>, page 11, as accessed 28th June 2010.
- [59] Capacity development for CDM project, UNEP RISOE Centre(2007), Guidebook to financing CDM Projects. Available at <http://www.cd4cdm.org/Guidebooks.htm>, page 12, as accessed 28th June 2010.
- [60] Capacity development for CDM project, UNEP RISOE Centre(2007), Guidebook to financing CDM Projects. Available at <http://www.cd4cdm.org/Guidebooks.htm>, page 13, as accessed 28th June 2010.
- [61] Capacity development for CDM project, UNEP RISOE Centre(2005), Baseline methodologies for clean development mechanism projects guidebook. Available at <http://www.cd4cdm.org/Guidebooks.htm>, as accessed 30th June 2010.
- [62] UNFCCC, Tool to calculate the emission factor for an electricity system. EB 50 Report, Annex 14(2009). Available at <http://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-07-v2.pdf>, as accessed 30<sup>th</sup> June 2010.
- [63] US Department of Energy, Energy Information Administration (2007). Electricity emission factors. Available at [http://www.eia.doe.gov/oiaf/1605/pdf/Appendix%20F\\_r071023.pdf](http://www.eia.doe.gov/oiaf/1605/pdf/Appendix%20F_r071023.pdf), accessed 1<sup>st</sup> July 2010.
- [64] Green A.G. (2008), A quantitative analysis of the cost effectiveness of project types in the CDM Pipeline. Masters Thesis in Environmental and natural resource economics, University of Copenhagen, page 30. Available at <http://www.cd4cdm.org/Publications/CostEffectivenessProjectsCDMPipeline.pdf>, accessed 27<sup>th</sup> June 2010.
- [65] Capacity development for CDM project, UNEP RISOE Centre(2007), Guidebook to financing CDM Projects. Available at <http://www.cd4cdm.org/Guidebooks.htm>, page 55, as accessed 28th June 2010.

- [66] HOMER Software version 2.68 Beta Component Library.
- [67] UNEP, Solar and Wind Energy Resource Assessment program (SWERA), accessed 17th September 2010. Available at <http://swera.unep.net>.
- [68] Yang H., Zhou W., Lu L., Fang Z. (2008), Optimal sizing method for stand-alone hybrid solar wind system with LPSP technology by using genetic algorithm. Solar Energy 2008;82(4):354-67.



## 6.0. APPENDICES.

**Table 6. 1. The 200 most feasible Optimization Results**

| PV (kW) | V82 | E33 | 6FM200D | Converter (kW) | Initial capital \$ | Total NPC \$ | COE (\$/kWh) |
|---------|-----|-----|---------|----------------|--------------------|--------------|--------------|
| 4070    | 2   |     | 15000   | 1500           | 11,584,000         | 19,738,794   | 0.201        |
| 3885    | 2   | 2   | 15000   | 1500           | 11,835,000         | 19,815,170   | 0.202        |
| 4440    | 1   | 2   | 15000   | 1500           | 11,346,000         | 19,903,396   | 0.203        |
| 4255    | 1   | 4   | 15000   | 1500           | 11,597,000         | 19,936,786   | 0.202        |
| 4440    | 2   | 2   | 10000   | 1500           | 11,916,000         | 19,963,058   | 0.203        |
| 4440    | 1   | 7   | 10000   | 1500           | 11,916,000         | 19,998,020   | 0.203        |
| 4070    | 2   | 1   | 15000   | 1500           | 11,848,000         | 20,004,150   | 0.203        |
| 4070    | 1   | 6   | 15000   | 1500           | 11,848,000         | 20,011,362   | 0.203        |
| 4070    |     | 11  | 15000   | 1500           | 11,848,000         | 20,012,548   | 0.204        |
| 3885    | 1   | 8   | 15000   | 1500           | 12,099,000         | 20,081,870   | 0.204        |
| 3885    | 2   | 3   | 15000   | 1500           | 12,099,000         | 20,082,430   | 0.203        |
| 4255    | 2   | 4   | 10000   | 1500           | 12,167,000         | 20,084,304   | 0.204        |
| 4255    | 1   | 9   | 10000   | 1500           | 12,167,000         | 20,108,452   | 0.204        |
| 3700    |     | 15  | 15000   | 1500           | 12,350,000         | 20,153,432   | 0.205        |
| 4440    | 1   | 3   | 15000   | 1500           | 11,610,000         | 20,156,608   | 0.204        |
| 3700    | 1   | 10  | 15000   | 1500           | 12,350,000         | 20,158,620   | 0.205        |
| 3700    | 2   | 5   | 15000   | 1500           | 12,350,000         | 20,169,092   | 0.205        |
| 4440    |     | 8   | 15000   | 1500           | 11,610,000         | 20,183,110   | 0.205        |
| 3700    | 3   |     | 15000   | 1500           | 12,350,000         | 20,191,870   | 0.205        |
| 4255    | 2   |     | 15000   | 1500           | 11,861,000         | 20,193,304   | 0.203        |
| 4255    | 1   | 5   | 15000   | 1500           | 11,861,000         | 20,206,962   | 0.204        |
| 4440    | 2   | 3   | 10000   | 1500           | 12,180,000         | 20,209,410   | 0.204        |
| 4255    |     | 10  | 15000   | 1500           | 11,861,000         | 20,215,128   | 0.205        |
| 4070    | 3   | 1   | 10000   | 1500           | 12,418,000         | 20,215,580   | 0.205        |
| 4070    | 2   | 6   | 10000   | 1500           | 12,418,000         | 20,216,914   | 0.205        |
| 4070    | 1   | 11  | 10000   | 1500           | 12,418,000         | 20,227,762   | 0.205        |
| 4440    | 1   | 8   | 10000   | 1500           | 12,180,000         | 20,241,594   | 0.205        |
| 4070    | 2   | 2   | 15000   | 1500           | 12,112,000         | 20,266,476   | 0.204        |

|      |   |    |       |      |               |               |       |
|------|---|----|-------|------|---------------|---------------|-------|
| 3515 | 2 | 7  | 15000 | 1500 | \$ 12,601,000 | \$ 20,273,406 | 0.206 |
| 4070 | 1 | 7  | 15000 | 1500 | \$ 12,112,000 | \$ 20,273,582 | 0.205 |
| 3515 | 3 | 2  | 15000 | 1500 | \$ 12,601,000 | \$ 20,309,862 | 0.206 |
| 3885 | 2 | 1  | 20000 | 1500 | \$ 12,321,000 | \$ 20,319,932 | 0.207 |
| 4255 | 3 |    | 10000 | 1500 | \$ 12,431,000 | \$ 20,324,878 | 0.205 |
| 4255 | 2 | 5  | 10000 | 1500 | \$ 12,431,000 | \$ 20,335,510 | 0.205 |
| 3885 | 1 | 9  | 15000 | 1500 | \$ 12,363,000 | \$ 20,345,784 | 0.205 |
| 3885 | 2 | 4  | 15000 | 1500 | \$ 12,363,000 | \$ 20,348,748 | 0.205 |
| 4255 | 1 | 10 | 10000 | 1500 | \$ 12,431,000 | \$ 20,356,730 | 0.206 |
| 3885 | 2 | 8  | 10000 | 1500 | \$ 12,669,000 | \$ 20,358,374 | 0.206 |
| 3885 | 3 | 3  | 10000 | 1500 | \$ 12,669,000 | \$ 20,368,656 | 0.207 |
| 4255 |   | 15 | 10000 | 1500 | \$ 12,431,000 | \$ 20,383,092 | 0.206 |
| 3330 | 2 | 9  | 15000 | 1500 | \$ 12,852,000 | \$ 20,400,470 | 0.208 |
| 3700 | 2 | 3  | 20000 | 1500 | \$ 12,572,000 | \$ 20,409,278 | 0.208 |
| 4440 | 1 | 4  | 15000 | 1500 | \$ 11,874,000 | \$ 20,410,976 | 0.205 |
| 3700 | 1 | 11 | 15000 | 1500 | \$ 12,614,000 | \$ 20,425,832 | 0.206 |
| 4440 | 1 | 1  | 20000 | 1500 | \$ 11,832,000 | \$ 20,432,998 | 0.208 |
| 4440 |   | 9  | 15000 | 1500 | \$ 11,874,000 | \$ 20,433,008 | 0.206 |
| 3330 | 3 | 4  | 15000 | 1500 | \$ 12,852,000 | \$ 20,440,728 | 0.208 |
| 3700 | 2 | 6  | 15000 | 1500 | \$ 12,614,000 | \$ 20,442,096 | 0.206 |
| 4255 | 2 | 1  | 15000 | 1500 | \$ 12,125,000 | \$ 20,449,072 | 0.205 |
| 4255 | 1 | 3  | 20000 | 1500 | \$ 12,083,000 | \$ 20,452,812 | 0.208 |
| 4440 | 2 | 4  | 10000 | 1500 | \$ 12,444,000 | \$ 20,453,530 | 0.206 |
| 4255 | 1 | 6  | 15000 | 1500 | \$ 12,125,000 | \$ 20,463,740 | 0.205 |
| 3700 | 3 | 1  | 15000 | 1500 | \$ 12,614,000 | \$ 20,466,882 | 0.206 |
| 3700 |   | 20 | 10000 | 1500 | \$ 12,920,000 | \$ 20,477,714 | 0.208 |
| 4070 | 2 | 7  | 10000 | 1500 | \$ 12,682,000 | \$ 20,477,902 | 0.207 |
| 4255 |   | 11 | 15000 | 1500 | \$ 12,125,000 | \$ 20,480,360 | 0.206 |
| 4070 | 3 | 2  | 10000 | 1500 | \$ 12,682,000 | \$ 20,481,832 | 0.207 |
| 4440 | 1 | 9  | 10000 | 1500 | \$ 12,444,000 | \$ 20,483,284 | 0.206 |

|      |   |    |       |      |               |               |       |
|------|---|----|-------|------|---------------|---------------|-------|
| 3700 | 1 | 15 | 10000 | 1500 | \$ 12,920,000 | \$ 20,488,318 | 0.208 |
| 4070 | 2 |    | 20000 | 1500 | \$ 12,334,000 | \$ 20,500,706 | 0.207 |
| 3700 | 2 | 10 | 10000 | 1500 | \$ 12,920,000 | \$ 20,503,298 | 0.208 |
| 4070 | 1 | 5  | 20000 | 1500 | \$ 12,334,000 | \$ 20,505,140 | 0.208 |
| 3515 | 2 | 5  | 20000 | 1500 | \$ 12,823,000 | \$ 20,506,298 | 0.209 |
| 4070 |   | 10 | 20000 | 1500 | \$ 12,334,000 | \$ 20,514,152 | 0.209 |
| 3700 | 3 | 5  | 10000 | 1500 | \$ 12,920,000 | \$ 20,520,050 | 0.208 |
| 4070 | 2 | 3  | 15000 | 1500 | \$ 12,376,000 | \$ 20,526,296 | 0.206 |
| 4070 | 1 | 8  | 15000 | 1500 | \$ 12,376,000 | \$ 20,531,522 | 0.206 |
| 3515 | 3 |    | 20000 | 1500 | \$ 12,823,000 | \$ 20,535,926 | 0.209 |
| 3700 | 4 |    | 10000 | 1500 | \$ 12,920,000 | \$ 20,540,506 | 0.209 |
| 3515 | 2 | 8  | 15000 | 1500 | \$ 12,865,000 | \$ 20,553,232 | 0.208 |
| 3885 | 2 | 2  | 20000 | 1500 | \$ 12,585,000 | \$ 20,583,380 | 0.208 |
| 3885 | 1 | 7  | 20000 | 1500 | \$ 12,585,000 | \$ 20,584,448 | 0.209 |
| 4255 | 3 | 1  | 10000 | 1500 | \$ 12,695,000 | \$ 20,585,876 | 0.207 |
| 3515 | 3 | 3  | 15000 | 1500 | \$ 12,865,000 | \$ 20,588,154 | 0.208 |
| 4255 | 2 | 6  | 10000 | 1500 | \$ 12,695,000 | \$ 20,590,406 | 0.207 |
| 4255 | 1 | 11 | 10000 | 1500 | \$ 12,695,000 | \$ 20,604,644 | 0.207 |
| 3885 | 1 | 10 | 15000 | 1500 | \$ 12,627,000 | \$ 20,607,420 | 0.207 |
| 3885 |   | 15 | 15000 | 1500 | \$ 12,627,000 | \$ 20,608,766 | 0.207 |
| 3885 | 2 | 5  | 15000 | 1500 | \$ 12,627,000 | \$ 20,610,112 | 0.207 |
| 3330 |   | 20 | 15000 | 1500 | \$ 13,116,000 | \$ 20,613,674 | 0.209 |
| 3885 | 3 |    | 15000 | 1500 | \$ 12,627,000 | \$ 20,623,604 | 0.207 |
| 3885 | 2 | 9  | 10000 | 1500 | \$ 12,933,000 | \$ 20,625,514 | 0.208 |
| 4440 | 2 |    | 15000 | 1500 | \$ 12,138,000 | \$ 20,633,586 | 0.206 |
| 3885 | 3 | 4  | 10000 | 1500 | \$ 12,933,000 | \$ 20,637,188 | 0.209 |
| 3330 | 1 | 15 | 15000 | 1500 | \$ 13,116,000 | \$ 20,648,024 | 0.209 |
| 4440 | 1 | 5  | 15000 | 1500 | \$ 12,138,000 | \$ 20,657,020 | 0.206 |
| 3330 | 3 | 2  | 20000 | 1500 | \$ 13,074,000 | \$ 20,662,550 | 0.21  |
| 3700 | 1 | 9  | 20000 | 1500 | \$ 12,836,000 | \$ 20,666,284 | 0.209 |

|      |   |    |       |      |               |               |       |
|------|---|----|-------|------|---------------|---------------|-------|
| 3700 | 2 | 4  | 20000 | 1500 | \$ 12,836,000 | \$ 20,676,960 | 0.209 |
| 4440 | 1 | 2  | 20000 | 1500 | \$ 12,096,000 | \$ 20,681,350 | 0.209 |
| 3330 | 2 | 10 | 15000 | 1500 | \$ 13,116,000 | \$ 20,683,644 | 0.21  |
| 4440 |   | 10 | 15000 | 1500 | \$ 12,138,000 | \$ 20,684,684 | 0.207 |
| 4440 | 3 |    | 10000 | 1500 | \$ 12,708,000 | \$ 20,686,794 | 0.208 |
| 4255 | 2 | 2  | 15000 | 1500 | \$ 12,389,000 | \$ 20,698,924 | 0.206 |
| 4440 | 2 | 5  | 10000 | 1500 | \$ 12,708,000 | \$ 20,700,138 | 0.208 |
| 4255 | 1 | 4  | 20000 | 1500 | \$ 12,347,000 | \$ 20,703,166 | 0.209 |
| 3700 | 2 | 7  | 15000 | 1500 | \$ 12,878,000 | \$ 20,712,066 | 0.208 |
| 4255 | 1 | 7  | 15000 | 1500 | \$ 12,389,000 | \$ 20,714,028 | 0.207 |
| 4440 |   | 7  | 20000 | 1500 | \$ 12,096,000 | \$ 20,715,010 | 0.21  |
| 3330 | 3 | 5  | 15000 | 1500 | \$ 13,116,000 | \$ 20,721,832 | 0.21  |
| 4440 | 1 | 10 | 10000 | 1500 | \$ 12,708,000 | \$ 20,724,192 | 0.208 |
| 4255 |   | 9  | 20000 | 1500 | \$ 12,347,000 | \$ 20,727,584 | 0.21  |
| 3700 | 3 | 2  | 15000 | 1500 | \$ 12,878,000 | \$ 20,739,938 | 0.208 |
| 4070 | 2 | 8  | 10000 | 1500 | \$ 12,946,000 | \$ 20,740,480 | 0.209 |
| 4070 | 3 | 3  | 10000 | 1500 | \$ 12,946,000 | \$ 20,747,360 | 0.209 |
| 4440 |   | 15 | 10000 | 1500 | \$ 12,708,000 | \$ 20,756,168 | 0.208 |
| 3515 | 1 | 11 | 20000 | 1500 | \$ 13,087,000 | \$ 20,757,432 | 0.211 |
| 4070 | 2 | 1  | 20000 | 1500 | \$ 12,598,000 | \$ 20,758,854 | 0.209 |
| 3330 | 4 |    | 15000 | 1500 | \$ 13,116,000 | \$ 20,763,562 | 0.21  |
| 4070 | 1 | 6  | 20000 | 1500 | \$ 12,598,000 | \$ 20,766,374 | 0.209 |
| 3885 | 2 |    | 25000 | 1500 | \$ 12,807,000 | \$ 20,768,612 | 0.212 |
| 3700 | 2 | 11 | 10000 | 1500 | \$ 13,184,000 | \$ 20,773,114 | 0.21  |
| 4070 |   | 11 | 20000 | 1500 | \$ 12,598,000 | \$ 20,774,470 | 0.21  |
| 4070 | 2 | 4  | 15000 | 1500 | \$ 12,640,000 | \$ 20,779,530 | 0.207 |
| 3515 | 2 | 6  | 20000 | 1500 | \$ 13,087,000 | \$ 20,782,214 | 0.21  |
| 4070 | 1 | 9  | 15000 | 1500 | \$ 12,640,000 | \$ 20,785,168 | 0.208 |
| 3700 | 3 | 6  | 10000 | 1500 | \$ 13,184,000 | \$ 20,791,880 | 0.211 |
| 3700 | 4 | 1  | 10000 | 1500 | \$ 13,184,000 | \$ 20,813,736 | 0.211 |

|      |   |    |       |      |               |               |       |
|------|---|----|-------|------|---------------|---------------|-------|
| 3515 | 3 | 1  | 20000 | 1500 | \$ 13,087,000 | \$ 20,814,926 | 0.211 |
| 3515 | 2 | 9  | 15000 | 1500 | \$ 13,129,000 | \$ 20,827,298 | 0.21  |
| 3885 | 1 | 8  | 20000 | 1500 | \$ 12,849,000 | \$ 20,844,024 | 0.21  |
| 4255 | 3 | 2  | 10000 | 1500 | \$ 12,959,000 | \$ 20,847,766 | 0.209 |
| 3885 | 2 | 3  | 20000 | 1500 | \$ 12,849,000 | \$ 20,848,220 | 0.21  |
| 4255 | 2 | 7  | 10000 | 1500 | \$ 12,959,000 | \$ 20,848,526 | 0.209 |
| 3700 | 2 | 2  | 25000 | 1500 | \$ 13,058,000 | \$ 20,856,074 | 0.212 |
| 3885 | 1 | 11 | 15000 | 1500 | \$ 12,891,000 | \$ 20,863,172 | 0.209 |
| 3515 | 3 | 4  | 15000 | 1500 | \$ 13,129,000 | \$ 20,863,872 | 0.21  |
| 3145 | 3 | 7  | 15000 | 1500 | \$ 13,367,000 | \$ 20,868,840 | 0.212 |
| 4440 | 2 | 1  | 15000 | 1500 | \$ 12,402,000 | \$ 20,870,074 | 0.207 |
| 3885 | 2 | 6  | 15000 | 1500 | \$ 12,891,000 | \$ 20,872,086 | 0.209 |
| 3885 |   | 20 | 10000 | 1500 | \$ 13,197,000 | \$ 20,877,856 | 0.21  |
| 3885 | 1 | 15 | 10000 | 1500 | \$ 13,197,000 | \$ 20,882,964 | 0.21  |
| 3885 | 3 | 1  | 15000 | 1500 | \$ 12,891,000 | \$ 20,891,720 | 0.209 |
| 3885 | 2 | 10 | 10000 | 1500 | \$ 13,197,000 | \$ 20,892,620 | 0.21  |
| 4440 | 1 | 6  | 15000 | 1500 | \$ 12,402,000 | \$ 20,895,478 | 0.208 |
| 4255 | 1 | 2  | 25000 | 1500 | \$ 12,569,000 | \$ 20,903,264 | 0.213 |
| 3330 | 2 | 8  | 20000 | 1500 | \$ 13,338,000 | \$ 20,903,510 | 0.212 |
| 3885 | 3 | 5  | 10000 | 1500 | \$ 13,197,000 | \$ 20,906,270 | 0.211 |
| 3145 | 4 | 2  | 15000 | 1500 | \$ 13,367,000 | \$ 20,914,768 | 0.212 |
| 4440 | 1 | 3  | 20000 | 1500 | \$ 12,360,000 | \$ 20,921,130 | 0.21  |
| 3885 | 4 |    | 10000 | 1500 | \$ 13,197,000 | \$ 20,923,170 | 0.211 |
| 3700 |   | 15 | 20000 | 1500 | \$ 13,100,000 | \$ 20,924,010 | 0.211 |
| 4440 |   | 11 | 15000 | 1500 | \$ 12,402,000 | \$ 20,924,324 | 0.208 |
| 3700 | 1 | 10 | 20000 | 1500 | \$ 13,100,000 | \$ 20,931,852 | 0.211 |
| 4255 | 2 |    | 20000 | 1500 | \$ 12,611,000 | \$ 20,940,610 | 0.209 |
| 4440 | 3 | 1  | 10000 | 1500 | \$ 12,972,000 | \$ 20,943,430 | 0.21  |
| 3700 | 2 | 5  | 20000 | 1500 | \$ 13,100,000 | \$ 20,944,588 | 0.211 |
| 4255 | 2 | 3  | 15000 | 1500 | \$ 12,653,000 | \$ 20,945,170 | 0.208 |

|      |   |    |       |      |               |               |       |
|------|---|----|-------|------|---------------|---------------|-------|
| 3330 | 3 | 3  | 20000 | 1500 | \$ 13,338,000 | \$ 20,946,124 | 0.212 |
| 3515 | 3 | 8  | 10000 | 1500 | \$ 13,435,000 | \$ 20,948,666 | 0.212 |
| 4440 | 2 | 6  | 10000 | 1500 | \$ 12,972,000 | \$ 20,951,904 | 0.21  |
| 4255 | 1 | 5  | 20000 | 1500 | \$ 12,611,000 | \$ 20,955,936 | 0.21  |
| 4440 |   | 8  | 20000 | 1500 | \$ 12,360,000 | \$ 20,956,880 | 0.211 |
| 4255 | 1 | 8  | 15000 | 1500 | \$ 12,653,000 | \$ 20,957,406 | 0.208 |
| 4070 | 1 | 4  | 25000 | 1500 | \$ 12,820,000 | \$ 20,958,834 | 0.213 |
| 3330 | 2 | 11 | 15000 | 1500 | \$ 13,380,000 | \$ 20,962,442 | 0.212 |
| 3700 | 3 |    | 20000 | 1500 | \$ 13,100,000 | \$ 20,966,410 | 0.211 |
| 4440 | 1 | 11 | 10000 | 1500 | \$ 12,972,000 | \$ 20,969,032 | 0.21  |
| 4255 |   | 10 | 20000 | 1500 | \$ 12,611,000 | \$ 20,972,136 | 0.211 |
| 3515 | 4 | 3  | 10000 | 1500 | \$ 13,435,000 | \$ 20,975,818 | 0.213 |
| 3700 | 2 | 8  | 15000 | 1500 | \$ 13,142,000 | \$ 20,981,742 | 0.21  |
| 3145 | 1 | 15 | 20000 | 1500 | \$ 13,589,000 | \$ 20,984,280 | 0.214 |
| 3330 | 3 | 6  | 15000 | 1500 | \$ 13,380,000 | \$ 21,002,086 | 0.212 |
| 4070 | 2 | 9  | 10000 | 1500 | \$ 13,210,000 | \$ 21,004,098 | 0.211 |
| 4070 | 3 | 4  | 10000 | 1500 | \$ 13,210,000 | \$ 21,012,152 | 0.211 |
| 3700 | 3 | 3  | 15000 | 1500 | \$ 13,142,000 | \$ 21,012,664 | 0.21  |
| 4070 | 2 | 2  | 20000 | 1500 | \$ 12,862,000 | \$ 21,017,170 | 0.21  |
| 4070 | 1 | 7  | 20000 | 1500 | \$ 12,862,000 | \$ 21,022,690 | 0.211 |
| 3145 | 2 | 10 | 20000 | 1500 | \$ 13,589,000 | \$ 21,030,848 | 0.214 |
| 3885 |   | 11 | 25000 | 1500 | \$ 13,071,000 | \$ 21,031,440 | 0.214 |
| 4070 | 2 | 5  | 15000 | 1500 | \$ 12,904,000 | \$ 21,033,014 | 0.209 |
| 3330 |   | 25 | 10000 | 1500 | \$ 13,686,000 | \$ 21,033,030 | 0.213 |
| 4070 | 1 | 10 | 15000 | 1500 | \$ 12,904,000 | \$ 21,035,448 | 0.209 |
| 3885 | 1 | 6  | 25000 | 1500 | \$ 13,071,000 | \$ 21,035,638 | 0.214 |
| 3515 |   | 20 | 15000 | 1500 | \$ 13,393,000 | \$ 21,041,750 | 0.211 |
| 3885 | 2 | 1  | 25000 | 1500 | \$ 13,071,000 | \$ 21,041,914 | 0.213 |
| 4070 | 3 |    | 15000 | 1500 | \$ 12,904,000 | \$ 21,042,988 | 0.209 |
| 4070 |   | 15 | 15000 | 1500 | \$ 12,904,000 | \$ 21,042,990 | 0.21  |

|      |   |    |       |      |               |               |       |
|------|---|----|-------|------|---------------|---------------|-------|
| 3330 | 4 | 1  | 15000 | 1500 | \$ 13,380,000 | \$ 21,043,756 | 0.212 |
| 3515 | 2 | 7  | 20000 | 1500 | \$ 13,351,000 | \$ 21,057,454 | 0.212 |
| 3330 | 1 | 20 | 10000 | 1500 | \$ 13,686,000 | \$ 21,058,198 | 0.214 |
| 3700 | 3 | 7  | 10000 | 1500 | \$ 13,448,000 | \$ 21,063,482 | 0.213 |
| 3515 | 1 | 15 | 15000 | 1500 | \$ 13,393,000 | \$ 21,067,502 | 0.212 |
| 3145 | 3 | 5  | 20000 | 1500 | \$ 13,589,000 | \$ 21,080,528 | 0.214 |
| 3330 | 2 | 15 | 10000 | 1500 | \$ 13,686,000 | \$ 21,085,802 | 0.214 |
| 3700 | 4 | 2  | 10000 | 1500 | \$ 13,448,000 | \$ 21,087,278 | 0.213 |
| 3515 | 3 | 2  | 20000 | 1500 | \$ 13,351,000 | \$ 21,093,668 | 0.212 |
| 3330 | 3 | 1  | 25000 | 1500 | \$ 13,560,000 | \$ 21,094,522 | 0.215 |
| 3515 | 2 | 10 | 15000 | 1500 | \$ 13,393,000 | \$ 21,101,714 | 0.212 |
| 3885 | 1 | 9  | 20000 | 1500 | \$ 13,113,000 | \$ 21,105,630 | 0.212 |
| 4440 | 2 | 2  | 15000 | 1500 | \$ 12,666,000 | \$ 21,106,370 | 0.209 |
| 4255 | 2 | 8  | 10000 | 1500 | \$ 13,223,000 | \$ 21,106,476 | 0.211 |
| 3885 | 2 | 4  | 20000 | 1500 | \$ 13,113,000 | \$ 21,107,784 | 0.211 |
| 4255 | 3 | 3  | 10000 | 1500 | \$ 13,223,000 | \$ 21,108,842 | 0.211 |
| 3330 | 3 | 10 | 10000 | 1500 | \$ 13,686,000 | \$ 21,115,828 | 0.215 |
| 3700 | 1 | 8  | 25000 | 1500 | \$ 13,322,000 | \$ 21,117,254 | 0.214 |
| 4440 | 1 | 7  | 15000 | 1500 | \$ 12,666,000 | \$ 21,126,760 | 0.209 |
| 3700 | 2 | 3  | 25000 | 1500 | \$ 13,322,000 | \$ 21,128,650 | 0.214 |
| 3145 | 4 |    | 20000 | 1500 | \$ 13,589,000 | \$ 21,134,166 | 0.214 |
| 3885 | 2 | 7  | 15000 | 1500 | \$ 13,155,000 | \$ 21,135,624 | 0.21  |